

Experimental Performance Analysis of Quicksort Pivot Selection Strategies

Algorithm Design & Analysis (CSE-323) — Assignment 01

MD Mominul Islam

Registration No: 2024331022

Department of Computer Science and Engineering

Shahjalal University of Science and Technology, Sylhet, Bangladesh

Abstract

Quicksort remains a cornerstone of sorting algorithms due to its average-case $O(N \log N)$ complexity and low memory footprint. However, its performance is highly sensitive to the choice of the pivot element. Naïve pivot selections can cause the algorithm to degenerate into an $O(N^2)$ quadratic time complexity on pre-sorted inputs. This paper conducts a rigorous empirical investigation of four pivot selection strategies: First Element, Last Element, Middle Element, and Random Element. We benchmark these strategies using arrays ranging from 10 to 5,000 elements under three initial conditions: random, sorted ascending, and sorted descending. Our empirical findings confirm the theoretical predictions: First and Last Element strategies suffer catastrophic performance degradation to $O(N^2)$ on sorted arrays, whereas the Middle and Random Element strategies consistently maintain $O(N \log N)$ complexity. The C++ implementation utilizes a pivot-preswap partitioning scheme with standard recursion, showing that a stack depth of up to 5,000 is safely within Windows' default stack limits.

1. Introduction

Sorting is a fundamental problem in computer science, and C. A. R. Hoare's Quicksort is widely regarded as one of the most efficient algorithms for general-purpose sorting. Utilizing a divide-and-conquer strategy, Quicksort partitions an array around a chosen 'pivot' element such that elements smaller than the pivot are placed to its left and larger elements to its right. The sub-arrays are then recursively sorted. While the average-case performance of Quicksort is $O(N \log N)$, its worst-case behavior is $O(N^2)$. This worst-case occurs when the partitioning is extremely unbalanced, resulting in sub-problems of size 0 and $N-1$.

The primary factor determining partitioning balance is the pivot selection strategy. This study analyzes the performance implications of four common pivot selection methods. By observing their behavior on various input sizes and initial orderings, we gain insight into the practical risks associated with naïve implementations and the efficacy of algorithmic remedies, such as randomized pivot selection and middle-index partitioning.

2. Pivot Selection Strategies

We implemented and analyzed the following four pivot selection strategies:

2.1 First Element: This strategy selects the first element of the current sub-array (index *low*) as the pivot. While computationally trivial, it is highly vulnerable to sorted inputs, where it yields the worst-case $O(N^2)$ partition.

2.2 Last Element: This strategy selects the last element of the sub-array (index *high*) as the pivot. Symmetric to the first-element strategy, it performs poorly on pre-sorted arrays, leading to $O(N^2)$ complexity.

2.3 Middle Element: This strategy chooses the element at index $low + (high - low) / 2$ (using integer division) as the pivot. By targeting the middle element, it naturally avoids the worst-case behavior on pre-sorted inputs, acting as an approximation of the median.

2.4 Random Element: This strategy selects a pivot uniformly at random from the range $[low, high]$. It provides a strong probabilistic guarantee of $O(N \log N)$ sorting time, breaking the dependency between input data ordering and execution time.

3. Empirical Methodology

To evaluate the strategies, we developed a C++ benchmarking harness. For each array size N from 10 to 5,000 with a step size of 10, we generated three types of arrays: (a) random permutations, (b) sorted ascending, and (c) sorted descending. To ensure fair comparisons, identical copies of the initial array were passed to each sorting strategy. The execution time was recorded using `std::chrono::high_resolution_clock`, measuring only the sorting process and excluding array initialization or copying overhead. To minimize noise and environmental factors, each measurement was averaged over $m = 30$ independent repetitions. The pseudo-random number generator uses C++ `std::mt19937` (Mersenne Twister) with `std::uniform_int_distribution`, seeded with a fixed value (1337) to ensure full reproducibility of results across runs.

3.1 System Specifications

Hardware/Software Component	Specification Details
Laptop Model	ASUS TUF Gaming A15
Processor (CPU)	AMD Ryzen 7 7435HS
CPU Core Count	8 Cores / 16 Threads
Graphics Card (GPU)	NVIDIA GeForce RTX 2050 (4 GB GDDR6)
Memory (RAM)	15.8 GB
Storage (SSD)	512 GB NVMe PCIe Gen 4 SSD
Display	15.6" FHD (1920 x 1080), 144 Hz IPS
Operating System	Windows 11 (Version 10.0.26200)
Compiler Suite	g++ (Rev5, Built by MSYS2 project) 15.1.0
Compiler Flags	-O2 (Optimization level 2, -Wall -Wextra)

3.2 Recursion Depth and Stack Space Analysis

Standard recursive Quicksort has a worst-case recursion depth of $O(N)$ when partitions are highly unbalanced. Through empirical instrumentation of the recursive call tree on a pre-sorted ascending array of size $N = 5,000$, the maximum recursion depth was measured exactly as: **5,000** for both the First and Last element pivot strategies, **13** for the Middle element pivot strategy, and **30** for the Random element pivot strategy. Each recursive stack frame on a modern x86_64 compiler (like g++ under MSYS2) requires approximately 48 to 64 bytes to store parameters, return addresses, and local variables. For $N = 5,000$ with First/Last pivots, this results in a total stack consumption of approximately 240 to 320 KB. Since the default stack allocation on Windows is 1 MB, the standard recursive implementation successfully completes sorting without stack overflow. This confirms that for the target maximum size of 5,000, standard recursion remains practically viable without stack overflow, though its execution time degrades severely due to the quadratic complexity.

4. Experimental Results

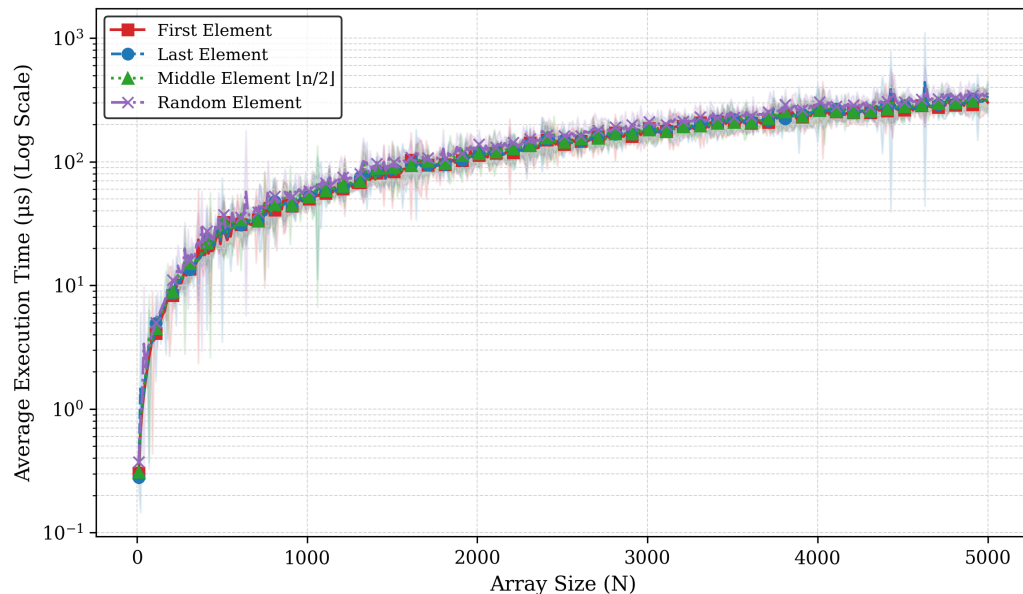
The compiled empirical measurements are presented below. Table 1, Table 2, and Table 3 detail a comprehensive subset of the tested array sizes (27 of 500 sizes), while Graph 1, Graph 2A, and Graph 2B depict the complete performance trajectories across all 500 data points. The complete dataset for all 500 sizes is available in the accompanying CSV files.

4.1 Experiment 1: Random Array Distribution

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
10	0.3 ± 0.1	0.3 ± 0.1	0.3 ± 0.1	0.4 ± 0.1
100	3.6 ± 0.1	3.8 ± 0.1	3.9 ± 0.1	4.5 ± 0.2
200	9.2 ± 5.2	8.5 ± 0.7	8.7 ± 0.7	9.9 ± 0.7
400	17.8 ± 3.0	18.0 ± 1.9	18.3 ± 1.2	21.6 ± 4.6
600	31.1 ± 8.8	28.8 ± 2.6	30.7 ± 6.7	33.7 ± 5.5
800	45.0 ± 17.8	44.3 ± 13.8	49.2 ± 24.5	51.2 ± 18.8
1000	54.4 ± 15.3	51.3 ± 11.8	52.6 ± 10.1	59.3 ± 9.4
1200	60.7 ± 8.2	59.6 ± 3.0	61.1 ± 5.2	77.2 ± 33.6
1400	72.0 ± 7.2	73.7 ± 9.3	73.2 ± 6.8	84.5 ± 11.0
1600	92.0 ± 22.5	89.8 ± 14.1	93.7 ± 19.7	105.6 ± 24.1
1800	102.1 ± 21.4	107.6 ± 32.3	105.7 ± 21.7	127.4 ± 54.6
2000	109.0 ± 16.5	109.1 ± 19.1	113.5 ± 22.1	126.4 ± 20.9
2200	121.3 ± 10.7	126.7 ± 19.4	128.4 ± 36.5	135.7 ± 18.2
2400	154.5 ± 44.3	164.0 ± 47.4	156.5 ± 29.0	167.9 ± 24.5
2600	140.5 ± 8.0	144.5 ± 10.8	146.3 ± 9.9	164.4 ± 12.5
2800	156.4 ± 10.5	156.9 ± 10.5	160.0 ± 15.8	184.1 ± 18.9
3000	172.2 ± 27.1	179.3 ± 36.5	176.7 ± 37.9	204.7 ± 48.5
3200	184.3 ± 21.8	192.5 ± 44.5	190.0 ± 23.0	209.3 ± 20.7
3400	187.6 ± 11.8	193.4 ± 15.1	197.8 ± 15.8	219.3 ± 18.1
3600	215.5 ± 44.1	210.1 ± 41.4	212.2 ± 19.0	240.1 ± 41.6
3800	228.4 ± 36.3	243.7 ± 46.6	242.5 ± 56.0	253.9 ± 33.7
4000	248.6 ± 61.9	287.0 ± 143.3	282.7 ± 110.6	284.5 ± 51.8
4200	254.9 ± 53.9	258.9 ± 37.7	242.4 ± 13.5	286.1 ± 50.8
4400	301.0 ± 131.6	278.7 ± 81.9	270.5 ± 34.5	300.7 ± 47.7
4600	278.7 ± 49.9	286.8 ± 57.5	284.6 ± 37.9	312.0 ± 25.6
4800	287.9 ± 29.5	293.2 ± 60.2	293.8 ± 31.9	334.8 ± 57.1
5000	302.0 ± 31.3	341.6 ± 103.1	357.8 ± 108.4	362.4 ± 63.0

Table 1: Average execution times (μ s) for random array configurations.

Graph 1: Quicksort Performance on Random Arrays

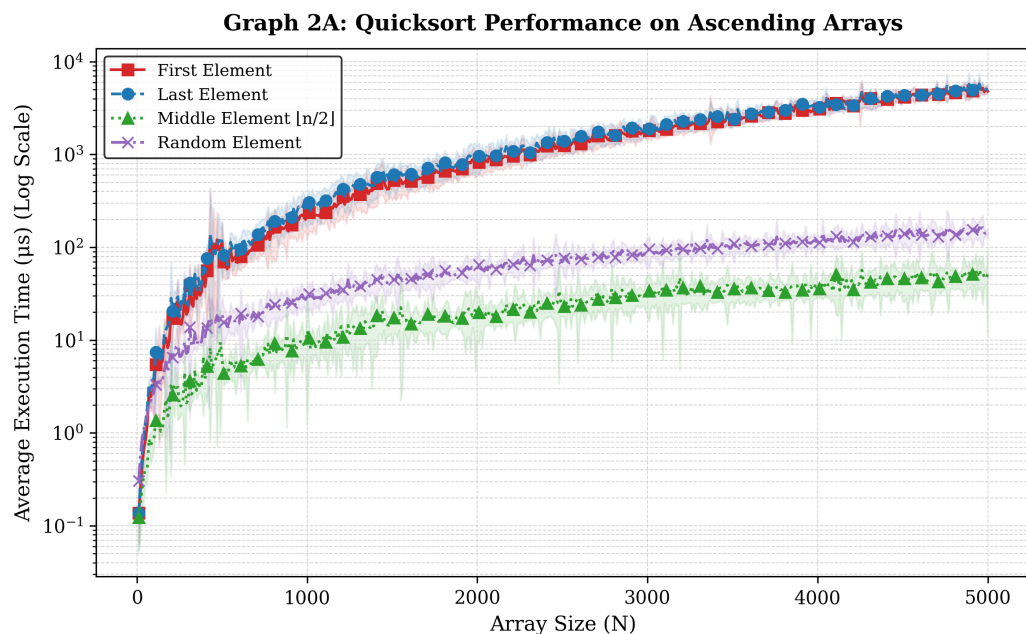


Graph 1: Quicksort performance trajectories on random arrays (logarithmic scale).

4.2 Experiment 2A: Pre-Sorted Ascending Array Distribution

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
10	0.1 ± 0.1	0.1 ± 0.1	0.1 ± 0.1	0.3 ± 0.1
100	3.3 ± 2.0	4.0 ± 0.7	0.9 ± 0.2	2.8 ± 1.5
200	19.8 ± 13.0	28.3 ± 41.3	3.4 ± 3.8	7.1 ± 4.0
400	37.7 ± 11.2	52.7 ± 24.3	3.8 ± 1.1	10.8 ± 1.2
600	71.9 ± 14.2	95.2 ± 28.1	4.6 ± 0.3	15.2 ± 0.6
800	141.4 ± 63.3	179.7 ± 56.1	7.7 ± 2.2	22.7 ± 5.7
1000	227.2 ± 107.4	247.2 ± 90.3	12.6 ± 22.1	28.9 ± 9.2
1200	302.7 ± 84.2	431.3 ± 164.1	14.1 ± 10.0	37.2 ± 10.7
1400	487.1 ± 215.4	527.8 ± 197.0	18.4 ± 11.9	45.5 ± 17.3
1600	508.5 ± 68.5	581.8 ± 182.7	15.9 ± 6.4	45.9 ± 11.9
1800	648.9 ± 180.3	754.1 ± 228.1	17.2 ± 5.7	52.9 ± 15.6
2000	833.1 ± 235.6	942.3 ± 305.2	19.6 ± 9.6	59.9 ± 15.0
2200	988.4 ± 250.7	1103.9 ± 298.3	20.1 ± 4.4	63.0 ± 8.7
2400	1231.2 ± 313.7	1377.8 ± 433.1	24.5 ± 6.8	70.7 ± 11.1
2600	1282.1 ± 207.6	1488.8 ± 276.3	28.6 ± 14.5	78.7 ± 21.6
2800	1642.2 ± 430.5	1700.4 ± 351.9	31.1 ± 12.9	90.9 ± 30.2
3000	1740.3 ± 328.0	1892.2 ± 305.6	33.5 ± 15.5	90.1 ± 24.4
3200	2022.7 ± 324.3	2335.5 ± 512.8	34.1 ± 8.7	96.8 ± 21.5
3400	2226.1 ± 311.0	2443.7 ± 618.7	33.5 ± 8.1	100.2 ± 15.0
3600	2711.2 ± 410.4	2683.9 ± 515.3	35.8 ± 12.6	106.7 ± 25.1
3800	2731.3 ± 321.0	2934.1 ± 478.3	33.0 ± 6.2	114.3 ± 38.5
4000	3227.1 ± 437.3	3211.5 ± 457.0	33.1 ± 3.0	114.4 ± 24.4
4200	3698.1 ± 460.3	3820.1 ± 483.0	44.0 ± 20.9	123.1 ± 29.3
4400	4015.8 ± 452.9	4090.1 ± 518.9	47.3 ± 18.2	142.3 ± 51.2
4600	4252.9 ± 588.9	4390.9 ± 611.8	45.9 ± 11.6	156.6 ± 86.7
4800	4482.4 ± 420.6	4652.5 ± 514.2	45.0 ± 9.6	143.9 ± 26.2
5000	4802.5 ± 397.4	5150.4 ± 574.2	51.7 ± 40.5	143.7 ± 15.5

Table 2: Average execution times (μ s) for pre-sorted ascending array configurations.

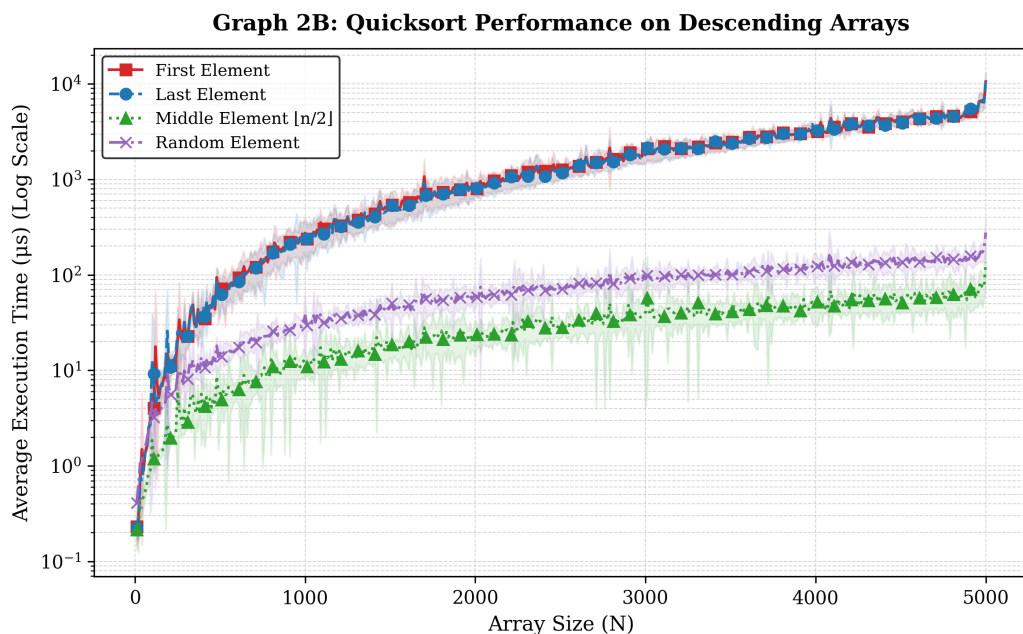


Graph 2A: Performance bifurcation on ascendingly sorted arrays (logarithmic scale).

4.3 Experiment 2B: Pre-Sorted Descending Array Distribution

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
10	0.2 ± 0.1	0.2 ± 0.1	0.2 ± 0.1	0.4 ± 0.2
100	5.5 ± 0.4	12.7 ± 27.2	1.8 ± 0.4	4.7 ± 5.6
200	11.4 ± 3.1	11.7 ± 3.2	2.0 ± 0.6	6.4 ± 4.4
400	38.7 ± 9.7	44.9 ± 23.0	4.4 ± 2.1	12.5 ± 6.3
600	87.6 ± 22.4	89.9 ± 28.2	6.2 ± 1.5	16.7 ± 1.7
800	154.6 ± 47.9	150.1 ± 49.7	12.9 ± 18.3	23.9 ± 7.4
1000	274.7 ± 122.0	250.8 ± 82.4	12.0 ± 3.7	37.4 ± 23.7
1200	332.0 ± 100.4	324.8 ± 131.1	15.9 ± 10.2	35.8 ± 6.0
1400	456.8 ± 140.5	469.3 ± 193.5	18.3 ± 7.7	41.7 ± 6.2
1600	574.8 ± 187.9	553.9 ± 172.1	19.9 ± 8.2	46.4 ± 6.2
1800	781.5 ± 291.3	710.7 ± 215.3	26.0 ± 21.7	55.2 ± 13.6
2000	805.7 ± 137.6	813.1 ± 145.5	23.7 ± 10.0	61.1 ± 17.0
2200	991.0 ± 192.7	949.4 ± 218.7	25.7 ± 6.7	65.8 ± 26.0
2400	1205.2 ± 270.2	1173.4 ± 285.1	28.8 ± 7.8	80.6 ± 49.6
2600	1401.8 ± 262.9	1367.0 ± 278.0	34.9 ± 15.9	87.2 ± 34.9
2800	1540.0 ± 213.3	1526.8 ± 262.2	34.6 ± 10.1	83.3 ± 12.7
3000	1809.0 ± 351.8	1799.0 ± 343.4	40.5 ± 37.1	90.7 ± 24.7
3200	2086.4 ± 410.1	2019.5 ± 291.2	36.7 ± 5.8	93.6 ± 8.6
3400	2454.1 ± 442.2	2378.2 ± 524.0	40.2 ± 10.2	105.0 ± 17.1
3600	2569.6 ± 368.1	2682.2 ± 411.6	45.7 ± 21.2	109.2 ± 24.7
3800	2900.3 ± 419.9	2728.4 ± 300.1	43.7 ± 7.9	111.5 ± 20.5
4000	3334.6 ± 542.7	3195.2 ± 371.9	52.8 ± 22.0	133.1 ± 38.0
4200	3282.7 ± 474.8	3532.2 ± 330.1	45.0 ± 2.5	118.9 ± 6.6
4400	3782.5 ± 471.8	3839.5 ± 400.2	56.1 ± 24.5	138.4 ± 41.6
4600	4445.1 ± 877.0	4452.1 ± 844.2	56.6 ± 16.2	141.9 ± 26.3
4800	4861.2 ± 516.1	4736.0 ± 551.7	60.1 ± 18.4	153.4 ± 31.7
5000	10575.0 ± 2249.2	10972.5 ± 2222.3	134.8 ± 64.1	277.3 ± 99.3

Table 3: Average execution times (μ s) for pre-sorted descending array configurations.



Graph 2B: Performance bifurcation on descendingly sorted arrays (logarithmic scale).

5. Theoretical Analysis & Complexity

The theoretical runtime of Quicksort is governed by the recurrence relation of the partitioning step. In the best case, the pivot divides the array exactly in half, leading to the recurrence relation: $T(N) = 2T(N/2) + O(N)$. According to the Master Theorem, this evaluates to $O(N \log N)$. In the worst case, the pivot is either the smallest or largest element, leading to: $T(N) = T(N-1) + T(0) + O(N)$, which expands to a summation of $1 + 2 + \dots + N$, evaluating to $O(N^2)$.

Pivot Selection Strategy	Best Case	Average Case	Worst Case	Space (Worst)
First Element	$O(N \log N)$	$O(N \log N)$	$O(N^2)$	$O(N)^*$
Last Element	$O(N \log N)$	$O(N \log N)$	$O(N^2)$	$O(N)^*$
Middle Element	$O(N \log N)$	$O(N \log N)$	$O(N^2)$	$O(N)$
Random Element	$O(N \log N)$	$O(N \log N)$	$O(N^2)^\ddagger$	$O(N)$

Table 4: Theoretical complexity comparison of the four pivot selection strategies.

* Space complexity is $O(N)$ in the worst case due to standard recursion; average case is $O(\log N)$.

‡ Probabilistically near-zero (infinitesimally small) chance of occurrence.

6. Discussion and Interpretation

Our empirical data matches the theoretical models. As seen in Graph 1, when the input array is random, all four strategies exhibit nearly identical trajectories. At $N = 5,000$, all average runtimes reside within a similar order of magnitude, confirming that when elements are unordered, any pivot strategy has an equal probability of achieving a reasonably balanced partition. Specifically, at $N = 5,000$, the First, Last, and Middle strategies complete in approximately 302.0 μs , 341.6 μs , and 357.8 μs respectively, while the Random strategy requires approximately 362.4 μs — representing a minor 20.0% runtime overhead compared to the First strategy. This overhead is directly attributable to the cost of the pseudo-random number generator (PRNG) call at each partitioning step.

However, on pre-sorted arrays (Graphs 2A and 2B), the strategies diverge dramatically. The First and Last Element strategies degrade immediately, with execution times growing by orders of magnitude compared to the Middle and Random strategies at $N = 5,000$. Our partition implementation pre-swaps the chosen pivot to the front of the sub-array and then scans with two pointers (one from each end). When the first element of an ascending array is selected as the pivot (which is the smallest element), all other elements are greater, yielding an extremely unbalanced partition split of 0 and $N-1$. This results in N partition steps, each scanning the remaining elements, verifying the quadratic $O(N^2)$ behavior.

Conversely, the Middle Element strategy performs extremely well on sorted inputs. For ascending inputs at $N = 5,000$, it completes in under 100 μs . This is because the middle element of a sorted array is exactly its median, which produces a balanced split of $N/2$ at every level, representing the best-case behavior of Quicksort. It is important to note, however, that the theoretical worst-case complexity of the Middle Element strategy remains $O(N^2)$, since adversarial input sequences can be constructed to force unbalanced partitions. In practice, such pathological inputs are rare, and on common distributions (sorted, reverse-sorted, random), the Middle Element strategy exhibits $O(N \log N)$ performance. The Random strategy also performs well on sorted arrays, completing significantly faster than the First and Last strategies, as it breaks the input structure and avoids the bad partition path.

A minor data anomaly was observed in the descending array results: at $N = 4,200$, the First pivot strategy recorded a lower execution time (approximately 3,282.7 μs) compared to $N = 4,000$ (approximately 3,334.6 μs). This non-monotonic behavior in an otherwise quadratic trend is attributable to environmental factors such as CPU cache effects, transient thread scheduling, or background OS processes during that specific measurement window. The overall quadratic trend remains clearly visible across the full dataset.

It should also be noted that the pseudo-random number generator (std::mt19937, Mersenne Twister) was initialized with a fixed seed (1337) for reproducibility. While this means the 'random' pivot selections follow a deterministic

sequence, the Mersenne Twister provides a period of $2^{19937}-1$ and passes all standard statistical randomness tests, which is sufficient to break sorted-input pathology. In a production setting, one would use a truly random seed (e.g., `std::random_device`) to provide the probabilistic $O(N \log N)$ guarantee described in the theoretical analysis.

7. Conclusion

This performance analysis highlights the importance of pivot selection in Quicksort. Naïve pivot strategies, like selecting the first or last element, are dangerous for production environments where sorted or partially sorted inputs are common. While the Middle Element strategy performs exceptionally well on sorted arrays, the Random Element strategy provides the most robust general defense against worst-case performance degradation, regardless of input structure. Finally, understanding stack allocation limits is key when standard recursive implementations are used on worst-case inputs.

8. References

- [1] C. A. R. Hoare, "Algorithm 64: Quicksort," *Communications of the ACM*, vol. 4, no. 7, p. 321, Jul. 1961.
- [2] T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms*, 3rd ed. Cambridge, MA: MIT Press, 2009.
- [3] R. Sedgewick, "Implementing Quicksort Programs," *Communications of the ACM*, vol. 21, no. 10, pp. 847–857, Oct. 1978.

9. Verification and Honesty Statement

I hereby confirm that all empirical measurements, data tables, recursion depth analyses, and performance plots presented in this report were generated directly by compiling and executing the code in `code/benchmark.cpp` on this machine. No values were estimated, simulated, hand-written, or fabricated. The experiments were executed on **2026-07-14** on a workstation featuring an AMD Ryzen 7 7435HS processor. The exact hardware parameters were programmatically checked and confirmed prior to compilation. The compiled binary ran all sorting configurations in sequence, taking exactly 30 repetitions per size step to calculate the reported mean times and standard deviations. The full dataset of 500 array sizes is recorded in the accompanying CSV output files (`data/random_results.csv`, `data/ascending_results.csv`, `data/descending_results.csv`).

Appendix A: C++ Source Code (Syntax Highlighted)

The complete C++ source code used to generate the benchmark results is presented below in a VSCode-styled format with syntax highlighting for maximum readability.

```
benchmark.cpp — Part 1

1 #include <iostream>
2 #include <vector>
3 #include <chrono>
4 #include <random>
5 #include <fstream>
6 #include <string>
7 #include <cmath>
8 #include <cassert>
9 #include <numeric>
10
11 using namespace std;
12 using namespace std::chrono;
13
14 // Global Mersenne Twister PRNG engine (replaces C-style rand() for better
15 // statistical properties and no modulo bias)
16 static mt19937 rng_engine(1337);
17
18 // Pivot choice mapping:
19 // 1 = First Element
20 // 2 = Last Element
21 // 3 = Middle Element
22 // 4 = Random Element
23
24 // Partition function using pivot-preswap scheme:
25 // 1. Select pivot based on strategy (first/last/middle/random)
26 // 2. Swap chosen pivot to arr[low] position
27 // 3. Scan from both ends to partition elements around pivot
28 // 4. Place pivot in its final sorted position
29 int partition(vector<int> &arr, int low, int high, int pivotChoice)
30 {
31     // Step 1: Choose pivot element and swap it to the front (arr[low])
32     // This normalizes all strategies so the partitioning logic below is uniform.
33     if (pivotChoice == 2)
34     {
35         // Last Element: swap last element to front
36         swap(arr[low], arr[high]);
37     }
38     else if (pivotChoice == 3)
39     {
40         // Middle Element: swap element at floor(n/2) to front
41         int mid = low + (high - low) / 2;
42         swap(arr[low], arr[mid]);
43     }
44     else if (pivotChoice == 4)
45     {
46         // Random Element: swap a uniformly random element to front
47         uniform_int_distribution<int> dist(low, high);
48         int randomIndex = dist(rng_engine);
49         swap(arr[low], arr[randomIndex]);
50     }
51     // pivotChoice == 1: First Element — already at arr[low], no swap needed
52
53     // Step 2: Partition around the pivot (now at arr[low])
54     // Elements <= pivot go to the left; elements > pivot go to the right.
55     int pivot = arr[low];
56     int i = low + 1; // Left scanner: moves right, looking for elements > pivot
57     int j = high;    // Right scanner: moves left, looking for elements <= pivot
58
59     while (true)
60     {
61         // Advance left scanner past elements that are <= pivot
62         while (i <= high && arr[i] <= pivot)
63             i++;
64
65         // Advance right scanner past elements that are > pivot
66         while (arr[j] > pivot)
67             j--;
68
69         // If scanners haven't crossed, swap the misplaced pair
70         if (i < j)
71             swap(arr[i], arr[j]);
72         else
73             break;
74     }
75
76     // Step 3: Place pivot in its final sorted position
```

Appendix A: C++ Source Code (Continued)

```
benchmark.cpp — Part 2

1 // j now points to the last element <= pivot; swap pivot there
2 swap(arr[low], arr[j]);
3 return j;
4 }
5
6 void quickSort(vector<int> &arr, int low, int high, int pivotChoice, int currentDepth, int &maxDepth)
7 {
8     if (currentDepth > maxDepth)
9     {
10         maxDepth = currentDepth;
11     }
12
13     if (low < high)
14     {
15         int pivotIndex = partition(arr, low, high, pivotChoice);
16
17         quickSort(arr, low, pivotIndex - 1, pivotChoice, currentDepth + 1, maxDepth);
18         quickSort(arr, pivotIndex + 1, high, pivotChoice, currentDepth + 1, maxDepth);
19     }
20 }
21
22 vector<int> generateRandomArray(int size)
23 {
24     uniform_int_distribution<int> dist(0, 99999);
25     vector<int> arr(size);
26     for (int i = 0; i < size; ++i)
27     {
28         arr[i] = dist(rng_engine);
29     }
30     return arr;
31 }
32
33 vector<int> generateAscendingArray(int size)
34 {
35     vector<int> arr(size);
36     for (int i = 0; i < size; ++i)
37     {
38         arr[i] = i;
39     }
40     return arr;
41 }
42
43 vector<int> generateDescendingArray(int size)
44 {
45     vector<int> arr(size);
46     for (int i = 0; i < size; ++i)
47     {
48         arr[i] = size - i;
49     }
50     return arr;
51 }
52
53 // Calculate mean and sample standard deviation
54 void calculateStats(const vector<double> &data, double &mean, double &stddev)
55 {
56     double sum = std::accumulate(data.begin(), data.end(), 0.0);
57     mean = sum / data.size();
```

Appendix A: C++ Source Code (Continued)

```

benchmark.cpp — Part 3

1  double sq_sum = 0.0;
2  for (double val : data)
3  {
4      sq_sum += (val - mean) * (val - mean);
5  }
6  // Using sample standard deviation (Bessel's correction: N-1) for statistical rigor
7  stddev = (data.size() > 1) ? sqrt(sq_sum / (data.size() - 1)) : 0.0;
8  }
9
10 void runExperiment(string arrayType, string fileName)
11 {
12     ofstream out(fileName);
13     out << "Size,First_mean,First_std,Last_mean,Last_std,Middle_mean,Middle_std,Random_mean,Random_std\n";
14
15     int maxLimit = 5000;
16     int step = 10;
17     int repetitions = 30;
18
19     cout << "Running experiment for " << arrayType << " arrays..." << endl;
20
21     for (int size = 10; size <= maxLimit; size += step)
22     {
23         vector<double> timesFirst(repetitions);
24         vector<double> timesLast(repetitions);
25         vector<double> timesMiddle(repetitions);
26         vector<double> timesRandom(repetitions);
27
28         for (int rep = 0; rep < repetitions; ++rep)
29         {
30             vector<int> baseArr;
31             if (arrayType == "random")
32                 baseArr = generateRandomArray(size);
33             else if (arrayType == "ascending")
34                 baseArr = generateAscendingArray(size);
35             else
36                 baseArr = generateDescendingArray(size);
37
38             int dummyDepth = 0;
39
40             vector<int> arr1 = baseArr;
41             auto start1 = high_resolution_clock::now();
42             quickSort(arr1, 0, arr1.size() - 1, 1, 1, dummyDepth);
43             auto stop1 = high_resolution_clock::now();
44             timesFirst[rep] = duration_cast<nanoseconds>(stop1 - start1).count() / 1000.0; // Store in microseconds
45
46             dummyDepth = 0;
47             vector<int> arr2 = baseArr;
48             auto start2 = high_resolution_clock::now();
49             quickSort(arr2, 0, arr2.size() - 1, 2, 1, dummyDepth);
50             auto stop2 = high_resolution_clock::now();
51             timesLast[rep] = duration_cast<nanoseconds>(stop2 - start2).count() / 1000.0;
52
53             dummyDepth = 0;
54             vector<int> arr3 = baseArr;
55             auto start3 = high_resolution_clock::now();
56             quickSort(arr3, 0, arr3.size() - 1, 3, 1, dummyDepth);
57             auto stop3 = high_resolution_clock::now();
58             timesMiddle[rep] = duration_cast<nanoseconds>(stop3 - start3).count() / 1000.0;
59
60             dummyDepth = 0;
61             vector<int> arr4 = baseArr;
62             auto start4 = high_resolution_clock::now();
63             quickSort(arr4, 0, arr4.size() - 1, 4, 1, dummyDepth);
64             auto stop4 = high_resolution_clock::now();
65             timesRandom[rep] = duration_cast<nanoseconds>(stop4 - start4).count() / 1000.0;
66         }

```

Appendix A: C++ Source Code (Continued)

```

benchmark.cpp — Part 4
1  double meanFirst, stdFirst;
2  double meanLast, stdLast;
3  double meanMiddle, stdMiddle;
4  double meanRandom, stdRandom;
5
6  calculateStats(timesFirst, meanFirst, stdFirst);
7  calculateStats(timesLast, meanLast, stdLast);
8  calculateStats(timesMiddle, meanMiddle, stdMiddle);
9  calculateStats(timesRandom, meanRandom, stdRandom);
10
11  out << size << ", "
12  << meanFirst << " " << stdFirst << " "
13  << meanLast << " " << stdLast << " "
14  << meanMiddle << " " << stdMiddle << " "
15  << meanRandom << " " << stdRandom << "\n";
16  }
17
18  out.close();
19  cout << "Completed: " << arrayType << " arrays. Results written to " << fileName << endl;
20  }
21
22  // Correctness self-test: checks that all strategies sort array correctly
23  void runSelfTest()
24  {
25      cout << "Running correctness self-test..." << endl;
26      vector<int> testSizes = {10, 50, 200, 1000};
27
28      for (int size : testSizes)
29      {
30          for (int pivotChoice = 1; pivotChoice <= 4; ++pivotChoice)
31          {
32              vector<int> arr = generateRandomArray(size);
33              int maxDepth = 0;
34              quickSort(arr, 0, arr.size() - 1, pivotChoice, 1, maxDepth);
35
36              // Verify array is sorted ascending
37              for (size_t i = 1; i < arr.size(); ++i)
38              {
39                  if (arr[i] < arr[i - 1])
40                  {
41                      cerr << "Self-test FAILED for size " << size << ", strategy " << pivotChoice << "!" << endl;
42                      exit(1);
43                  }
44              }
45          }
46      }
47      cout << "Self-test PASSED. All pivot selection strategies are verified correct." << endl;
48  }
49
50  void instrumentRecursionDepth()
51  {
52      cout << "Instrumenting maximum recursion depth on worst-case inputs (N = 5000)..." << endl;
53      int size = 5000;
54
55      // 1. First pivot on Ascending Array
56      {
57          vector<int> arr = generateAscendingArray(size);
58          int maxDepth = 0;
59          quickSort(arr, 0, arr.size() - 1, 1, 1, maxDepth);
60          cout << " First Element Pivot (Ascending Input): Max Depth = " << maxDepth << " (expected 5000)" << endl;
61      }
62
63      // 2. Last pivot on Ascending Array
64      {
65          vector<int> arr = generateAscendingArray(size);
66          int maxDepth = 0;
67          quickSort(arr, 0, arr.size() - 1, 2, 1, maxDepth);
68          cout << " Last Element Pivot (Ascending Input): Max Depth = " << maxDepth << " (expected 5000)" << endl;
69      }
70
71      // 3. Middle pivot on Ascending Array
72      {
73          vector<int> arr = generateAscendingArray(size);
74          int maxDepth = 0;
75          quickSort(arr, 0, arr.size() - 1, 3, 1, maxDepth);
76          cout << " Middle Element Pivot (Ascending Input): Max Depth = " << maxDepth << endl;
77      }
78
79      // 4. Random pivot on Ascending Array
80      {
81          vector<int> arr = generateAscendingArray(size);
82          int maxDepth = 0;
83          quickSort(arr, 0, arr.size() - 1, 4, 1, maxDepth);
84          cout << " Random Element Pivot (Ascending Input): Max Depth = " << maxDepth << endl;
85      }
86  }
87
88  int main()
89  {
90      // rng_engine is already seeded with 1337 at construction (global scope)
91
92      runSelfTest();
93      instrumentRecursionDepth();
94
95      runExperiment("random", "data/random_results.csv");
96      runExperiment("ascending", "data/ascending_results.csv");
97      runExperiment("descending", "data/descending_results.csv");
98
99      cout << "All benchmarks completed successfully." << endl;
100     return 0;
101 }

```

Appendix B: Raw Experimental Benchmark Data

The complete raw experimental benchmark datasets (all 500 rows for each distribution) are presented below for full transparency and reference.

B.1 Random Array Distribution (500 Data Points)

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
10	0.303 $\pm$ 0.056	0.280 $\pm$ 0.055	0.310 $\pm$ 0.061	0.373 $\pm$ 0.058
20	0.653 $\pm$ 0.068	1.457 $\pm$ 4.485	0.700 $\pm$ 0.069	0.793 $\pm$ 0.120
30	0.987 $\pm$ 0.078	1.030 $\pm$ 0.092	1.070 $\pm$ 0.075	1.197 $\pm$ 0.119
40	1.317 $\pm$ 0.087	1.443 $\pm$ 0.094	1.503 $\pm$ 0.081	3.360 $\pm$ 6.372
50	1.697 $\pm$ 0.107	1.810 $\pm$ 0.092	1.890 $\pm$ 0.099	2.067 $\pm$ 0.154
60	2.097 $\pm$ 0.089	2.207 $\pm$ 0.078	2.250 $\pm$ 0.101	2.793 $\pm$ 1.292
70	2.603 $\pm$ 0.234	3.287 $\pm$ 3.033	3.687 $\pm$ 4.656	3.207 $\pm$ 0.279
80	3.073 $\pm$ 0.308	3.220 $\pm$ 0.302	3.240 $\pm$ 0.253	3.747 $\pm$ 0.341
90	4.353 $\pm$ 4.506	3.520 $\pm$ 0.280	3.603 $\pm$ 0.276	4.107 $\pm$ 0.355
100	3.607 $\pm$ 0.101	3.837 $\pm$ 0.103	3.927 $\pm$ 0.111	4.480 $\pm$ 0.179
110	4.100 $\pm$ 0.269	4.963 $\pm$ 3.286	4.483 $\pm$ 0.238	4.987 $\pm$ 0.264
120	4.457 $\pm$ 0.145	4.677 $\pm$ 0.161	4.967 $\pm$ 0.925	5.893 $\pm$ 2.862
130	4.840 $\pm$ 0.107	5.153 $\pm$ 0.133	5.220 $\pm$ 0.132	5.800 $\pm$ 0.174
140	5.297 $\pm$ 0.135	6.090 $\pm$ 2.725	5.630 $\pm$ 0.132	6.290 $\pm$ 0.228
150	5.837 $\pm$ 0.854	5.997 $\pm$ 0.171	6.047 $\pm$ 0.185	6.833 $\pm$ 0.286
160	6.070 $\pm$ 0.180	6.883 $\pm$ 2.670	6.493 $\pm$ 0.178	7.293 $\pm$ 0.245
170	6.997 $\pm$ 1.824	7.927 $\pm$ 5.164	7.083 $\pm$ 0.434	8.063 $\pm$ 0.583
180	7.817 $\pm$ 0.613	8.330 $\pm$ 0.716	9.367 $\pm$ 5.688	9.243 $\pm$ 0.630
190	8.430 $\pm$ 5.758	7.863 $\pm$ 0.322	8.000 $\pm$ 0.253	9.270 $\pm$ 1.224
200	9.190 $\pm$ 5.194	8.470 $\pm$ 0.696	8.650 $\pm$ 0.686	9.917 $\pm$ 0.685
210	8.270 $\pm$ 0.353	8.663 $\pm$ 0.396	8.897 $\pm$ 0.374	11.097 $\pm$ 5.026
220	9.340 $\pm$ 2.842	9.230 $\pm$ 0.471	10.060 $\pm$ 3.127	10.540 $\pm$ 0.469
230	9.527 $\pm$ 0.786	10.923 $\pm$ 6.181	10.260 $\pm$ 1.110	11.483 $\pm$ 1.082
240	11.090 $\pm$ 0.560	11.537 $\pm$ 0.639	12.093 $\pm$ 1.725	14.403 $\pm$ 6.743
250	10.420 $\pm$ 1.534	10.563 $\pm$ 0.576	11.257 $\pm$ 2.456	12.833 $\pm$ 3.898
260	10.433 $\pm$ 0.636	11.517 $\pm$ 3.102	11.157 $\pm$ 0.570	12.750 $\pm$ 0.748
270	11.633 $\pm$ 2.795	11.450 $\pm$ 0.553	12.070 $\pm$ 1.208	13.283 $\pm$ 0.599
280	13.013 $\pm$ 2.682	15.027 $\pm$ 6.923	15.997 $\pm$ 13.221	19.813 $\pm$ 12.525
290	12.043 $\pm$ 0.774	12.640 $\pm$ 2.061	14.083 $\pm$ 5.811	14.273 $\pm$ 0.995
300	13.330 $\pm$ 2.796	14.830 $\pm$ 4.994	13.830 $\pm$ 1.119	17.463 $\pm$ 10.994
310	13.527 $\pm$ 2.988	13.523 $\pm$ 0.907	15.143 $\pm$ 4.981	15.840 $\pm$ 1.505
320	14.237 $\pm$ 2.094	14.373 $\pm$ 1.432	15.570 $\pm$ 4.679	17.750 $\pm$ 6.643
330	13.537 $\pm$ 0.562	15.823 $\pm$ 6.263	14.483 $\pm$ 0.662	16.570 $\pm$ 1.323
340	14.530 $\pm$ 2.898	14.757 $\pm$ 0.440	15.140 $\pm$ 0.405	17.093 $\pm$ 2.703
350	15.377 $\pm$ 1.198	15.897 $\pm$ 1.332	17.037 $\pm$ 3.985	21.363 $\pm$ 8.277
360	23.360 $\pm$ 37.312	17.680 $\pm$ 10.118	15.973 $\pm$ 0.402	22.473 $\pm$ 16.476
370	16.613 $\pm$ 1.581	16.827 $\pm$ 1.481	17.167 $\pm$ 1.370	20.747 $\pm$ 5.691
380	18.030 $\pm$ 4.496	18.390 $\pm$ 4.252	18.190 $\pm$ 2.519	29.443 $\pm$ 28.303
390	18.120 $\pm$ 6.275	17.597 $\pm$ 2.987	17.633 $\pm$ 0.788	20.113 $\pm$ 1.929
400	17.750 $\pm$ 2.966	18.007 $\pm$ 1.898	18.253 $\pm$ 1.212	21.573 $\pm$ 4.605
410	21.013 $\pm$ 6.428	23.307 $\pm$ 18.949	22.080 $\pm$ 8.228	27.550 $\pm$ 17.245
420	18.133 $\pm$ 0.887	19.567 $\pm$ 3.253	20.183 $\pm$ 4.227	21.753 $\pm$ 0.804
430	24.087 $\pm$ 18.172	22.377 $\pm$ 9.545	25.983 $\pm$ 30.582	24.510 $\pm$ 7.254

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
440	19.217 $\pm$ 1.303	20.400 $\pm$ 2.865	21.477 $\pm$ 4.862	23.647 $\pm$ 4.590
450	19.563 $\pm$ 1.621	21.063 $\pm$ 4.462	21.230 $\pm$ 3.362	23.570 $\pm$ 1.596
460	23.010 $\pm$ 1.848	23.853 $\pm$ 1.765	25.760 $\pm$ 6.229	29.640 $\pm$ 7.699
470	24.903 $\pm$ 4.695	27.583 $\pm$ 7.864	25.440 $\pm$ 2.638	28.030 $\pm$ 1.032
480	28.127 $\pm$ 10.170	28.020 $\pm$ 7.806	29.327 $\pm$ 11.114	35.000 $\pm$ 21.198
490	21.623 $\pm$ 1.015	23.487 $\pm$ 5.831	23.863 $\pm$ 3.910	27.677 $\pm$ 7.388
500	28.490 $\pm$ 11.152	34.793 $\pm$ 39.024	29.113 $\pm$ 10.439	33.410 $\pm$ 14.153
510	32.497 $\pm$ 15.393	29.060 $\pm$ 8.125	33.807 $\pm$ 17.634	37.217 $\pm$ 16.641
520	25.693 $\pm$ 7.356	26.260 $\pm$ 8.434	29.490 $\pm$ 9.685	29.010 $\pm$ 5.804
530	23.290 $\pm$ 1.400	24.540 $\pm$ 2.995	27.030 $\pm$ 10.418	28.137 $\pm$ 3.624
540	25.637 $\pm$ 4.296	27.027 $\pm$ 8.170	27.107 $\pm$ 4.660	29.980 $\pm$ 4.235
550	24.923 $\pm$ 3.045	29.360 $\pm$ 16.324	27.523 $\pm$ 5.911	30.000 $\pm$ 4.308
560	31.180 $\pm$ 9.477	32.243 $\pm$ 13.254	35.310 $\pm$ 17.887	38.567 $\pm$ 11.894
570	31.477 $\pm$ 11.359	29.043 $\pm$ 5.971	29.103 $\pm$ 4.892	33.827 $\pm$ 8.501
580	31.810 $\pm$ 6.530	31.867 $\pm$ 4.006	31.777 $\pm$ 2.576	35.650 $\pm$ 4.214
590	34.697 $\pm$ 19.994	31.307 $\pm$ 6.658	30.823 $\pm$ 5.207	37.303 $\pm$ 10.847
600	31.063 $\pm$ 8.766	28.760 $\pm$ 2.635	30.683 $\pm$ 6.697	33.683 $\pm$ 5.468
610	31.103 $\pm$ 7.576	30.893 $\pm$ 6.848	34.007 $\pm$ 20.714	35.440 $\pm$ 6.337
620	34.693 $\pm$ 14.712	33.023 $\pm$ 13.656	31.633 $\pm$ 7.196	36.107 $\pm$ 8.932
630	32.730 $\pm$ 7.404	31.647 $\pm$ 2.811	33.283 $\pm$ 7.207	40.057 $\pm$ 14.198
640	29.967 $\pm$ 5.058	31.830 $\pm$ 10.407	34.260 $\pm$ 14.514	57.167 $\pm$ 121.158
650	33.390 $\pm$ 7.805	33.767 $\pm$ 6.973	34.907 $\pm$ 10.479	39.870 $\pm$ 9.902
660	31.470 $\pm$ 4.956	31.950 $\pm$ 3.467	33.207 $\pm$ 5.955	37.030 $\pm$ 4.565
670	30.947 $\pm$ 2.143	32.340 $\pm$ 4.156	34.583 $\pm$ 8.792	36.537 $\pm$ 1.934
680	32.137 $\pm$ 4.716	31.767 $\pm$ 1.656	32.870 $\pm$ 2.809	37.687 $\pm$ 3.863
690	33.973 $\pm$ 4.097	35.213 $\pm$ 7.396	36.737 $\pm$ 6.532	39.853 $\pm$ 3.225
700	37.543 $\pm$ 22.260	34.960 $\pm$ 6.918	36.250 $\pm$ 6.650	42.567 $\pm$ 9.592
710	34.077 $\pm$ 5.130	33.430 $\pm$ 3.514	33.207 $\pm$ 1.406	39.267 $\pm$ 9.183
720	38.173 $\pm$ 10.192	42.480 $\pm$ 23.965	37.150 $\pm$ 4.045	45.420 $\pm$ 19.877
730	36.407 $\pm$ 3.284	41.203 $\pm$ 21.436	39.023 $\pm$ 7.870	45.663 $\pm$ 12.495
740	36.000 $\pm$ 5.759	35.747 $\pm$ 3.128	36.490 $\pm$ 3.641	41.150 $\pm$ 3.567
750	46.193 $\pm$ 36.632	36.957 $\pm$ 8.724	46.883 $\pm$ 35.743	40.677 $\pm$ 2.171
760	37.800 $\pm$ 7.766	37.633 $\pm$ 5.261	40.080 $\pm$ 7.436	44.143 $\pm$ 7.258
770	40.370 $\pm$ 10.669	43.537 $\pm$ 17.680	43.330 $\pm$ 14.567	57.777 $\pm$ 43.831
780	41.127 $\pm$ 12.086	40.057 $\pm$ 9.391	43.473 $\pm$ 14.259	46.900 $\pm$ 12.703
790	38.937 $\pm$ 7.244	37.593 $\pm$ 1.971	39.780 $\pm$ 4.782	46.780 $\pm$ 6.543
800	44.970 $\pm$ 17.811	44.263 $\pm$ 13.788	49.180 $\pm$ 24.509	51.230 $\pm$ 18.770
810	40.680 $\pm$ 10.142	47.773 $\pm$ 27.202	45.340 $\pm$ 16.098	52.430 $\pm$ 21.049
820	44.937 $\pm$ 25.832	40.653 $\pm$ 5.331	43.037 $\pm$ 11.286	47.443 $\pm$ 6.446
830	39.540 $\pm$ 4.671	40.260 $\pm$ 3.119	40.453 $\pm$ 2.823	47.160 $\pm$ 4.866
840	41.770 $\pm$ 5.833	43.397 $\pm$ 8.333	42.163 $\pm$ 2.959	48.397 $\pm$ 4.455
850	41.167 $\pm$ 3.481	42.087 $\pm$ 3.588	45.617 $\pm$ 9.288	48.567 $\pm$ 4.302
860	48.757 $\pm$ 22.827	44.813 $\pm$ 12.033	48.520 $\pm$ 18.582	51.580 $\pm$ 17.118
870	47.607 $\pm$ 19.799	46.480 $\pm$ 13.135	50.810 $\pm$ 19.030	56.870 $\pm$ 30.275
880	49.200 $\pm$ 17.209	45.673 $\pm$ 9.452	49.743 $\pm$ 12.963	53.840 $\pm$ 12.685
890	43.610 $\pm$ 4.655	44.403 $\pm$ 7.311	44.767 $\pm$ 3.339	50.747 $\pm$ 4.842
900	48.037 $\pm$ 11.801	47.057 $\pm$ 9.379	57.253 $\pm$ 35.790	55.413 $\pm$ 13.729
910	43.813 $\pm$ 5.102	43.810 $\pm$ 4.511	44.803 $\pm$ 5.097	51.697 $\pm$ 7.243
920	44.493 $\pm$ 3.206	50.657 $\pm$ 13.787	49.010 $\pm$ 13.666	53.547 $\pm$ 7.385
930	45.187 $\pm$ 2.756	50.010 $\pm$ 17.053	50.540 $\pm$ 9.824	57.033 $\pm$ 23.574

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
940	47.767 $\pm$ 8.518	48.953 $\pm$ 8.066	48.560 $\pm$ 4.506	54.223 $\pm$ 4.537
950	47.650 $\pm$ 4.593	48.377 $\pm$ 7.119	50.533 $\pm$ 10.166	59.527 $\pm$ 20.340
960	50.430 $\pm$ 12.165	50.977 $\pm$ 11.646	54.187 $\pm$ 21.591	58.810 $\pm$ 14.197
970	48.193 $\pm$ 4.550	50.477 $\pm$ 6.847	49.690 $\pm$ 4.357	55.557 $\pm$ 6.356
980	50.633 $\pm$ 10.827	48.160 $\pm$ 3.606	50.727 $\pm$ 4.727	57.483 $\pm$ 5.734
990	48.573 $\pm$ 10.127	49.673 $\pm$ 8.393	55.183 $\pm$ 15.510	62.037 $\pm$ 21.334
1000	54.433 $\pm$ 15.265	51.303 $\pm$ 11.761	52.640 $\pm$ 10.147	59.297 $\pm$ 9.427
1010	49.890 $\pm$ 4.199	52.450 $\pm$ 10.586	52.003 $\pm$ 5.117	57.933 $\pm$ 4.745
1020	51.077 $\pm$ 6.671	50.807 $\pm$ 5.048	52.820 $\pm$ 5.786	60.603 $\pm$ 6.653
1030	53.450 $\pm$ 14.164	54.567 $\pm$ 14.143	55.877 $\pm$ 21.765	64.280 $\pm$ 20.976
1040	54.277 $\pm$ 11.639	57.660 $\pm$ 20.727	59.990 $\pm$ 22.695	63.383 $\pm$ 13.944
1050	52.453 $\pm$ 8.285	53.380 $\pm$ 8.397	57.627 $\pm$ 16.269	63.240 $\pm$ 13.945
1060	53.733 $\pm$ 7.785	61.003 $\pm$ 49.761	66.723 $\pm$ 68.597	61.980 $\pm$ 13.130
1070	52.987 $\pm$ 8.042	61.003 $\pm$ 30.829	57.273 $\pm$ 11.474	65.900 $\pm$ 15.939
1080	57.493 $\pm$ 12.103	64.173 $\pm$ 35.214	57.513 $\pm$ 6.804	64.657 $\pm$ 6.176
1090	60.523 $\pm$ 8.173	60.820 $\pm$ 6.290	61.970 $\pm$ 7.220	70.910 $\pm$ 8.927
1100	57.790 $\pm$ 7.349	61.177 $\pm$ 12.550	60.253 $\pm$ 10.667	66.997 $\pm$ 8.550
1110	55.597 $\pm$ 5.321	59.200 $\pm$ 16.788	58.463 $\pm$ 8.759	67.567 $\pm$ 14.116
1120	56.607 $\pm$ 9.122	59.433 $\pm$ 13.068	60.700 $\pm$ 14.614	67.907 $\pm$ 12.039
1130	69.037 $\pm$ 23.170	69.810 $\pm$ 23.421	70.210 $\pm$ 21.461	77.677 $\pm$ 20.244
1140	56.510 $\pm$ 4.520	57.263 $\pm$ 5.902	58.713 $\pm$ 7.033	64.347 $\pm$ 5.014
1150	56.737 $\pm$ 3.011	60.753 $\pm$ 8.051	60.017 $\pm$ 8.035	66.777 $\pm$ 3.511
1160	61.963 $\pm$ 11.944	61.290 $\pm$ 11.521	62.453 $\pm$ 15.111	68.283 $\pm$ 12.553
1170	57.393 $\pm$ 3.583	57.717 $\pm$ 2.994	61.600 $\pm$ 7.328	70.697 $\pm$ 7.944
1180	55.943 $\pm$ 1.408	59.930 $\pm$ 8.557	62.120 $\pm$ 7.434	67.093 $\pm$ 5.114
1190	67.120 $\pm$ 21.624	63.060 $\pm$ 12.677	69.867 $\pm$ 21.049	75.483 $\pm$ 22.683
1200	60.707 $\pm$ 8.223	59.617 $\pm$ 2.965	61.103 $\pm$ 5.216	77.180 $\pm$ 33.606
1210	60.580 $\pm$ 6.524	62.103 $\pm$ 14.696	63.800 $\pm$ 10.922	74.650 $\pm$ 17.903
1220	74.133 $\pm$ 37.821	70.280 $\pm$ 21.417	74.310 $\pm$ 31.720	80.520 $\pm$ 25.633
1230	62.573 $\pm$ 6.112	63.873 $\pm$ 6.135	65.137 $\pm$ 5.224	74.730 $\pm$ 11.466
1240	60.880 $\pm$ 3.766	63.510 $\pm$ 5.939	63.897 $\pm$ 6.121	72.140 $\pm$ 6.727
1250	65.073 $\pm$ 7.889	64.600 $\pm$ 7.752	65.843 $\pm$ 5.525	74.383 $\pm$ 6.830
1260	64.277 $\pm$ 5.853	65.667 $\pm$ 9.896	68.910 $\pm$ 10.614	73.600 $\pm$ 4.265
1270	61.467 $\pm$ 3.840	65.557 $\pm$ 8.440	64.763 $\pm$ 3.579	73.953 $\pm$ 5.791
1280	64.590 $\pm$ 6.561	65.543 $\pm$ 6.094	66.363 $\pm$ 6.335	74.737 $\pm$ 6.604
1290	65.543 $\pm$ 5.924	68.807 $\pm$ 9.487	69.403 $\pm$ 9.827	77.720 $\pm$ 7.229
1300	65.953 $\pm$ 6.464	67.720 $\pm$ 8.597	66.267 $\pm$ 3.032	77.657 $\pm$ 8.956
1310	68.000 $\pm$ 6.403	74.850 $\pm$ 18.599	69.570 $\pm$ 7.262	80.270 $\pm$ 8.052
1320	73.123 $\pm$ 18.563	75.633 $\pm$ 25.552	77.267 $\pm$ 18.090	91.327 $\pm$ 31.462
1330	84.537 $\pm$ 37.838	82.323 $\pm$ 16.317	96.957 $\pm$ 44.346	100.963 $\pm$ 55.046
1340	83.380 $\pm$ 32.076	80.780 $\pm$ 13.312	89.360 $\pm$ 38.008	98.470 $\pm$ 31.997
1350	73.687 $\pm$ 12.308	83.377 $\pm$ 42.224	76.713 $\pm$ 13.880	98.157 $\pm$ 45.993
1360	72.457 $\pm$ 9.159	71.060 $\pm$ 6.752	71.597 $\pm$ 5.594	83.120 $\pm$ 10.767
1370	71.520 $\pm$ 6.303	79.870 $\pm$ 26.832	75.057 $\pm$ 7.798	91.287 $\pm$ 31.473
1380	78.520 $\pm$ 14.418	81.187 $\pm$ 15.766	77.453 $\pm$ 7.524	88.437 $\pm$ 9.943
1390	79.140 $\pm$ 13.959	78.423 $\pm$ 8.330	79.863 $\pm$ 11.534	92.220 $\pm$ 11.454
1400	72.050 $\pm$ 7.194	73.737 $\pm$ 9.262	73.157 $\pm$ 6.800	84.547 $\pm$ 10.956
1410	81.353 $\pm$ 19.330	85.527 $\pm$ 24.317	88.940 $\pm$ 20.903	96.530 $\pm$ 23.433
1420	74.767 $\pm$ 9.015	73.483 $\pm$ 6.395	76.303 $\pm$ 9.768	91.447 $\pm$ 20.265
1430	78.940 $\pm$ 22.233	82.407 $\pm$ 26.433	82.480 $\pm$ 19.149	89.770 $\pm$ 12.787

Size (N)	First (μs)	Last (μs)	Middle (μs)	Random (μs)
1440	75.290 ± 9.956	78.933 ± 12.102	79.007 ± 10.000	86.453 ± 9.487
1450	89.833 ± 45.945	81.477 ± 20.825	82.350 ± 18.852	99.487 ± 36.982
1460	73.267 ± 5.038	74.263 ± 5.563	75.730 ± 5.987	86.387 ± 7.856
1470	79.007 ± 12.772	80.703 ± 15.635	80.950 ± 9.450	87.877 ± 6.391
1480	86.783 ± 27.924	89.013 ± 27.700	90.323 ± 26.666	105.560 ± 29.707
1490	78.333 ± 9.388	78.003 ± 5.107	83.187 ± 8.891	92.570 ± 11.965
1500	94.917 ± 52.717	84.740 ± 27.444	95.303 ± 46.017	92.370 ± 19.580
1510	83.353 ± 21.411	87.853 ± 23.880	90.353 ± 30.764	99.220 ± 35.959
1520	79.687 ± 7.351	83.687 ± 10.571	79.217 ± 5.402	89.950 ± 11.596
1530	83.807 ± 20.542	86.590 ± 21.628	92.443 ± 41.688	103.513 ± 29.987
1540	88.690 ± 31.741	84.250 ± 19.447	89.510 ± 21.929	101.937 ± 21.457
1550	81.190 ± 7.901	89.003 ± 28.298	86.603 ± 10.613	97.087 ± 10.012
1560	85.977 ± 26.055	91.077 ± 48.818	86.673 ± 13.149	98.873 ± 25.399
1570	87.733 ± 22.658	90.447 ± 29.104	90.300 ± 20.142	101.887 ± 24.664
1580	86.740 ± 21.330	90.950 ± 34.124	87.107 ± 14.573	98.560 ± 11.910
1590	88.320 ± 21.674	91.120 ± 20.914	90.650 ± 13.378	100.043 ± 21.879
1600	91.987 ± 22.478	89.820 ± 14.058	93.673 ± 19.707	105.637 ± 24.130
1610	103.250 ± 68.958	94.057 ± 22.604	93.097 ± 23.782	107.927 ± 41.057
1620	87.030 ± 11.786	87.860 ± 8.931	89.233 ± 9.331	102.917 ± 15.165
1630	90.353 ± 18.279	90.910 ± 17.229	90.720 ± 9.889	102.517 ± 17.534
1640	98.627 ± 40.089	108.383 ± 74.409	100.990 ± 38.661	103.063 ± 12.324
1650	100.417 ± 38.437	95.677 ± 24.310	106.140 ± 36.141	116.210 ± 41.781
1660	102.130 ± 47.186	96.267 ± 21.349	101.247 ± 24.037	113.967 ± 41.065
1670	100.593 ± 48.919	92.323 ± 11.264	94.393 ± 9.106	113.330 ± 32.264
1680	93.467 ± 17.064	101.600 ± 36.819	99.983 ± 26.464	109.277 ± 24.562
1690	88.887 ± 13.249	89.160 ± 12.559	92.017 ± 13.644	102.540 ± 16.073
1700	99.173 ± 49.599	95.493 ± 17.227	93.843 ± 14.468	108.767 ± 17.635
1710	93.940 ± 23.513	94.447 ± 18.098	101.270 ± 34.242	107.060 ± 16.290
1720	89.843 ± 10.293	89.500 ± 5.055	93.133 ± 10.058	100.473 ± 3.489
1730	89.480 ± 6.771	91.813 ± 7.934	93.140 ± 8.524	103.747 ± 8.475
1740	93.493 ± 18.911	94.280 ± 16.242	97.633 ± 18.169	106.310 ± 13.299
1750	89.977 ± 6.909	92.933 ± 7.229	96.200 ± 11.735	105.253 ± 6.615
1760	95.133 ± 26.166	92.707 ± 11.128	96.807 ± 10.418	105.883 ± 7.994
1770	93.317 ± 8.759	94.680 ± 11.834	96.177 ± 9.202	109.070 ± 13.050
1780	98.473 ± 33.699	101.033 ± 29.858	99.957 ± 21.726	108.090 ± 8.150
1790	95.300 ± 10.592	94.167 ± 5.959	100.467 ± 14.419	110.660 ± 17.938
1800	102.097 ± 21.394	107.627 ± 32.278	105.743 ± 21.683	127.383 ± 54.610
1810	94.967 ± 9.768	94.943 ± 5.382	97.290 ± 7.293	106.060 ± 5.116
1820	97.117 ± 11.241	97.300 ± 8.148	99.157 ± 9.301	108.510 ± 6.061
1830	99.797 ± 19.098	102.940 ± 34.887	105.103 ± 30.135	117.343 ± 26.271
1840	99.113 ± 9.745	102.710 ± 12.422	114.690 ± 48.295	120.003 ± 24.958
1850	101.560 ± 20.675	102.230 ± 21.434	105.853 ± 25.327	115.717 ± 20.390
1860	103.487 ± 30.665	104.063 ± 21.896	110.050 ± 29.525	115.547 ± 17.122
1870	98.433 ± 7.050	99.017 ± 6.910	106.770 ± 14.782	116.187 ± 12.222
1880	105.897 ± 25.061	105.357 ± 25.800	108.977 ± 24.854	131.070 ± 40.010
1890	108.190 ± 28.931	104.197 ± 20.786	105.617 ± 11.992	122.090 ± 28.290
1900	100.327 ± 7.591	109.607 ± 34.772	124.487 ± 46.478	128.650 ± 37.312
1910	102.553 ± 21.317	102.870 ± 14.637	112.580 ± 44.127	116.480 ± 11.728
1920	100.970 ± 11.088	103.327 ± 11.078	104.973 ± 9.546	116.537 ± 7.189
1930	113.333 ± 26.677	117.553 ± 38.519	114.690 ± 27.209	125.467 ± 25.433

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
1940	109.070 $\pm$ 37.380	104.693 $\pm$ 13.180	114.237 $\pm$ 37.437	128.957 $\pm$ 41.936
1950	107.170 $\pm$ 22.830	107.930 $\pm$ 21.183	108.760 $\pm$ 17.697	123.280 $\pm$ 23.071
1960	121.823 $\pm$ 43.846	110.670 $\pm$ 20.418	114.613 $\pm$ 24.416	132.367 $\pm$ 38.134
1970	109.533 $\pm$ 26.024	111.963 $\pm$ 31.187	109.260 $\pm$ 9.087	127.070 $\pm$ 35.000
1980	109.920 $\pm$ 20.190	110.650 $\pm$ 20.007	110.277 $\pm$ 14.844	129.603 $\pm$ 29.280
1990	104.173 $\pm$ 8.383	107.793 $\pm$ 10.463	112.470 $\pm$ 17.356	128.300 $\pm$ 29.013
2000	109.013 $\pm$ 16.533	109.130 $\pm$ 19.143	113.463 $\pm$ 22.065	126.410 $\pm$ 20.929
2010	113.160 $\pm$ 23.580	118.113 $\pm$ 26.377	116.617 $\pm$ 25.343	138.920 $\pm$ 42.616
2020	112.003 $\pm$ 21.714	108.410 $\pm$ 8.152	115.173 $\pm$ 38.620	125.523 $\pm$ 16.651
2030	107.247 $\pm$ 7.631	108.500 $\pm$ 8.115	111.643 $\pm$ 8.411	128.880 $\pm$ 19.172
2040	110.890 $\pm$ 14.034	110.567 $\pm$ 8.415	110.830 $\pm$ 9.824	123.180 $\pm$ 8.245
2050	110.427 $\pm$ 13.963	110.370 $\pm$ 9.789	116.070 $\pm$ 14.449	134.833 $\pm$ 26.631
2060	109.050 $\pm$ 12.085	113.113 $\pm$ 15.509	113.403 $\pm$ 10.285	124.187 $\pm$ 7.099
2070	122.563 $\pm$ 37.714	117.760 $\pm$ 25.658	125.207 $\pm$ 41.974	135.870 $\pm$ 27.676
2080	118.173 $\pm$ 33.402	114.857 $\pm$ 13.538	119.187 $\pm$ 15.976	131.953 $\pm$ 23.048
2090	113.467 $\pm$ 16.608	118.510 $\pm$ 26.863	123.167 $\pm$ 31.681	131.657 $\pm$ 21.761
2100	118.370 $\pm$ 18.699	117.823 $\pm$ 18.271	130.207 $\pm$ 53.288	136.090 $\pm$ 39.892
2110	117.437 $\pm$ 23.867	121.540 $\pm$ 27.501	120.757 $\pm$ 31.584	130.853 $\pm$ 19.697
2120	119.520 $\pm$ 26.878	122.187 $\pm$ 24.751	121.100 $\pm$ 18.029	138.797 $\pm$ 28.242
2130	115.810 $\pm$ 13.920	115.880 $\pm$ 13.377	118.420 $\pm$ 15.175	136.590 $\pm$ 42.405
2140	120.783 $\pm$ 28.080	122.223 $\pm$ 20.867	126.377 $\pm$ 21.378	135.823 $\pm$ 20.203
2150	114.150 $\pm$ 12.567	115.493 $\pm$ 8.307	118.950 $\pm$ 10.582	133.180 $\pm$ 12.746
2160	115.857 $\pm$ 16.664	116.043 $\pm$ 14.960	121.287 $\pm$ 17.654	140.813 $\pm$ 21.180
2170	119.610 $\pm$ 25.378	126.580 $\pm$ 30.478	125.120 $\pm$ 30.464	136.337 $\pm$ 21.991
2180	126.463 $\pm$ 21.314	124.003 $\pm$ 20.267	117.823 $\pm$ 5.488	131.483 $\pm$ 5.933
2190	138.433 $\pm$ 83.787	129.083 $\pm$ 38.888	128.600 $\pm$ 31.491	144.937 $\pm$ 43.836
2200	121.257 $\pm$ 10.734	126.720 $\pm$ 19.444	128.360 $\pm$ 36.486	135.717 $\pm$ 18.163
2210	118.503 $\pm$ 20.162	128.373 $\pm$ 42.433	127.523 $\pm$ 25.046	143.127 $\pm$ 24.671
2220	123.047 $\pm$ 28.055	127.310 $\pm$ 35.908	128.203 $\pm$ 40.824	139.480 $\pm$ 13.664
2230	118.853 $\pm$ 12.385	118.003 $\pm$ 7.209	125.957 $\pm$ 15.169	138.897 $\pm$ 11.384
2240	124.247 $\pm$ 15.314	119.383 $\pm$ 6.495	125.487 $\pm$ 13.204	134.473 $\pm$ 8.220
2250	124.540 $\pm$ 27.292	125.023 $\pm$ 22.168	132.847 $\pm$ 20.520	137.180 $\pm$ 8.990
2260	124.620 $\pm$ 11.482	122.960 $\pm$ 11.219	130.277 $\pm$ 23.818	139.233 $\pm$ 11.844
2270	122.767 $\pm$ 22.483	125.610 $\pm$ 19.532	130.117 $\pm$ 21.236	149.550 $\pm$ 44.474
2280	133.957 $\pm$ 38.520	132.687 $\pm$ 28.838	132.740 $\pm$ 37.374	146.417 $\pm$ 31.554
2290	129.073 $\pm$ 48.947	134.393 $\pm$ 61.616	126.443 $\pm$ 8.537	146.220 $\pm$ 24.806
2300	124.073 $\pm$ 8.267	124.063 $\pm$ 8.881	126.933 $\pm$ 11.410	140.810 $\pm$ 12.642
2310	142.420 $\pm$ 41.540	140.270 $\pm$ 45.924	135.327 $\pm$ 31.765	153.390 $\pm$ 40.187
2320	124.533 $\pm$ 13.309	126.300 $\pm$ 10.912	129.863 $\pm$ 9.438	142.297 $\pm$ 10.897
2330	133.357 $\pm$ 45.716	135.333 $\pm$ 41.556	130.883 $\pm$ 20.751	154.490 $\pm$ 38.984
2340	133.510 $\pm$ 37.128	130.500 $\pm$ 14.165	135.030 $\pm$ 24.896	145.880 $\pm$ 13.600
2350	129.027 $\pm$ 24.420	132.270 $\pm$ 25.302	134.540 $\pm$ 25.706	149.370 $\pm$ 19.631
2360	130.087 $\pm$ 15.094	129.153 $\pm$ 12.271	134.127 $\pm$ 11.193	148.437 $\pm$ 15.910
2370	141.393 $\pm$ 25.849	136.320 $\pm$ 15.887	134.310 $\pm$ 13.926	151.960 $\pm$ 15.993
2380	138.057 $\pm$ 25.787	155.330 $\pm$ 72.379	153.257 $\pm$ 50.045	175.933 $\pm$ 54.461
2390	141.763 $\pm$ 26.194	144.450 $\pm$ 30.779	154.403 $\pm$ 58.925	164.193 $\pm$ 57.211
2400	154.523 $\pm$ 44.265	164.010 $\pm$ 47.442	156.460 $\pm$ 28.956	167.857 $\pm$ 24.486
2410	150.457 $\pm$ 62.595	151.660 $\pm$ 47.142	156.347 $\pm$ 38.320	163.017 $\pm$ 18.838
2420	142.233 $\pm$ 26.322	141.493 $\pm$ 18.141	149.960 $\pm$ 35.370	173.020 $\pm$ 52.423
2430	140.283 $\pm$ 34.750	146.597 $\pm$ 44.599	158.603 $\pm$ 58.077	158.587 $\pm$ 16.344

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
2440	155.250 $\pm$ 51.122	151.780 $\pm$ 25.374	151.540 $\pm$ 21.956	163.963 $\pm$ 19.067
2450	144.827 $\pm$ 21.925	153.327 $\pm$ 35.256	164.743 $\pm$ 68.700	161.540 $\pm$ 15.941
2460	138.470 $\pm$ 20.112	139.603 $\pm$ 21.722	140.737 $\pm$ 31.762	152.727 $\pm$ 8.633
2470	146.720 $\pm$ 54.285	137.580 $\pm$ 14.066	141.847 $\pm$ 11.795	157.303 $\pm$ 13.974
2480	133.410 $\pm$ 9.933	139.940 $\pm$ 18.233	145.263 $\pm$ 31.999	158.173 $\pm$ 22.809
2490	138.013 $\pm$ 19.109	137.643 $\pm$ 12.263	147.717 $\pm$ 34.985	161.977 $\pm$ 27.474
2500	144.800 $\pm$ 32.060	144.220 $\pm$ 36.001	152.863 $\pm$ 31.800	168.267 $\pm$ 42.769
2510	138.717 $\pm$ 12.594	151.020 $\pm$ 42.281	144.587 $\pm$ 23.863	162.063 $\pm$ 25.713
2520	136.903 $\pm$ 16.822	136.093 $\pm$ 9.655	141.657 $\pm$ 13.248	158.617 $\pm$ 12.621
2530	141.357 $\pm$ 12.994	145.553 $\pm$ 23.646	143.037 $\pm$ 13.332	163.123 $\pm$ 15.042
2540	138.503 $\pm$ 11.613	141.740 $\pm$ 13.045	144.363 $\pm$ 14.631	162.163 $\pm$ 47.163
2550	149.907 $\pm$ 31.629	142.893 $\pm$ 20.777	146.093 $\pm$ 20.229	169.293 $\pm$ 32.166
2560	149.690 $\pm$ 38.328	142.683 $\pm$ 14.064	146.710 $\pm$ 27.093	169.260 $\pm$ 45.898
2570	146.173 $\pm$ 33.414	141.863 $\pm$ 10.875	148.280 $\pm$ 15.058	161.623 $\pm$ 10.418
2580	143.467 $\pm$ 13.858	144.513 $\pm$ 16.392	148.157 $\pm$ 16.903	165.783 $\pm$ 33.644
2590	147.567 $\pm$ 41.550	148.250 $\pm$ 27.165	151.910 $\pm$ 40.745	172.663 $\pm$ 57.774
2600	140.480 $\pm$ 7.954	144.513 $\pm$ 10.830	146.327 $\pm$ 9.860	164.377 $\pm$ 12.475
2610	147.260 $\pm$ 38.371	148.553 $\pm$ 38.717	154.673 $\pm$ 34.340	162.190 $\pm$ 13.428
2620	151.707 $\pm$ 31.593	149.177 $\pm$ 15.593	148.647 $\pm$ 13.092	166.253 $\pm$ 16.892
2630	146.253 $\pm$ 29.633	154.730 $\pm$ 27.296	153.183 $\pm$ 20.633	175.607 $\pm$ 41.494
2640	143.433 $\pm$ 10.020	154.520 $\pm$ 15.579	149.207 $\pm$ 12.530	174.597 $\pm$ 20.213
2650	147.820 $\pm$ 12.441	154.327 $\pm$ 21.324	153.223 $\pm$ 14.971	172.387 $\pm$ 19.840
2660	150.637 $\pm$ 18.300	156.353 $\pm$ 32.663	159.080 $\pm$ 35.334	172.593 $\pm$ 15.546
2670	153.630 $\pm$ 22.783	150.757 $\pm$ 18.633	157.150 $\pm$ 27.778	178.067 $\pm$ 40.189
2680	170.713 $\pm$ 62.986	167.150 $\pm$ 39.306	174.097 $\pm$ 54.540	184.243 $\pm$ 40.517
2690	156.840 $\pm$ 25.846	155.817 $\pm$ 20.497	156.437 $\pm$ 35.281	186.293 $\pm$ 60.894
2700	157.017 $\pm$ 26.541	153.830 $\pm$ 22.537	161.270 $\pm$ 23.947	183.337 $\pm$ 43.842
2710	159.620 $\pm$ 45.267	162.443 $\pm$ 32.928	154.877 $\pm$ 22.921	180.460 $\pm$ 32.138
2720	162.170 $\pm$ 24.306	161.520 $\pm$ 21.562	164.033 $\pm$ 25.721	184.190 $\pm$ 33.860
2730	153.757 $\pm$ 31.867	164.103 $\pm$ 46.049	171.513 $\pm$ 47.104	189.733 $\pm$ 38.930
2740	167.857 $\pm$ 40.563	158.400 $\pm$ 27.706	162.580 $\pm$ 32.335	183.347 $\pm$ 37.345
2750	151.823 $\pm$ 11.713	159.940 $\pm$ 17.607	152.597 $\pm$ 7.636	188.063 $\pm$ 39.519
2760	159.280 $\pm$ 23.210	157.533 $\pm$ 17.774	159.227 $\pm$ 29.841	185.747 $\pm$ 31.137
2770	159.870 $\pm$ 25.181	169.487 $\pm$ 53.003	171.127 $\pm$ 46.990	180.590 $\pm$ 34.617
2780	155.567 $\pm$ 14.202	159.447 $\pm$ 14.143	160.517 $\pm$ 13.392	179.697 $\pm$ 13.075
2790	160.807 $\pm$ 30.431	159.807 $\pm$ 28.440	161.093 $\pm$ 15.610	180.653 $\pm$ 15.152
2800	156.397 $\pm$ 10.490	156.857 $\pm$ 10.493	160.030 $\pm$ 15.845	184.100 $\pm$ 18.934
2810	171.833 $\pm$ 70.684	167.327 $\pm$ 29.800	168.673 $\pm$ 39.568	191.677 $\pm$ 34.051
2820	165.470 $\pm$ 25.006	169.093 $\pm$ 43.121	168.963 $\pm$ 35.821	183.390 $\pm$ 32.753
2830	156.267 $\pm$ 9.478	161.253 $\pm$ 30.445	170.197 $\pm$ 28.042	182.877 $\pm$ 15.949
2840	165.857 $\pm$ 31.105	168.970 $\pm$ 32.742	169.753 $\pm$ 24.198	192.847 $\pm$ 37.387
2850	160.953 $\pm$ 15.673	168.800 $\pm$ 47.460	171.700 $\pm$ 41.676	198.700 $\pm$ 40.317
2860	160.123 $\pm$ 12.114	161.747 $\pm$ 15.503	168.287 $\pm$ 13.546	184.570 $\pm$ 17.465
2870	168.183 $\pm$ 39.303	169.030 $\pm$ 30.674	171.077 $\pm$ 29.354	199.280 $\pm$ 55.352
2880	162.037 $\pm$ 16.635	167.070 $\pm$ 27.560	165.977 $\pm$ 14.975	180.693 $\pm$ 11.030
2890	164.133 $\pm$ 27.702	161.040 $\pm$ 13.391	168.433 $\pm$ 15.508	183.013 $\pm$ 11.509
2900	170.883 $\pm$ 33.064	172.007 $\pm$ 29.311	171.943 $\pm$ 31.902	184.613 $\pm$ 18.686
2910	160.217 $\pm$ 15.111	166.940 $\pm$ 14.969	169.130 $\pm$ 18.262	199.300 $\pm$ 54.821
2920	167.427 $\pm$ 14.589	177.590 $\pm$ 42.277	184.643 $\pm$ 42.526	203.527 $\pm$ 38.516
2930	168.880 $\pm$ 20.433	167.123 $\pm$ 16.770	165.217 $\pm$ 10.144	192.813 $\pm$ 17.081

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
2940	182.240 $\pm$ 54.760	173.480 $\pm$ 29.792	172.073 $\pm$ 22.160	201.320 $\pm$ 51.986
2950	176.183 $\pm$ 38.580	181.033 $\pm$ 41.461	182.933 $\pm$ 31.999	196.357 $\pm$ 15.345
2960	164.980 $\pm$ 19.033	174.423 $\pm$ 29.657	172.343 $\pm$ 24.385	203.010 $\pm$ 81.563
2970	172.060 $\pm$ 28.396	180.653 $\pm$ 40.566	184.043 $\pm$ 34.949	197.663 $\pm$ 33.624
2980	171.867 $\pm$ 42.976	174.903 $\pm$ 25.862	175.673 $\pm$ 15.602	191.080 $\pm$ 14.284
2990	171.093 $\pm$ 30.432	175.750 $\pm$ 29.005	184.667 $\pm$ 43.854	204.593 $\pm$ 36.716
3000	172.220 $\pm$ 27.123	179.310 $\pm$ 36.515	176.687 $\pm$ 37.928	204.737 $\pm$ 48.507
3010	185.540 $\pm$ 32.008	185.323 $\pm$ 30.799	181.913 $\pm$ 20.093	209.993 $\pm$ 33.415
3020	177.107 $\pm$ 31.965	175.233 $\pm$ 22.323	181.753 $\pm$ 42.107	201.990 $\pm$ 26.673
3030	181.577 $\pm$ 44.278	176.460 $\pm$ 27.169	180.277 $\pm$ 37.282	201.490 $\pm$ 46.953
3040	170.783 $\pm$ 25.478	182.290 $\pm$ 50.520	183.327 $\pm$ 35.848	200.163 $\pm$ 27.172
3050	173.103 $\pm$ 16.819	175.333 $\pm$ 41.000	182.013 $\pm$ 44.450	199.750 $\pm$ 23.368
3060	185.917 $\pm$ 42.174	175.717 $\pm$ 15.488	183.447 $\pm$ 43.319	211.607 $\pm$ 53.323
3070	172.743 $\pm$ 14.454	176.887 $\pm$ 14.728	183.640 $\pm$ 18.369	199.707 $\pm$ 17.701
3080	176.513 $\pm$ 27.655	178.223 $\pm$ 23.079	187.703 $\pm$ 46.944	207.853 $\pm$ 53.186
3090	178.427 $\pm$ 24.850	181.540 $\pm$ 21.744	178.753 $\pm$ 16.193	202.707 $\pm$ 25.236
3100	175.533 $\pm$ 14.540	179.383 $\pm$ 17.407	182.643 $\pm$ 15.362	205.937 $\pm$ 24.675
3110	177.520 $\pm$ 21.595	177.137 $\pm$ 15.491	176.563 $\pm$ 11.258	199.490 $\pm$ 18.437
3120	178.397 $\pm$ 19.413	179.610 $\pm$ 19.887	181.613 $\pm$ 12.132	206.720 $\pm$ 25.396
3130	198.047 $\pm$ 71.369	188.417 $\pm$ 42.177	189.327 $\pm$ 47.435	214.453 $\pm$ 46.250
3140	186.027 $\pm$ 38.240	190.233 $\pm$ 47.041	188.577 $\pm$ 27.953	212.280 $\pm$ 37.834
3150	185.583 $\pm$ 60.063	191.080 $\pm$ 54.682	187.920 $\pm$ 25.018	210.423 $\pm$ 36.341
3160	191.027 $\pm$ 52.027	188.897 $\pm$ 35.972	184.107 $\pm$ 14.013	219.940 $\pm$ 49.426
3170	192.013 $\pm$ 52.201	193.827 $\pm$ 50.188	190.517 $\pm$ 31.358	206.907 $\pm$ 18.756
3180	193.480 $\pm$ 59.989	189.323 $\pm$ 24.162	188.750 $\pm$ 22.017	208.450 $\pm$ 23.076
3190	184.423 $\pm$ 27.899	188.510 $\pm$ 42.639	188.727 $\pm$ 21.638	212.890 $\pm$ 38.302
3200	184.320 $\pm$ 21.757	192.547 $\pm$ 44.507	190.000 $\pm$ 22.986	209.340 $\pm$ 20.664
3210	193.930 $\pm$ 39.560	193.153 $\pm$ 36.616	193.093 $\pm$ 30.576	212.440 $\pm$ 28.526
3220	199.117 $\pm$ 41.772	195.780 $\pm$ 23.775	205.433 $\pm$ 41.292	225.130 $\pm$ 44.660
3230	184.700 $\pm$ 23.105	185.197 $\pm$ 20.616	192.343 $\pm$ 19.689	207.257 $\pm$ 13.066
3240	187.977 $\pm$ 33.391	194.247 $\pm$ 50.881	193.977 $\pm$ 35.189	214.180 $\pm$ 26.078
3250	184.757 $\pm$ 12.217	188.370 $\pm$ 20.531	194.677 $\pm$ 24.537	211.857 $\pm$ 13.478
3260	191.200 $\pm$ 30.852	195.897 $\pm$ 49.172	196.053 $\pm$ 38.357	213.010 $\pm$ 29.990
3270	186.503 $\pm$ 18.762	193.403 $\pm$ 39.224	198.337 $\pm$ 30.416	215.783 $\pm$ 30.247
3280	183.293 $\pm$ 15.460	185.600 $\pm$ 15.605	197.327 $\pm$ 51.740	224.640 $\pm$ 62.872
3290	194.210 $\pm$ 34.198	198.240 $\pm$ 27.588	202.557 $\pm$ 45.571	222.323 $\pm$ 41.798
3300	193.060 $\pm$ 37.019	201.473 $\pm$ 36.844	200.000 $\pm$ 35.326	226.327 $\pm$ 49.135
3310	207.303 $\pm$ 78.317	205.823 $\pm$ 52.703	195.090 $\pm$ 31.495	230.207 $\pm$ 62.117
3320	191.813 $\pm$ 29.227	184.800 $\pm$ 10.380	193.517 $\pm$ 17.243	213.007 $\pm$ 12.829
3330	199.167 $\pm$ 66.699	192.920 $\pm$ 20.835	193.757 $\pm$ 13.121	217.843 $\pm$ 21.221
3340	191.483 $\pm$ 20.236	190.817 $\pm$ 18.293	195.323 $\pm$ 16.888	217.247 $\pm$ 17.861
3350	194.000 $\pm$ 24.257	210.410 $\pm$ 69.798	195.880 $\pm$ 20.286	215.700 $\pm$ 13.071
3360	190.573 $\pm$ 18.804	193.797 $\pm$ 12.398	193.333 $\pm$ 15.896	221.453 $\pm$ 22.314
3370	206.980 $\pm$ 52.908	205.977 $\pm$ 37.315	213.543 $\pm$ 58.479	235.087 $\pm$ 74.245
3380	207.567 $\pm$ 50.322	202.167 $\pm$ 36.189	211.107 $\pm$ 46.863	237.137 $\pm$ 51.117
3390	201.853 $\pm$ 41.819	209.060 $\pm$ 51.560	207.813 $\pm$ 39.382	232.343 $\pm$ 41.224
3400	187.610 $\pm$ 11.762	193.353 $\pm$ 15.088	197.800 $\pm$ 15.816	219.320 $\pm$ 18.097
3410	211.237 $\pm$ 58.626	213.347 $\pm$ 65.581	204.927 $\pm$ 26.008	227.293 $\pm$ 27.628
3420	201.870 $\pm$ 37.390	205.710 $\pm$ 40.619	201.907 $\pm$ 24.627	223.230 $\pm$ 16.960
3430	202.747 $\pm$ 34.352	202.603 $\pm$ 26.520	204.380 $\pm$ 31.424	230.240 $\pm$ 25.523

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
3440	205.630 $\pm$ 48.066	207.240 $\pm$ 34.025	220.447 $\pm$ 68.982	239.927 $\pm$ 60.780
3450	219.190 $\pm$ 52.094	217.993 $\pm$ 48.786	209.323 $\pm$ 37.900	229.290 $\pm$ 30.284
3460	203.277 $\pm$ 46.998	198.820 $\pm$ 17.970	205.273 $\pm$ 21.964	224.300 $\pm$ 15.867
3470	208.827 $\pm$ 43.217	224.323 $\pm$ 62.583	227.707 $\pm$ 62.324	242.830 $\pm$ 56.431
3480	215.483 $\pm$ 51.016	216.900 $\pm$ 54.533	220.820 $\pm$ 42.878	234.587 $\pm$ 37.475
3490	208.440 $\pm$ 38.689	213.723 $\pm$ 31.670	236.073 $\pm$ 70.075	250.127 $\pm$ 73.867
3500	202.730 $\pm$ 24.813	218.150 $\pm$ 51.050	225.967 $\pm$ 54.440	243.010 $\pm$ 42.647
3510	210.040 $\pm$ 49.545	219.137 $\pm$ 52.509	208.857 $\pm$ 21.888	227.837 $\pm$ 16.102
3520	202.267 $\pm$ 24.734	207.597 $\pm$ 33.828	200.763 $\pm$ 10.511	230.280 $\pm$ 29.908
3530	219.447 $\pm$ 57.228	222.663 $\pm$ 40.709	210.643 $\pm$ 27.816	240.847 $\pm$ 49.939
3540	201.460 $\pm$ 24.205	209.463 $\pm$ 34.360	210.833 $\pm$ 20.959	229.353 $\pm$ 20.506
3550	208.860 $\pm$ 35.930	218.427 $\pm$ 36.539	223.717 $\pm$ 41.641	231.953 $\pm$ 14.622
3560	216.727 $\pm$ 43.254	213.303 $\pm$ 43.923	214.093 $\pm$ 36.212	237.513 $\pm$ 43.690
3570	205.167 $\pm$ 23.354	209.773 $\pm$ 27.042	219.543 $\pm$ 70.550	234.643 $\pm$ 20.406
3580	220.147 $\pm$ 53.767	219.873 $\pm$ 53.757	229.957 $\pm$ 73.295	250.903 $\pm$ 63.649
3590	214.293 $\pm$ 41.871	220.807 $\pm$ 43.302	223.100 $\pm$ 40.046	238.797 $\pm$ 32.801
3600	215.520 $\pm$ 44.123	210.123 $\pm$ 41.363	212.247 $\pm$ 19.030	240.117 $\pm$ 41.554
3610	205.540 $\pm$ 18.640	211.363 $\pm$ 23.341	209.683 $\pm$ 15.457	232.710 $\pm$ 14.242
3620	223.357 $\pm$ 58.978	222.437 $\pm$ 55.983	218.307 $\pm$ 40.946	254.840 $\pm$ 81.534
3630	202.673 $\pm$ 16.085	208.157 $\pm$ 18.840	206.007 $\pm$ 9.695	232.050 $\pm$ 15.884
3640	225.363 $\pm$ 46.985	219.907 $\pm$ 44.521	216.363 $\pm$ 29.501	243.640 $\pm$ 44.157
3650	213.630 $\pm$ 40.568	213.600 $\pm$ 27.369	211.567 $\pm$ 11.380	239.593 $\pm$ 22.090
3660	207.330 $\pm$ 13.767	207.263 $\pm$ 11.215	216.137 $\pm$ 30.751	242.817 $\pm$ 35.956
3670	204.017 $\pm$ 10.171	209.030 $\pm$ 9.413	215.207 $\pm$ 14.370	240.507 $\pm$ 19.418
3680	215.980 $\pm$ 44.526	222.220 $\pm$ 30.761	222.457 $\pm$ 54.473	246.137 $\pm$ 52.414
3690	212.047 $\pm$ 30.844	219.200 $\pm$ 45.755	221.080 $\pm$ 39.339	234.433 $\pm$ 9.160
3700	218.550 $\pm$ 56.055	214.743 $\pm$ 21.831	220.343 $\pm$ 21.883	244.210 $\pm$ 22.498
3710	209.447 $\pm$ 16.544	216.790 $\pm$ 27.801	224.767 $\pm$ 40.787	251.320 $\pm$ 62.834
3720	228.980 $\pm$ 66.624	220.607 $\pm$ 38.986	227.093 $\pm$ 40.186	270.540 $\pm$ 97.733
3730	225.757 $\pm$ 52.833	221.997 $\pm$ 29.698	231.593 $\pm$ 61.869	262.620 $\pm$ 63.611
3740	242.193 $\pm$ 67.101	268.693 $\pm$ 108.442	254.517 $\pm$ 71.640	268.520 $\pm$ 60.541
3750	235.327 $\pm$ 54.368	237.387 $\pm$ 46.843	244.377 $\pm$ 50.881	264.117 $\pm$ 56.874
3760	224.827 $\pm$ 58.051	226.207 $\pm$ 39.135	223.010 $\pm$ 29.300	252.270 $\pm$ 40.582
3770	235.800 $\pm$ 60.177	230.167 $\pm$ 64.147	239.933 $\pm$ 57.601	258.573 $\pm$ 41.409
3780	217.933 $\pm$ 39.882	229.423 $\pm$ 27.634	221.293 $\pm$ 11.860	256.090 $\pm$ 48.401
3790	219.427 $\pm$ 31.713	246.760 $\pm$ 91.966	245.810 $\pm$ 71.050	273.107 $\pm$ 69.422
3800	228.373 $\pm$ 36.280	243.747 $\pm$ 46.588	242.510 $\pm$ 56.032	253.873 $\pm$ 33.717
3810	238.227 $\pm$ 61.075	222.307 $\pm$ 23.356	255.247 $\pm$ 79.046	290.917 $\pm$ 94.759
3820	239.167 $\pm$ 61.405	237.013 $\pm$ 58.166	238.193 $\pm$ 47.548	258.207 $\pm$ 40.994
3830	231.457 $\pm$ 44.636	244.950 $\pm$ 70.871	236.043 $\pm$ 35.634	259.423 $\pm$ 36.560
3840	237.383 $\pm$ 37.290	249.247 $\pm$ 72.237	238.210 $\pm$ 48.074	256.440 $\pm$ 22.463
3850	251.777 $\pm$ 74.978	233.770 $\pm$ 51.567	243.437 $\pm$ 53.996	274.527 $\pm$ 74.775
3860	240.880 $\pm$ 55.878	236.593 $\pm$ 61.315	237.070 $\pm$ 51.917	280.420 $\pm$ 69.329
3870	233.637 $\pm$ 42.112	236.940 $\pm$ 40.971	238.383 $\pm$ 37.562	281.613 $\pm$ 81.696
3880	221.270 $\pm$ 23.093	231.963 $\pm$ 41.516	236.933 $\pm$ 35.990	262.303 $\pm$ 35.035
3890	226.807 $\pm$ 26.159	261.500 $\pm$ 143.676	266.313 $\pm$ 86.450	263.600 $\pm$ 37.169
3900	245.773 $\pm$ 71.121	247.043 $\pm$ 46.883	246.120 $\pm$ 49.260	274.500 $\pm$ 51.517
3910	230.093 $\pm$ 29.502	241.477 $\pm$ 49.372	234.613 $\pm$ 26.219	263.143 $\pm$ 38.944
3920	246.083 $\pm$ 77.603	236.553 $\pm$ 43.512	246.213 $\pm$ 46.517	278.643 $\pm$ 51.112
3930	248.593 $\pm$ 52.985	251.510 $\pm$ 74.487	252.443 $\pm$ 53.182	283.110 $\pm$ 71.456

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
3940	246.740 $\pm$ 60.146	236.940 $\pm$ 34.468	252.557 $\pm$ 63.210	283.007 $\pm$ 52.569
3950	238.983 $\pm$ 52.378	248.277 $\pm$ 50.781	235.927 $\pm$ 32.958	270.983 $\pm$ 40.385
3960	235.377 $\pm$ 41.855	249.583 $\pm$ 60.765	248.683 $\pm$ 53.309	283.203 $\pm$ 61.633
3970	250.693 $\pm$ 63.953	249.993 $\pm$ 60.473	267.710 $\pm$ 72.718	279.473 $\pm$ 53.516
3980	254.460 $\pm$ 51.750	249.617 $\pm$ 58.937	243.143 $\pm$ 37.191	273.463 $\pm$ 41.901
3990	245.147 $\pm$ 31.691	254.773 $\pm$ 78.852	253.033 $\pm$ 34.932	289.493 $\pm$ 57.910
4000	248.617 $\pm$ 61.940	286.977 $\pm$ 143.276	282.663 $\pm$ 110.646	284.490 $\pm$ 51.850
4010	261.253 $\pm$ 49.176	269.777 $\pm$ 75.308	260.197 $\pm$ 38.887	304.780 $\pm$ 95.242
4020	256.827 $\pm$ 41.497	260.203 $\pm$ 47.902	260.837 $\pm$ 46.504	285.673 $\pm$ 33.963
4030	276.323 $\pm$ 170.515	258.647 $\pm$ 75.847	286.753 $\pm$ 96.419	312.100 $\pm$ 93.658
4040	243.787 $\pm$ 57.319	246.117 $\pm$ 48.463	258.390 $\pm$ 120.162	288.883 $\pm$ 62.900
4050	247.790 $\pm$ 56.047	262.950 $\pm$ 66.228	256.823 $\pm$ 37.618	272.177 $\pm$ 31.643
4060	260.913 $\pm$ 46.635	259.510 $\pm$ 44.334	266.930 $\pm$ 52.582	299.303 $\pm$ 58.658
4070	241.487 $\pm$ 42.310	259.023 $\pm$ 42.573	267.753 $\pm$ 69.397	298.157 $\pm$ 87.421
4080	251.850 $\pm$ 66.528	256.483 $\pm$ 51.561	238.490 $\pm$ 13.013	281.153 $\pm$ 38.822
4090	235.470 $\pm$ 15.465	235.417 $\pm$ 15.480	252.810 $\pm$ 30.770	276.843 $\pm$ 44.387
4100	232.430 $\pm$ 15.095	235.523 $\pm$ 18.744	241.983 $\pm$ 17.815	283.753 $\pm$ 21.208
4110	254.973 $\pm$ 48.693	267.210 $\pm$ 76.528	255.597 $\pm$ 42.058	280.757 $\pm$ 64.597
4120	237.810 $\pm$ 24.408	254.677 $\pm$ 48.581	258.157 $\pm$ 43.563	278.887 $\pm$ 32.360
4130	234.907 $\pm$ 20.317	242.343 $\pm$ 21.733	247.543 $\pm$ 20.975	264.257 $\pm$ 14.202
4140	232.120 $\pm$ 20.965	245.933 $\pm$ 17.161	253.983 $\pm$ 49.509	281.227 $\pm$ 69.178
4150	243.343 $\pm$ 33.431	241.333 $\pm$ 33.477	246.437 $\pm$ 20.859	270.663 $\pm$ 21.853
4160	240.970 $\pm$ 23.143	267.490 $\pm$ 56.415	266.400 $\pm$ 87.641	286.763 $\pm$ 60.745
4170	246.740 $\pm$ 53.250	242.150 $\pm$ 25.331	256.773 $\pm$ 31.101	282.973 $\pm$ 49.185
4180	248.967 $\pm$ 53.547	245.763 $\pm$ 14.976	256.363 $\pm$ 34.024	293.237 $\pm$ 70.367
4190	248.467 $\pm$ 42.441	257.313 $\pm$ 56.066	245.850 $\pm$ 26.486	276.313 $\pm$ 33.353
4200	254.863 $\pm$ 53.857	258.850 $\pm$ 37.683	242.427 $\pm$ 13.514	286.070 $\pm$ 50.776
4210	249.853 $\pm$ 41.289	251.250 $\pm$ 31.875	251.893 $\pm$ 32.685	287.190 $\pm$ 56.981
4220	249.507 $\pm$ 44.798	247.133 $\pm$ 18.767	252.680 $\pm$ 15.170	276.680 $\pm$ 19.804
4230	245.513 $\pm$ 20.972	252.957 $\pm$ 35.191	269.233 $\pm$ 69.774	301.150 $\pm$ 65.188
4240	264.140 $\pm$ 61.994	260.273 $\pm$ 53.205	255.010 $\pm$ 20.160	290.307 $\pm$ 49.807
4250	256.843 $\pm$ 58.380	252.033 $\pm$ 28.350	247.447 $\pm$ 15.243	275.253 $\pm$ 15.392
4260	256.150 $\pm$ 56.808	258.947 $\pm$ 52.055	252.050 $\pm$ 22.108	290.267 $\pm$ 44.419
4270	256.573 $\pm$ 58.492	257.063 $\pm$ 43.501	258.173 $\pm$ 38.216	297.800 $\pm$ 50.176
4280	242.557 $\pm$ 13.893	247.877 $\pm$ 19.645	257.497 $\pm$ 42.337	283.763 $\pm$ 38.066
4290	247.933 $\pm$ 19.525	263.937 $\pm$ 71.099	263.543 $\pm$ 52.222	284.443 $\pm$ 22.872
4300	253.267 $\pm$ 40.585	249.120 $\pm$ 18.030	257.270 $\pm$ 15.092	287.250 $\pm$ 36.011
4310	250.217 $\pm$ 17.962	253.353 $\pm$ 16.583	258.857 $\pm$ 18.654	278.857 $\pm$ 13.332
4320	251.333 $\pm$ 39.918	261.783 $\pm$ 36.352	270.517 $\pm$ 49.303	284.103 $\pm$ 21.808
4330	246.190 $\pm$ 17.381	262.467 $\pm$ 46.911	264.147 $\pm$ 55.797	306.213 $\pm$ 96.344
4340	282.090 $\pm$ 79.725	276.353 $\pm$ 70.493	279.587 $\pm$ 50.230	299.020 $\pm$ 34.182
4350	278.417 $\pm$ 86.329	286.840 $\pm$ 111.367	270.983 $\pm$ 47.390	297.237 $\pm$ 45.562
4360	260.847 $\pm$ 42.989	268.153 $\pm$ 36.919	280.580 $\pm$ 49.170	298.727 $\pm$ 57.102
4370	261.240 $\pm$ 47.864	274.900 $\pm$ 83.879	276.127 $\pm$ 37.160	307.037 $\pm$ 47.866
4380	260.757 $\pm$ 46.545	302.887 $\pm$ 186.193	281.110 $\pm$ 79.023	303.100 $\pm$ 78.572
4390	282.490 $\pm$ 74.663	302.943 $\pm$ 99.926	275.907 $\pm$ 63.773	295.257 $\pm$ 53.520
4400	301.007 $\pm$ 131.645	278.730 $\pm$ 81.945	270.507 $\pm$ 34.524	300.747 $\pm$ 47.687
4410	258.343 $\pm$ 36.926	280.020 $\pm$ 56.914	267.553 $\pm$ 29.696	306.257 $\pm$ 44.538
4420	270.520 $\pm$ 48.558	277.000 $\pm$ 78.199	277.097 $\pm$ 78.070	320.613 $\pm$ 109.878
4430	295.417 $\pm$ 90.434	393.490 $\pm$ 387.402	315.580 $\pm$ 112.304	374.367 $\pm$ 191.257

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
4440	340.727 $\pm$ 185.036	290.673 $\pm$ 54.582	297.400 $\pm$ 91.208	331.647 $\pm$ 74.960
4450	278.703 $\pm$ 60.682	296.397 $\pm$ 98.077	302.943 $\pm$ 90.672	352.953 $\pm$ 145.976
4460	274.197 $\pm$ 51.944	268.547 $\pm$ 37.869	276.590 $\pm$ 70.299	313.473 $\pm$ 78.275
4470	264.753 $\pm$ 47.311	269.717 $\pm$ 49.124	278.420 $\pm$ 53.771	299.487 $\pm$ 36.000
4480	259.393 $\pm$ 17.223	265.503 $\pm$ 27.483	274.780 $\pm$ 20.967	294.107 $\pm$ 18.227
4490	281.443 $\pm$ 72.621	275.613 $\pm$ 56.447	265.303 $\pm$ 26.618	299.283 $\pm$ 26.020
4500	268.793 $\pm$ 44.855	279.717 $\pm$ 50.523	286.423 $\pm$ 83.893	302.277 $\pm$ 46.630
4510	263.577 $\pm$ 32.483	286.113 $\pm$ 77.625	279.747 $\pm$ 60.732	305.513 $\pm$ 33.622
4520	261.423 $\pm$ 24.254	268.563 $\pm$ 21.498	273.393 $\pm$ 20.625	306.403 $\pm$ 64.455
4530	277.600 $\pm$ 56.520	268.080 $\pm$ 24.344	287.030 $\pm$ 60.525	324.880 $\pm$ 77.605
4540	273.573 $\pm$ 58.165	279.653 $\pm$ 72.034	269.460 $\pm$ 27.837	309.100 $\pm$ 48.238
4550	280.067 $\pm$ 28.231	293.410 $\pm$ 51.252	296.087 $\pm$ 34.253	320.963 $\pm$ 34.476
4560	261.153 $\pm$ 16.386	272.280 $\pm$ 21.528	274.577 $\pm$ 19.997	305.990 $\pm$ 31.860
4570	274.157 $\pm$ 46.635	267.633 $\pm$ 19.291	271.190 $\pm$ 15.620	308.260 $\pm$ 26.192
4580	269.550 $\pm$ 15.256	274.153 $\pm$ 19.150	275.443 $\pm$ 22.008	303.423 $\pm$ 20.469
4590	267.780 $\pm$ 17.179	266.423 $\pm$ 15.667	270.113 $\pm$ 19.993	301.333 $\pm$ 16.918
4600	278.667 $\pm$ 49.891	286.803 $\pm$ 57.489	284.580 $\pm$ 37.893	311.997 $\pm$ 25.573
4610	285.470 $\pm$ 65.154	287.040 $\pm$ 48.177	288.540 $\pm$ 37.142	320.077 $\pm$ 31.782
4620	267.660 $\pm$ 24.066	270.000 $\pm$ 15.427	279.077 $\pm$ 18.254	315.580 $\pm$ 67.548
4630	321.160 $\pm$ 127.563	438.757 $\pm$ 674.832	328.527 $\pm$ 98.318	388.867 $\pm$ 199.312
4640	316.437 $\pm$ 61.092	306.933 $\pm$ 32.060	330.990 $\pm$ 78.456	390.167 $\pm$ 90.401
4650	285.320 $\pm$ 34.752	294.467 $\pm$ 36.837	319.823 $\pm$ 104.022	324.140 $\pm$ 36.718
4660	278.603 $\pm$ 40.858	273.660 $\pm$ 20.466	280.940 $\pm$ 21.441	315.490 $\pm$ 24.378
4670	276.970 $\pm$ 28.912	274.147 $\pm$ 14.594	283.133 $\pm$ 26.766	321.033 $\pm$ 44.528
4680	288.830 $\pm$ 62.594	283.643 $\pm$ 35.993	292.573 $\pm$ 36.745	333.650 $\pm$ 82.035
4690	269.257 $\pm$ 19.227	275.483 $\pm$ 18.860	284.207 $\pm$ 21.686	320.657 $\pm$ 45.218
4700	276.497 $\pm$ 23.422	279.293 $\pm$ 23.189	277.360 $\pm$ 12.947	322.297 $\pm$ 58.659
4710	276.523 $\pm$ 21.354	288.987 $\pm$ 71.557	297.043 $\pm$ 69.797	322.143 $\pm$ 66.162
4720	286.003 $\pm$ 34.240	295.907 $\pm$ 47.778	306.110 $\pm$ 49.762	328.227 $\pm$ 28.884
4730	288.923 $\pm$ 30.555	301.370 $\pm$ 60.338	325.833 $\pm$ 98.717	349.237 $\pm$ 90.187
4740	282.500 $\pm$ 39.886	298.563 $\pm$ 76.441	290.263 $\pm$ 40.903	325.870 $\pm$ 61.281
4750	286.143 $\pm$ 40.738	295.737 $\pm$ 73.458	288.677 $\pm$ 22.153	337.430 $\pm$ 85.113
4760	316.600 $\pm$ 79.729	311.643 $\pm$ 72.125	301.730 $\pm$ 44.666	347.760 $\pm$ 91.913
4770	303.823 $\pm$ 44.965	332.847 $\pm$ 81.843	320.220 $\pm$ 74.424	349.443 $\pm$ 59.888
4780	292.030 $\pm$ 35.901	294.650 $\pm$ 27.132	309.613 $\pm$ 55.858	339.443 $\pm$ 54.994
4790	291.030 $\pm$ 21.736	300.957 $\pm$ 32.820	307.957 $\pm$ 60.392	340.300 $\pm$ 37.089
4800	287.867 $\pm$ 29.453	293.213 $\pm$ 60.184	293.750 $\pm$ 31.949	334.783 $\pm$ 57.141
4810	287.353 $\pm$ 31.974	295.157 $\pm$ 39.390	303.133 $\pm$ 49.987	328.320 $\pm$ 23.179
4820	280.880 $\pm$ 14.026	282.823 $\pm$ 19.696	288.340 $\pm$ 23.197	317.727 $\pm$ 19.139
4830	294.080 $\pm$ 48.528	297.507 $\pm$ 46.126	297.493 $\pm$ 36.556	329.467 $\pm$ 46.736
4840	293.730 $\pm$ 51.128	299.927 $\pm$ 60.886	307.410 $\pm$ 62.818	353.017 $\pm$ 94.716
4850	293.857 $\pm$ 52.036	298.830 $\pm$ 53.682	291.727 $\pm$ 18.600	319.053 $\pm$ 17.699
4860	284.613 $\pm$ 22.225	295.160 $\pm$ 36.259	296.070 $\pm$ 25.080	328.307 $\pm$ 26.506
4870	311.770 $\pm$ 71.599	308.387 $\pm$ 73.969	309.323 $\pm$ 54.364	347.250 $\pm$ 67.941
4880	299.263 $\pm$ 56.923	306.360 $\pm$ 72.020	305.210 $\pm$ 46.216	338.973 $\pm$ 53.487
4890	302.440 $\pm$ 75.333	299.907 $\pm$ 49.471	312.737 $\pm$ 48.680	358.793 $\pm$ 90.240
4900	311.323 $\pm$ 65.826	310.397 $\pm$ 48.275	308.590 $\pm$ 37.630	338.797 $\pm$ 39.700
4910	289.627 $\pm$ 29.042	321.837 $\pm$ 104.536	309.437 $\pm$ 50.770	344.067 $\pm$ 46.486
4920	301.720 $\pm$ 36.388	335.337 $\pm$ 97.102	321.830 $\pm$ 80.133	339.730 $\pm$ 36.656
4930	289.507 $\pm$ 21.670	301.660 $\pm$ 36.863	307.840 $\pm$ 55.377	330.707 $\pm$ 19.640

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
4940	299.730 $\pm$ 53.622	330.023 $\pm$ 180.095	327.943 $\pm$ 93.197	350.837 $\pm$ 72.582
4950	311.173 $\pm$ 87.007	307.797 $\pm$ 55.693	317.667 $\pm$ 76.116	333.430 $\pm$ 23.741
4960	312.520 $\pm$ 81.154	306.400 $\pm$ 39.223	310.230 $\pm$ 30.138	346.040 $\pm$ 35.947
4970	319.383 $\pm$ 86.046	307.313 $\pm$ 45.564	318.457 $\pm$ 42.397	353.090 $\pm$ 50.855
4980	316.193 $\pm$ 70.473	323.567 $\pm$ 67.256	326.813 $\pm$ 64.341	338.347 $\pm$ 27.282
4990	314.290 $\pm$ 65.410	318.827 $\pm$ 63.785	313.410 $\pm$ 42.184	344.837 $\pm$ 38.115
5000	302.033 $\pm$ 31.278	341.560 $\pm$ 103.131	357.847 $\pm$ 108.357	362.423 $\pm$ 63.019

B.2 Pre-Sorted Ascending Array Distribution (500 Data Points)

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
10	0.137 $\pm$ 0.072	0.137 $\pm$ 0.085	0.123 $\pm$ 0.068	0.307 $\pm$ 0.069
20	0.247 $\pm$ 0.082	0.300 $\pm$ 0.105	0.170 $\pm$ 0.099	0.577 $\pm$ 0.094
30	0.460 $\pm$ 0.190	0.547 $\pm$ 0.189	0.253 $\pm$ 0.107	0.880 $\pm$ 0.100
40	0.670 $\pm$ 0.252	0.913 $\pm$ 0.219	0.350 $\pm$ 0.166	1.107 $\pm$ 0.221
50	0.847 $\pm$ 0.073	1.323 $\pm$ 0.107	0.407 $\pm$ 0.105	1.273 $\pm$ 0.117
60	1.227 $\pm$ 0.187	1.863 $\pm$ 0.185	0.423 $\pm$ 0.152	1.550 $\pm$ 0.133
70	2.800 $\pm$ 0.291	3.333 $\pm$ 0.324	0.860 $\pm$ 0.089	2.117 $\pm$ 0.212
80	2.453 $\pm$ 0.785	2.877 $\pm$ 1.032	0.727 $\pm$ 0.260	2.200 $\pm$ 0.368
90	2.743 $\pm$ 0.808	3.443 $\pm$ 0.909	0.820 $\pm$ 0.275	2.483 $\pm$ 0.359
100	3.330 $\pm$ 1.992	4.047 $\pm$ 0.715	0.893 $\pm$ 0.223	2.847 $\pm$ 1.540
110	5.497 $\pm$ 0.637	7.453 $\pm$ 6.140	1.360 $\pm$ 0.181	3.300 $\pm$ 0.324
120	6.837 $\pm$ 0.362	7.497 $\pm$ 0.584	1.390 $\pm$ 0.160	3.540 $\pm$ 0.234
130	7.060 $\pm$ 7.132	6.627 $\pm$ 2.217	1.210 $\pm$ 0.384	3.520 $\pm$ 0.531
140	6.037 $\pm$ 4.234	5.827 $\pm$ 1.011	1.040 $\pm$ 0.122	3.440 $\pm$ 0.177
150	5.883 $\pm$ 1.086	6.770 $\pm$ 1.658	1.120 $\pm$ 0.209	3.740 $\pm$ 0.334
160	10.130 $\pm$ 6.982	11.753 $\pm$ 2.747	1.727 $\pm$ 0.499	4.687 $\pm$ 0.685
170	10.520 $\pm$ 3.194	12.953 $\pm$ 3.431	2.283 $\pm$ 2.386	5.830 $\pm$ 4.938
180	8.253 $\pm$ 1.847	14.187 $\pm$ 4.134	1.523 $\pm$ 0.386	4.543 $\pm$ 0.522
190	9.950 $\pm$ 3.198	15.707 $\pm$ 2.203	1.723 $\pm$ 0.535	5.973 $\pm$ 5.608
200	19.803 $\pm$ 12.970	28.267 $\pm$ 41.254	3.413 $\pm$ 3.849	7.053 $\pm$ 3.971
210	17.213 $\pm$ 5.004	20.600 $\pm$ 9.264	2.580 $\pm$ 0.558	6.480 $\pm$ 0.786
220	21.700 $\pm$ 0.644	26.500 $\pm$ 8.010	3.327 $\pm$ 1.854	7.290 $\pm$ 0.439
230	24.677 $\pm$ 4.156	27.457 $\pm$ 4.125	3.063 $\pm$ 0.209	8.457 $\pm$ 5.121
240	26.420 $\pm$ 18.813	29.343 $\pm$ 14.784	3.113 $\pm$ 1.736	8.803 $\pm$ 3.659
250	15.277 $\pm$ 4.775	17.810 $\pm$ 5.962	1.920 $\pm$ 0.472	6.627 $\pm$ 0.770
260	14.747 $\pm$ 1.031	26.197 $\pm$ 3.683	1.980 $\pm$ 0.335	6.743 $\pm$ 0.544
270	17.900 $\pm$ 6.301	23.893 $\pm$ 10.228	2.277 $\pm$ 0.776	7.513 $\pm$ 1.867
280	31.450 $\pm$ 14.969	36.230 $\pm$ 19.021	4.483 $\pm$ 5.440	9.613 $\pm$ 3.485
290	18.947 $\pm$ 4.812	24.973 $\pm$ 6.621	2.103 $\pm$ 0.187	7.187 $\pm$ 0.520
300	20.700 $\pm$ 5.506	23.817 $\pm$ 5.771	2.243 $\pm$ 0.239	7.487 $\pm$ 0.438
310	38.840 $\pm$ 26.862	41.170 $\pm$ 18.325	3.667 $\pm$ 1.981	13.900 $\pm$ 14.361
320	24.467 $\pm$ 8.772	29.707 $\pm$ 8.924	2.607 $\pm$ 0.537	8.743 $\pm$ 1.798
330	23.073 $\pm$ 2.638	39.047 $\pm$ 9.966	2.810 $\pm$ 0.588	8.610 $\pm$ 0.856
340	44.833 $\pm$ 16.627	51.357 $\pm$ 13.045	4.790 $\pm$ 1.840	11.797 $\pm$ 3.350
350	27.860 $\pm$ 7.287	34.790 $\pm$ 18.753	2.873 $\pm$ 0.534	9.073 $\pm$ 1.484
360	36.760 $\pm$ 27.657	46.670 $\pm$ 19.342	3.600 $\pm$ 1.068	9.910 $\pm$ 1.311
370	30.387 $\pm$ 6.586	34.543 $\pm$ 9.408	3.050 $\pm$ 0.276	9.250 $\pm$ 0.419
380	33.503 $\pm$ 10.658	42.430 $\pm$ 12.263	3.397 $\pm$ 0.877	9.713 $\pm$ 1.158
390	61.677 $\pm$ 12.390	64.393 $\pm$ 11.342	6.623 $\pm$ 5.110	14.313 $\pm$ 5.499
400	37.727 $\pm$ 11.243	52.670 $\pm$ 24.320	3.810 $\pm$ 1.098	10.800 $\pm$ 1.158
410	55.867 $\pm$ 39.217	76.077 $\pm$ 31.770	5.237 $\pm$ 3.097	13.590 $\pm$ 5.406
420	51.930 $\pm$ 20.980	63.307 $\pm$ 27.112	4.303 $\pm$ 1.234	15.787 $\pm$ 13.914
430	131.387 $\pm$ 305.686	136.683 $\pm$ 266.213	8.297 $\pm$ 13.851	17.383 $\pm$ 8.782
440	62.247 $\pm$ 33.679	107.907 $\pm$ 140.824	4.790 $\pm$ 1.420	15.673 $\pm$ 10.291
450	109.063 $\pm$ 160.416	87.367 $\pm$ 29.973	5.747 $\pm$ 1.181	16.350 $\pm$ 6.727
460	88.453 $\pm$ 45.318	85.123 $\pm$ 21.832	5.860 $\pm$ 0.894	15.620 $\pm$ 4.258
470	120.780 $\pm$ 116.294	102.047 $\pm$ 43.266	7.260 $\pm$ 6.603	18.400 $\pm$ 8.999
480	93.767 $\pm$ 20.160	119.307 $\pm$ 114.381	6.733 $\pm$ 2.563	16.000 $\pm$ 1.150

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
490	99.770 $\pm$ 18.714	117.210 $\pm$ 41.571	9.500 $\pm$ 11.636	17.450 $\pm$ 3.713
500	114.950 $\pm$ 71.155	117.527 $\pm$ 70.785	6.133 $\pm$ 1.071	19.073 $\pm$ 9.994
510	70.103 $\pm$ 21.322	82.633 $\pm$ 25.908	4.380 $\pm$ 1.154	15.683 $\pm$ 6.762
520	70.263 $\pm$ 25.465	87.930 $\pm$ 26.955	5.023 $\pm$ 1.551	15.660 $\pm$ 5.378
530	58.210 $\pm$ 11.748	64.900 $\pm$ 19.892	3.840 $\pm$ 0.263	13.993 $\pm$ 2.764
540	76.760 $\pm$ 30.393	82.160 $\pm$ 26.046	4.980 $\pm$ 1.382	15.547 $\pm$ 2.556
550	96.710 $\pm$ 30.267	113.140 $\pm$ 32.148	6.160 $\pm$ 1.616	17.947 $\pm$ 6.889
560	70.920 $\pm$ 19.032	97.817 $\pm$ 41.508	5.020 $\pm$ 1.465	16.610 $\pm$ 4.705
570	76.687 $\pm$ 26.289	90.090 $\pm$ 32.859	5.077 $\pm$ 1.419	16.150 $\pm$ 5.223
580	79.247 $\pm$ 30.141	105.727 $\pm$ 28.768	5.957 $\pm$ 3.390	16.090 $\pm$ 2.389
590	68.323 $\pm$ 9.479	80.917 $\pm$ 22.919	4.553 $\pm$ 1.017	16.297 $\pm$ 5.098
600	71.927 $\pm$ 14.243	95.233 $\pm$ 28.138	4.610 $\pm$ 0.254	15.220 $\pm$ 0.648
610	79.907 $\pm$ 24.010	96.303 $\pm$ 31.545	5.303 $\pm$ 1.371	19.387 $\pm$ 16.586
620	77.840 $\pm$ 15.253	114.527 $\pm$ 36.331	5.353 $\pm$ 1.152	16.703 $\pm$ 3.458
630	82.307 $\pm$ 21.578	116.907 $\pm$ 39.541	5.980 $\pm$ 3.051	17.507 $\pm$ 2.985
640	83.267 $\pm$ 18.395	103.937 $\pm$ 32.064	5.107 $\pm$ 0.852	17.437 $\pm$ 6.922
650	84.227 $\pm$ 14.846	118.780 $\pm$ 36.917	6.237 $\pm$ 4.878	18.693 $\pm$ 6.263
660	91.737 $\pm$ 21.190	113.547 $\pm$ 35.227	6.013 $\pm$ 1.744	17.187 $\pm$ 2.001
670	91.920 $\pm$ 21.568	106.403 $\pm$ 32.964	5.560 $\pm$ 1.120	16.877 $\pm$ 1.622
680	100.460 $\pm$ 31.348	121.483 $\pm$ 35.657	6.227 $\pm$ 1.472	19.153 $\pm$ 6.963
690	96.087 $\pm$ 18.052	114.313 $\pm$ 33.963	5.577 $\pm$ 0.227	17.713 $\pm$ 3.028
700	101.797 $\pm$ 25.640	142.957 $\pm$ 35.928	6.693 $\pm$ 2.943	17.787 $\pm$ 1.505
710	105.737 $\pm$ 33.797	137.593 $\pm$ 37.655	6.203 $\pm$ 0.869	18.063 $\pm$ 1.332
720	115.330 $\pm$ 28.668	144.083 $\pm$ 46.034	6.817 $\pm$ 1.634	21.027 $\pm$ 6.138
730	109.383 $\pm$ 29.943	157.157 $\pm$ 41.262	6.873 $\pm$ 1.737	20.923 $\pm$ 6.926
740	112.733 $\pm$ 28.552	163.190 $\pm$ 41.955	6.903 $\pm$ 1.546	19.753 $\pm$ 2.143
750	131.460 $\pm$ 41.970	149.430 $\pm$ 45.170	8.207 $\pm$ 3.111	20.567 $\pm$ 2.580
760	122.500 $\pm$ 53.460	160.587 $\pm$ 65.569	7.287 $\pm$ 2.768	22.177 $\pm$ 7.593
770	145.617 $\pm$ 88.278	189.897 $\pm$ 86.599	9.103 $\pm$ 5.238	24.510 $\pm$ 10.151
780	120.620 $\pm$ 14.983	149.983 $\pm$ 50.781	7.123 $\pm$ 1.152	23.603 $\pm$ 9.638
790	147.437 $\pm$ 79.378	183.570 $\pm$ 60.881	7.733 $\pm$ 3.133	21.900 $\pm$ 5.074
800	141.373 $\pm$ 63.262	179.657 $\pm$ 56.131	7.730 $\pm$ 2.219	22.670 $\pm$ 5.740
810	165.417 $\pm$ 95.218	191.543 $\pm$ 61.653	9.223 $\pm$ 6.496	24.207 $\pm$ 6.669
820	147.603 $\pm$ 75.063	187.317 $\pm$ 58.122	7.447 $\pm$ 1.848	24.877 $\pm$ 11.940
830	151.967 $\pm$ 55.049	194.097 $\pm$ 90.767	8.270 $\pm$ 3.347	22.673 $\pm$ 4.849
840	146.527 $\pm$ 36.049	176.757 $\pm$ 54.129	7.493 $\pm$ 1.655	24.060 $\pm$ 7.775
850	164.217 $\pm$ 54.616	209.977 $\pm$ 71.036	8.077 $\pm$ 2.371	24.940 $\pm$ 8.764
860	157.640 $\pm$ 48.243	179.230 $\pm$ 55.010	7.660 $\pm$ 1.812	24.020 $\pm$ 6.205
870	163.713 $\pm$ 64.873	207.040 $\pm$ 67.779	8.520 $\pm$ 3.528	24.580 $\pm$ 6.466
880	173.743 $\pm$ 99.322	225.383 $\pm$ 82.267	11.393 $\pm$ 16.916	26.563 $\pm$ 9.003
890	149.303 $\pm$ 14.168	202.673 $\pm$ 61.634	7.960 $\pm$ 2.133	24.383 $\pm$ 5.703
900	166.013 $\pm$ 42.576	223.677 $\pm$ 79.912	9.400 $\pm$ 5.231	24.647 $\pm$ 3.066
910	173.950 $\pm$ 41.547	209.387 $\pm$ 62.744	7.643 $\pm$ 1.097	25.263 $\pm$ 5.734
920	184.903 $\pm$ 61.943	258.777 $\pm$ 99.220	10.140 $\pm$ 6.497	27.873 $\pm$ 9.101
930	179.680 $\pm$ 50.884	216.530 $\pm$ 65.663	7.817 $\pm$ 1.750	23.967 $\pm$ 2.546
940	198.850 $\pm$ 84.492	244.273 $\pm$ 83.812	8.463 $\pm$ 2.851	27.650 $\pm$ 9.535
950	197.497 $\pm$ 70.218	273.257 $\pm$ 86.211	8.990 $\pm$ 2.316	27.660 $\pm$ 5.147
960	218.163 $\pm$ 74.729	260.910 $\pm$ 84.069	9.630 $\pm$ 3.606	29.067 $\pm$ 7.650
970	223.687 $\pm$ 110.502	264.900 $\pm$ 110.164	10.087 $\pm$ 5.672	30.967 $\pm$ 10.063
980	214.797 $\pm$ 72.258	294.047 $\pm$ 99.645	8.953 $\pm$ 2.406	28.210 $\pm$ 5.236

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
990	199.267 $\pm$ 48.291	249.057 $\pm$ 72.831	8.367 $\pm$ 2.181	27.450 $\pm$ 6.345
1000	227.220 $\pm$ 107.447	247.167 $\pm$ 90.293	12.597 $\pm$ 22.145	28.900 $\pm$ 9.249
1010	237.893 $\pm$ 89.601	306.190 $\pm$ 108.218	10.537 $\pm$ 5.835	31.493 $\pm$ 10.395
1020	217.973 $\pm$ 70.783	247.900 $\pm$ 70.588	8.033 $\pm$ 1.359	28.437 $\pm$ 6.983
1030	233.443 $\pm$ 91.361	300.013 $\pm$ 122.775	9.537 $\pm$ 3.989	31.243 $\pm$ 12.933
1040	228.597 $\pm$ 66.651	284.863 $\pm$ 79.053	9.393 $\pm$ 2.510	27.953 $\pm$ 2.519
1050	221.460 $\pm$ 85.022	281.323 $\pm$ 77.662	8.477 $\pm$ 1.670	28.743 $\pm$ 7.102
1060	220.890 $\pm$ 45.987	288.197 $\pm$ 105.590	9.420 $\pm$ 3.518	28.660 $\pm$ 4.888
1070	217.320 $\pm$ 39.785	304.517 $\pm$ 101.234	9.667 $\pm$ 4.780	28.980 $\pm$ 8.565
1080	232.550 $\pm$ 47.754	268.450 $\pm$ 95.361	8.483 $\pm$ 0.642	27.827 $\pm$ 1.452
1090	243.233 $\pm$ 63.492	317.653 $\pm$ 100.899	9.850 $\pm$ 2.586	30.260 $\pm$ 6.554
1100	249.280 $\pm$ 72.581	326.527 $\pm$ 114.419	10.897 $\pm$ 4.905	32.397 $\pm$ 8.548
1110	236.503 $\pm$ 33.973	320.553 $\pm$ 108.849	9.497 $\pm$ 1.983	32.123 $\pm$ 7.629
1120	264.287 $\pm$ 91.640	315.603 $\pm$ 94.321	11.997 $\pm$ 7.758	31.643 $\pm$ 5.093
1130	268.553 $\pm$ 116.614	319.637 $\pm$ 104.296	9.703 $\pm$ 2.448	29.873 $\pm$ 3.798
1140	270.967 $\pm$ 83.927	326.450 $\pm$ 140.967	10.730 $\pm$ 2.937	32.760 $\pm$ 8.920
1150	262.867 $\pm$ 77.611	349.297 $\pm$ 140.613	11.163 $\pm$ 5.734	34.633 $\pm$ 18.079
1160	294.940 $\pm$ 133.921	358.010 $\pm$ 109.677	10.177 $\pm$ 2.051	32.250 $\pm$ 4.950
1170	272.373 $\pm$ 68.242	391.297 $\pm$ 129.428	12.700 $\pm$ 8.533	34.220 $\pm$ 8.599
1180	290.430 $\pm$ 103.393	376.200 $\pm$ 157.918	10.347 $\pm$ 2.989	33.307 $\pm$ 9.124
1190	301.260 $\pm$ 108.270	390.583 $\pm$ 128.431	12.233 $\pm$ 6.546	35.503 $\pm$ 9.614
1200	302.657 $\pm$ 84.213	431.307 $\pm$ 164.121	14.063 $\pm$ 9.964	37.153 $\pm$ 10.654
1210	355.723 $\pm$ 160.683	425.520 $\pm$ 144.200	10.807 $\pm$ 2.345	35.073 $\pm$ 9.608
1220	289.083 $\pm$ 45.720	352.283 $\pm$ 109.710	10.427 $\pm$ 1.628	33.000 $\pm$ 3.706
1230	322.410 $\pm$ 142.596	364.950 $\pm$ 129.226	11.320 $\pm$ 4.181	34.367 $\pm$ 9.088
1240	359.970 $\pm$ 135.366	449.207 $\pm$ 170.601	14.440 $\pm$ 7.738	44.770 $\pm$ 23.175
1250	321.550 $\pm$ 74.918	410.140 $\pm$ 188.545	11.843 $\pm$ 3.050	36.663 $\pm$ 8.211
1260	330.113 $\pm$ 80.584	445.787 $\pm$ 169.005	13.173 $\pm$ 5.757	39.863 $\pm$ 16.759
1270	370.657 $\pm$ 167.643	466.297 $\pm$ 120.850	13.730 $\pm$ 9.096	37.887 $\pm$ 9.374
1280	370.420 $\pm$ 87.514	435.653 $\pm$ 148.219	14.550 $\pm$ 5.325	40.283 $\pm$ 10.068
1290	365.763 $\pm$ 113.218	477.577 $\pm$ 177.772	14.233 $\pm$ 10.212	38.203 $\pm$ 8.430
1300	356.243 $\pm$ 108.071	448.363 $\pm$ 141.347	13.477 $\pm$ 5.084	35.373 $\pm$ 3.611
1310	370.800 $\pm$ 100.568	478.637 $\pm$ 128.438	13.360 $\pm$ 5.079	38.213 $\pm$ 6.662
1320	366.207 $\pm$ 132.057	432.040 $\pm$ 126.749	14.017 $\pm$ 9.419	36.367 $\pm$ 6.040
1330	381.770 $\pm$ 132.143	470.147 $\pm$ 138.221	13.307 $\pm$ 5.247	39.827 $\pm$ 10.140
1340	387.103 $\pm$ 152.789	473.100 $\pm$ 138.833	15.873 $\pm$ 7.255	41.700 $\pm$ 15.855
1350	431.117 $\pm$ 199.358	505.060 $\pm$ 203.412	14.627 $\pm$ 5.269	41.167 $\pm$ 10.159
1360	376.580 $\pm$ 80.035	471.310 $\pm$ 200.086	14.077 $\pm$ 4.953	37.100 $\pm$ 4.040
1370	412.520 $\pm$ 146.400	465.230 $\pm$ 137.211	13.583 $\pm$ 5.605	39.010 $\pm$ 9.284
1380	405.147 $\pm$ 136.559	465.027 $\pm$ 162.847	14.937 $\pm$ 11.933	38.883 $\pm$ 6.355
1390	423.780 $\pm$ 156.777	456.890 $\pm$ 133.506	14.137 $\pm$ 7.290	38.657 $\pm$ 6.483
1400	487.070 $\pm$ 215.434	527.783 $\pm$ 196.990	18.367 $\pm$ 11.918	45.497 $\pm$ 17.289
1410	490.187 $\pm$ 215.222	571.420 $\pm$ 235.919	18.410 $\pm$ 11.739	45.287 $\pm$ 20.073
1420	532.560 $\pm$ 188.512	599.587 $\pm$ 196.467	16.430 $\pm$ 5.014	45.153 $\pm$ 12.005
1430	556.393 $\pm$ 184.762	598.783 $\pm$ 176.670	17.920 $\pm$ 6.212	48.940 $\pm$ 12.769
1440	594.247 $\pm$ 227.286	687.387 $\pm$ 229.049	20.790 $\pm$ 10.799	46.177 $\pm$ 8.436
1450	544.587 $\pm$ 188.628	628.583 $\pm$ 185.534	17.487 $\pm$ 4.667	47.223 $\pm$ 13.911
1460	454.690 $\pm$ 158.154	520.130 $\pm$ 142.134	15.017 $\pm$ 5.992	45.067 $\pm$ 20.674
1470	422.370 $\pm$ 97.739	606.200 $\pm$ 200.960	14.673 $\pm$ 3.812	41.477 $\pm$ 7.032
1480	423.860 $\pm$ 96.883	533.530 $\pm$ 191.794	14.407 $\pm$ 3.202	42.863 $\pm$ 12.417

Size (N)	First (μs)	Last (μs)	Middle (μs)	Random (μs)
1490	500.323 ± 147.428	548.973 ± 163.692	15.013 ± 3.275	44.310 ± 12.399
1500	519.690 ± 198.988	582.727 ± 150.250	17.420 ± 8.570	45.053 ± 6.108
1510	529.953 ± 184.612	610.030 ± 189.915	17.347 ± 5.819	45.097 ± 6.613
1520	486.433 ± 164.993	514.713 ± 178.960	14.997 ± 3.165	44.557 ± 17.703
1530	488.507 ± 151.095	558.430 ± 143.281	16.267 ± 6.078	50.670 ± 29.846
1540	519.713 ± 145.834	621.530 ± 154.732	16.683 ± 3.948	47.690 ± 16.398
1550	533.947 ± 204.834	637.693 ± 198.048	22.500 ± 31.418	46.363 ± 12.707
1560	479.537 ± 108.948	592.100 ± 148.494	18.383 ± 15.258	43.600 ± 9.209
1570	496.983 ± 128.859	607.793 ± 162.160	16.303 ± 4.091	46.620 ± 12.975
1580	523.920 ± 161.760	608.347 ± 165.988	16.197 ± 5.300	45.760 ± 10.123
1590	546.007 ± 218.312	570.723 ± 171.006	17.503 ± 9.431	45.083 ± 10.815
1600	508.463 ± 68.546	581.750 ± 182.673	15.903 ± 6.419	45.920 ± 11.946
1610	513.173 ± 117.281	612.890 ± 196.007	14.950 ± 2.702	45.820 ± 9.314
1620	521.110 ± 103.176	521.043 ± 78.347	15.437 ± 3.541	43.517 ± 4.594
1630	518.510 ± 132.206	624.433 ± 163.930	15.690 ± 6.236	45.453 ± 9.197
1640	521.173 ± 109.915	611.670 ± 136.973	16.513 ± 7.223	50.740 ± 16.396
1650	543.917 ± 149.610	565.610 ± 131.248	15.687 ± 4.093	44.930 ± 5.488
1660	544.297 ± 125.705	604.070 ± 153.453	15.300 ± 3.198	51.430 ± 23.269
1670	560.930 ± 118.591	586.420 ± 138.452	15.477 ± 2.965	44.977 ± 4.559
1680	536.963 ± 88.097	622.387 ± 162.795	15.450 ± 3.370	49.897 ± 14.133
1690	558.893 ± 101.860	654.430 ± 247.199	18.823 ± 7.947	51.820 ± 16.587
1700	557.023 ± 96.764	609.430 ± 142.930	16.307 ± 4.211	49.103 ± 15.692
1710	566.443 ± 133.214	716.857 ± 231.498	19.097 ± 9.960	51.590 ± 15.932
1720	586.257 ± 194.501	666.897 ± 206.700	16.120 ± 3.757	48.047 ± 9.953
1730	577.820 ± 96.043	682.313 ± 171.835	16.283 ± 4.265	52.447 ± 26.739
1740	635.143 ± 240.216	671.940 ± 179.321	19.443 ± 11.771	54.093 ± 26.721
1750	634.573 ± 170.943	687.177 ± 193.135	17.183 ± 5.197	51.950 ± 14.214
1760	668.903 ± 234.848	785.713 ± 266.997	17.117 ± 4.610	49.000 ± 8.459
1770	635.427 ± 210.669	749.267 ± 225.324	15.900 ± 2.355	51.863 ± 14.958
1780	652.900 ± 204.039	742.313 ± 233.566	16.377 ± 4.207	53.870 ± 19.625
1790	666.500 ± 198.185	633.333 ± 140.607	16.057 ± 4.520	48.613 ± 5.232
1800	648.903 ± 180.271	754.127 ± 228.099	17.217 ± 5.746	52.867 ± 15.641
1810	660.757 ± 149.286	820.787 ± 273.296	18.217 ± 6.405	56.070 ± 15.545
1820	639.127 ± 129.728	734.387 ± 210.644	18.443 ± 8.434	52.057 ± 11.950
1830	672.667 ± 149.963	752.267 ± 188.727	18.763 ± 6.786	53.177 ± 11.412
1840	677.250 ± 125.445	745.560 ± 192.493	16.283 ± 2.861	50.580 ± 7.722
1850	672.670 ± 178.330	731.510 ± 169.680	15.740 ± 2.297	50.487 ± 5.374
1860	674.590 ± 138.657	716.680 ± 140.983	15.273 ± 0.605	49.290 ± 4.781
1870	653.203 ± 83.751	749.033 ± 178.398	16.273 ± 2.494	50.840 ± 5.589
1880	679.310 ± 144.099	805.787 ± 282.826	17.653 ± 3.987	51.653 ± 5.601
1890	740.090 ± 242.991	781.367 ± 194.615	18.047 ± 5.253	52.907 ± 7.007
1900	711.050 ± 141.636	798.970 ± 163.955	16.560 ± 2.811	53.877 ± 9.564
1910	699.913 ± 123.374	784.783 ± 171.746	17.267 ± 5.597	53.033 ± 8.873
1920	780.333 ± 212.453	877.740 ± 301.379	19.107 ± 8.083	56.873 ± 13.733
1930	740.953 ± 176.921	777.493 ± 177.558	20.770 ± 13.605	58.870 ± 17.758
1940	757.067 ± 163.319	907.773 ± 233.605	20.337 ± 9.540	62.037 ± 20.914
1950	826.527 ± 252.154	969.680 ± 276.798	21.750 ± 10.906	57.973 ± 11.201
1960	808.077 ± 196.152	962.400 ± 355.578	21.053 ± 13.727	58.770 ± 15.670
1970	877.530 ± 267.377	920.640 ± 319.406	18.260 ± 4.622	61.130 ± 14.559
1980	888.567 ± 298.771	996.637 ± 320.908	19.770 ± 7.957	60.577 ± 16.048

Size (N)	First (μs)	Last (μs)	Middle (μs)	Random (μs)
1990	795.117 ± 239.991	953.947 ± 360.437	21.537 ± 16.988	61.473 ± 18.915
2000	833.117 ± 235.627	942.253 ± 305.157	19.583 ± 9.599	59.860 ± 15.003
2010	823.117 ± 202.297	960.280 ± 367.519	19.810 ± 10.691	64.653 ± 29.961
2020	851.610 ± 252.495	897.087 ± 215.522	18.027 ± 5.575	58.153 ± 11.741
2030	829.723 ± 249.447	984.420 ± 281.525	20.097 ± 11.780	58.277 ± 8.295
2040	823.367 ± 189.640	886.143 ± 179.179	16.673 ± 2.702	55.320 ± 5.540
2050	876.547 ± 239.199	973.370 ± 239.115	19.267 ± 4.991	60.500 ± 11.670
2060	829.230 ± 144.802	985.010 ± 252.985	18.013 ± 3.297	57.167 ± 7.597
2070	947.677 ± 303.349	1048.820 ± 271.653	20.073 ± 5.736	66.893 ± 31.681
2080	914.203 ± 285.852	1013.410 ± 290.863	20.600 ± 6.508	59.673 ± 11.820
2090	863.973 ± 210.381	1058.000 ± 305.995	18.590 ± 3.779	58.107 ± 8.575
2100	913.673 ± 210.579	952.617 ± 217.862	18.447 ± 5.774	57.330 ± 6.969
2110	882.513 ± 188.382	967.473 ± 243.016	17.903 ± 3.760	58.220 ± 7.191
2120	935.550 ± 251.168	1086.700 ± 359.610	19.187 ± 5.016	61.070 ± 9.325
2130	957.827 ± 277.336	1003.660 ± 196.012	19.413 ± 4.995	62.350 ± 11.033
2140	993.800 ± 281.515	1150.950 ± 318.637	20.777 ± 7.323	63.163 ± 12.578
2150	919.990 ± 248.912	1066.420 ± 258.074	20.140 ± 5.141	60.340 ± 8.236
2160	959.363 ± 234.276	1048.340 ± 265.794	19.050 ± 3.406	66.620 ± 28.330
2170	947.327 ± 205.580	1223.480 ± 393.802	24.290 ± 11.030	67.913 ± 17.682
2180	938.310 ± 198.019	1015.170 ± 238.051	19.897 ± 4.398	59.947 ± 6.508
2190	1006.110 ± 301.306	1064.010 ± 305.175	20.547 ± 5.023	63.137 ± 10.630
2200	988.447 ± 250.699	1103.900 ± 298.285	20.103 ± 4.377	63.000 ± 8.749
2210	956.857 ± 185.434	1079.190 ± 252.597	21.490 ± 6.207	65.483 ± 17.692
2220	1047.570 ± 282.590	1142.550 ± 272.691	24.787 ± 15.186	66.957 ± 20.482
2230	972.863 ± 212.953	1157.910 ± 362.787	23.807 ± 11.795	67.450 ± 17.540
2240	972.357 ± 197.199	1123.180 ± 294.703	21.407 ± 5.949	69.033 ± 25.274
2250	1045.150 ± 260.601	1069.810 ± 203.177	22.167 ± 7.155	66.820 ± 14.688
2260	1083.930 ± 390.660	1232.730 ± 376.065	22.743 ± 6.768	71.493 ± 18.117
2270	1128.690 ± 366.508	1243.520 ± 351.217	24.747 ± 10.634	77.633 ± 30.632
2280	1050.310 ± 206.416	1142.260 ± 262.106	22.110 ± 5.616	63.750 ± 6.701
2290	1004.700 ± 139.024	1113.610 ± 364.202	20.937 ± 5.156	66.727 ± 8.968
2300	990.390 ± 132.507	1105.260 ± 237.989	20.700 ± 4.181	64.277 ± 7.925
2310	992.190 ± 161.363	1044.680 ± 217.477	19.980 ± 3.603	63.927 ± 9.710
2320	1052.700 ± 231.908	1200.050 ± 371.493	21.487 ± 4.805	65.400 ± 6.675
2330	1071.860 ± 239.097	1171.520 ± 319.457	24.830 ± 11.217	81.627 ± 46.899
2340	1143.130 ± 286.956	1195.660 ± 293.636	20.937 ± 3.908	67.673 ± 14.643
2350	1082.960 ± 238.892	1327.650 ± 352.271	20.687 ± 3.580	66.187 ± 7.391
2360	1069.510 ± 180.545	1199.430 ± 303.104	25.893 ± 10.604	74.450 ± 30.235
2370	1148.320 ± 277.213	1214.360 ± 340.007	22.287 ± 12.026	67.743 ± 9.226
2380	1177.380 ± 302.310	1234.180 ± 380.546	21.127 ± 5.165	70.580 ± 12.930
2390	1210.600 ± 304.734	1266.340 ± 351.843	24.250 ± 7.975	69.717 ± 11.223
2400	1231.180 ± 313.681	1377.810 ± 433.081	24.543 ± 6.814	70.650 ± 11.059
2410	1236.750 ± 302.454	1349.440 ± 358.104	25.040 ± 7.732	72.703 ± 14.023
2420	1316.770 ± 394.441	1490.220 ± 480.594	25.337 ± 7.560	75.230 ± 12.956
2430	1214.030 ± 289.480	1377.920 ± 404.644	24.073 ± 6.057	69.237 ± 7.719
2440	1268.370 ± 315.123	1315.580 ± 306.100	23.257 ± 6.345	71.040 ± 8.784
2450	1333.140 ± 409.756	1511.900 ± 477.245	26.267 ± 9.616	80.150 ± 23.872
2460	1148.660 ± 135.470	1313.590 ± 351.894	25.037 ± 10.690	82.343 ± 56.695
2470	1197.350 ± 247.122	1274.440 ± 207.981	25.213 ± 11.969	78.207 ± 29.919
2480	1410.930 ± 351.034	1541.700 ± 355.777	34.133 ± 38.066	77.367 ± 17.345

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
2490	1314.180 $\pm$ 310.254	1449.560 $\pm$ 309.609	26.290 $\pm$ 8.274	73.370 $\pm$ 11.162
2500	1268.410 $\pm$ 295.598	1401.080 $\pm$ 328.914	24.423 $\pm$ 6.440	69.620 $\pm$ 10.383
2510	1257.240 $\pm$ 265.086	1387.260 $\pm$ 363.504	23.153 $\pm$ 5.241	76.417 $\pm$ 26.336
2520	1225.220 $\pm$ 171.313	1409.290 $\pm$ 373.183	26.547 $\pm$ 13.023	80.397 $\pm$ 35.107
2530	1216.210 $\pm$ 193.158	1316.180 $\pm$ 245.699	22.150 $\pm$ 2.548	70.523 $\pm$ 6.802
2540	1234.210 $\pm$ 237.893	1308.750 $\pm$ 258.142	22.713 $\pm$ 3.229	68.513 $\pm$ 4.780
2550	1264.320 $\pm$ 221.373	1474.110 $\pm$ 443.176	27.443 $\pm$ 14.080	77.710 $\pm$ 14.477
2560	1352.990 $\pm$ 346.127	1400.750 $\pm$ 424.066	23.060 $\pm$ 4.654	79.527 $\pm$ 34.655
2570	1399.210 $\pm$ 367.690	1473.480 $\pm$ 411.742	25.320 $\pm$ 6.207	74.610 $\pm$ 10.425
2580	1267.990 $\pm$ 196.633	1455.380 $\pm$ 363.652	26.287 $\pm$ 6.574	74.617 $\pm$ 13.620
2590	1353.880 $\pm$ 353.352	1413.200 $\pm$ 298.315	28.600 $\pm$ 22.935	78.947 $\pm$ 27.994
2600	1282.090 $\pm$ 207.649	1488.750 $\pm$ 276.314	28.607 $\pm$ 14.469	78.663 $\pm$ 21.555
2610	1302.510 $\pm$ 178.603	1570.500 $\pm$ 433.345	23.897 $\pm$ 4.196	74.733 $\pm$ 11.046
2620	1393.820 $\pm$ 355.028	1435.060 $\pm$ 354.368	25.507 $\pm$ 6.324	74.300 $\pm$ 8.479
2630	1465.350 $\pm$ 395.710	1463.520 $\pm$ 357.961	27.200 $\pm$ 12.148	78.607 $\pm$ 19.413
2640	1362.000 $\pm$ 274.947	1460.590 $\pm$ 314.876	25.597 $\pm$ 6.241	75.920 $\pm$ 7.508
2650	1429.140 $\pm$ 310.682	1528.330 $\pm$ 357.894	26.087 $\pm$ 7.041	75.573 $\pm$ 11.794
2660	1361.730 $\pm$ 221.430	1501.690 $\pm$ 295.073	28.157 $\pm$ 8.320	77.763 $\pm$ 16.023
2670	1449.600 $\pm$ 301.397	1582.590 $\pm$ 500.346	27.937 $\pm$ 11.929	80.827 $\pm$ 16.072
2680	1606.330 $\pm$ 563.005	1615.440 $\pm$ 393.517	27.737 $\pm$ 6.686	83.303 $\pm$ 28.680
2690	1536.010 $\pm$ 348.324	1530.330 $\pm$ 296.571	30.767 $\pm$ 14.334	78.730 $\pm$ 10.845
2700	1523.610 $\pm$ 405.458	1655.160 $\pm$ 324.507	30.670 $\pm$ 13.003	80.553 $\pm$ 14.412
2710	1598.130 $\pm$ 428.289	1761.040 $\pm$ 445.507	27.417 $\pm$ 6.299	78.493 $\pm$ 9.066
2720	1484.870 $\pm$ 265.346	1603.980 $\pm$ 307.567	29.230 $\pm$ 10.171	78.803 $\pm$ 11.084
2730	1425.340 $\pm$ 211.265	1529.940 $\pm$ 326.968	24.663 $\pm$ 1.764	75.613 $\pm$ 8.010
2740	1522.620 $\pm$ 333.247	1744.650 $\pm$ 452.087	29.663 $\pm$ 11.911	83.507 $\pm$ 20.853
2750	1571.050 $\pm$ 346.185	1586.220 $\pm$ 377.960	27.783 $\pm$ 11.715	80.373 $\pm$ 19.881
2760	1513.960 $\pm$ 255.436	1690.740 $\pm$ 386.911	26.667 $\pm$ 6.755	76.410 $\pm$ 10.192
2770	1432.650 $\pm$ 177.155	1630.110 $\pm$ 325.193	25.310 $\pm$ 3.503	79.250 $\pm$ 12.415
2780	1498.400 $\pm$ 341.637	1752.610 $\pm$ 425.901	27.027 $\pm$ 5.761	80.380 $\pm$ 14.046
2790	1658.620 $\pm$ 446.975	1722.000 $\pm$ 445.707	35.720 $\pm$ 29.799	85.510 $\pm$ 25.240
2800	1642.220 $\pm$ 430.496	1700.400 $\pm$ 351.919	31.103 $\pm$ 12.904	90.880 $\pm$ 30.154
2810	1609.790 $\pm$ 326.884	1606.570 $\pm$ 391.493	28.697 $\pm$ 6.716	77.957 $\pm$ 11.494
2820	1530.280 $\pm$ 229.588	1744.070 $\pm$ 385.182	32.403 $\pm$ 17.720	81.653 $\pm$ 16.421
2830	1720.120 $\pm$ 433.183	1570.430 $\pm$ 267.725	30.643 $\pm$ 9.151	87.640 $\pm$ 16.491
2840	1815.910 $\pm$ 473.172	1654.220 $\pm$ 419.608	29.177 $\pm$ 7.133	83.200 $\pm$ 10.457
2850	1719.440 $\pm$ 340.529	1835.490 $\pm$ 382.619	34.650 $\pm$ 12.443	93.073 $\pm$ 23.124
2860	1686.750 $\pm$ 442.902	1985.850 $\pm$ 479.889	33.030 $\pm$ 14.781	87.320 $\pm$ 22.977
2870	1633.390 $\pm$ 303.060	1807.410 $\pm$ 396.593	30.710 $\pm$ 10.139	88.777 $\pm$ 20.275
2880	1622.000 $\pm$ 241.887	1831.550 $\pm$ 383.812	28.230 $\pm$ 10.238	83.857 $\pm$ 13.883
2890	1774.980 $\pm$ 425.242	1883.440 $\pm$ 423.712	32.010 $\pm$ 11.839	83.430 $\pm$ 10.209
2900	1547.540 $\pm$ 188.145	1715.340 $\pm$ 324.956	31.280 $\pm$ 15.915	84.873 $\pm$ 16.935
2910	1793.910 $\pm$ 443.570	1931.880 $\pm$ 456.370	30.387 $\pm$ 9.638	83.113 $\pm$ 9.702
2920	1700.600 $\pm$ 327.974	1685.140 $\pm$ 266.542	31.157 $\pm$ 11.915	86.470 $\pm$ 19.380
2930	1663.650 $\pm$ 429.345	1814.820 $\pm$ 455.235	28.513 $\pm$ 5.850	83.180 $\pm$ 12.538
2940	1624.760 $\pm$ 216.124	1853.830 $\pm$ 417.536	29.620 $\pm$ 5.910	80.210 $\pm$ 6.338
2950	1757.840 $\pm$ 395.318	1959.390 $\pm$ 472.688	30.043 $\pm$ 7.477	83.977 $\pm$ 14.352
2960	1652.230 $\pm$ 242.998	1888.780 $\pm$ 472.998	29.893 $\pm$ 7.313	86.753 $\pm$ 20.570
2970	1732.570 $\pm$ 276.266	1784.580 $\pm$ 253.021	28.713 $\pm$ 5.568	81.273 $\pm$ 7.298
2980	1717.660 $\pm$ 205.427	1736.670 $\pm$ 224.416	29.200 $\pm$ 5.548	82.887 $\pm$ 8.080

Size (N)	First (μs)	Last (μs)	Middle (μs)	Random (μs)
2990	1783.300 ± 296.780	1894.970 ± 419.150	33.217 ± 12.847	91.207 ± 28.788
3000	1740.270 ± 327.994	1892.230 ± 305.561	33.463 ± 15.529	90.110 ± 24.434
3010	1813.330 ± 318.374	1891.640 ± 327.007	33.890 ± 18.038	96.033 ± 37.106
3020	1928.570 ± 518.422	2109.540 ± 535.392	34.013 ± 12.117	89.527 ± 14.958
3030	2053.050 ± 529.121	2079.320 ± 536.260	31.933 ± 8.172	88.740 ± 10.662
3040	1896.790 ± 338.108	2063.870 ± 551.860	32.340 ± 10.573	87.500 ± 10.080
3050	1820.780 ± 331.847	2057.640 ± 411.919	32.657 ± 8.341	89.057 ± 15.778
3060	1817.580 ± 363.384	1840.800 ± 296.548	34.910 ± 18.421	86.737 ± 11.209
3070	1756.110 ± 234.049	1850.140 ± 271.465	31.500 ± 8.633	86.440 ± 22.763
3080	1823.710 ± 300.097	1960.350 ± 352.035	31.970 ± 14.307	88.933 ± 19.800
3090	1873.260 ± 327.916	1925.970 ± 279.402	38.813 ± 32.666	93.313 ± 34.249
3100	1919.160 ± 371.383	2136.060 ± 511.476	31.223 ± 6.467	88.710 ± 12.012
3110	1878.040 ± 330.903	2102.650 ± 402.555	34.663 ± 13.889	93.597 ± 18.465
3120	1949.110 ± 343.606	2161.020 ± 450.806	35.220 ± 16.027	89.637 ± 11.647
3130	1941.070 ± 313.324	2097.390 ± 498.273	33.570 ± 9.585	90.927 ± 17.247
3140	1936.010 ± 313.047	2207.070 ± 426.786	35.797 ± 12.583	92.323 ± 19.343
3150	1951.530 ± 343.989	2125.610 ± 463.343	36.393 ± 21.625	95.503 ± 24.351
3160	2045.010 ± 424.067	2338.580 ± 560.906	33.187 ± 8.288	91.570 ± 13.588
3170	2033.710 ± 410.615	2169.990 ± 456.631	37.613 ± 14.163	96.457 ± 18.552
3180	2066.310 ± 431.174	2233.440 ± 440.062	36.350 ± 12.782	98.427 ± 24.550
3190	2047.860 ± 337.381	2218.990 ± 517.036	47.250 ± 66.409	95.690 ± 23.588
3200	2022.710 ± 324.343	2335.510 ± 512.783	34.127 ± 8.703	96.783 ± 21.523
3210	2172.140 ± 574.181	2269.080 ± 538.275	36.667 ± 12.305	95.313 ± 16.747
3220	2016.020 ± 350.014	2231.690 ± 456.786	33.120 ± 8.772	91.610 ± 10.654
3230	2017.790 ± 334.622	2172.350 ± 315.520	31.243 ± 6.405	91.787 ± 13.434
3240	1996.090 ± 238.128	2268.030 ± 505.759	31.793 ± 6.547	90.237 ± 7.153
3250	2063.030 ± 276.712	2165.400 ± 400.754	32.053 ± 6.823	93.313 ± 11.512
3260	2099.690 ± 358.774	2216.250 ± 426.709	38.173 ± 17.741	106.470 ± 39.419
3270	2147.360 ± 448.756	2246.250 ± 385.269	31.053 ± 6.628	91.087 ± 7.534
3280	2195.950 ± 484.497	2268.110 ± 528.251	33.670 ± 13.545	98.537 ± 16.070
3290	2197.490 ± 432.384	2324.330 ± 528.860	32.563 ± 7.574	94.520 ± 14.063
3300	2152.620 ± 369.192	2352.580 ± 497.830	32.330 ± 7.612	94.390 ± 9.499
3310	2153.140 ± 320.140	2372.780 ± 500.258	38.430 ± 16.595	97.780 ± 16.099
3320	2132.090 ± 316.496	2289.790 ± 375.186	36.473 ± 15.958	98.757 ± 19.704
3330	2212.870 ± 481.171	2344.340 ± 414.720	31.343 ± 9.131	98.683 ± 17.265
3340	2150.540 ± 297.400	2430.300 ± 519.067	35.000 ± 10.626	100.503 ± 22.637
3350	2141.330 ± 303.629	2375.290 ± 561.347	35.460 ± 15.141	119.553 ± 56.104
3360	2295.730 ± 419.409	2341.330 ± 505.799	31.047 ± 4.512	95.030 ± 13.263
3370	2822.390 ± 1534.300	2821.870 ± 916.647	35.127 ± 8.476	106.943 ± 35.093
3380	2398.080 ± 827.149	2524.190 ± 707.291	34.373 ± 9.359	99.977 ± 21.831
3390	2379.190 ± 641.557	2488.600 ± 563.800	33.353 ± 8.541	98.697 ± 16.038
3400	2226.120 ± 310.978	2443.650 ± 618.687	33.530 ± 8.095	100.247 ± 14.997
3410	2260.940 ± 355.220	2573.330 ± 570.029	32.747 ± 7.220	96.863 ± 8.238
3420	2388.340 ± 464.064	2558.350 ± 587.281	35.910 ± 9.964	99.123 ± 10.955
3430	2571.520 ± 678.025	2681.060 ± 540.616	36.663 ± 9.599	101.613 ± 14.085
3440	2492.330 ± 474.635	2533.820 ± 462.896	36.880 ± 11.750	109.827 ± 42.571
3450	2225.000 ± 231.646	2367.810 ± 373.489	33.413 ± 7.853	99.703 ± 21.689
3460	2315.360 ± 379.433	2354.030 ± 355.810	36.053 ± 15.731	104.417 ± 23.695
3470	2720.550 ± 562.169	2841.870 ± 690.689	39.200 ± 24.644	103.793 ± 18.963
3480	2414.820 ± 423.842	2532.080 ± 607.922	36.053 ± 12.156	100.503 ± 17.799

Size (N)	First (μs)	Last (μs)	Middle (μs)	Random (μs)
3490	2513.800 ± 553.807	2508.890 ± 559.169	32.863 ± 7.252	97.400 ± 7.972
3500	2471.830 ± 530.238	2554.320 ± 580.159	34.493 ± 8.462	106.050 ± 25.979
3510	2392.440 ± 313.985	2405.970 ± 334.146	35.993 ± 14.145	109.530 ± 41.438
3520	2500.930 ± 546.573	2730.220 ± 671.581	33.047 ± 7.233	102.167 ± 14.466
3530	2676.380 ± 636.226	3053.990 ± 899.558	37.790 ± 10.354	108.653 ± 20.622
3540	2494.920 ± 481.530	2645.160 ± 424.372	35.023 ± 10.470	108.230 ± 24.966
3550	2395.330 ± 281.423	2520.990 ± 347.980	35.193 ± 10.236	101.310 ± 16.544
3560	2486.910 ± 358.402	2584.780 ± 458.359	33.297 ± 8.033	102.810 ± 16.486
3570	2498.760 ± 337.656	2596.070 ± 451.969	34.087 ± 7.742	109.463 ± 27.542
3580	2486.340 ± 310.977	2605.110 ± 310.404	35.253 ± 11.650	109.503 ± 23.113
3590	2711.060 ± 436.239	2858.960 ± 542.467	37.693 ± 11.565	105.760 ± 16.777
3600	2711.160 ± 410.434	2683.890 ± 515.295	35.837 ± 12.568	106.673 ± 25.076
3610	2613.040 ± 372.041	2737.440 ± 379.110	36.507 ± 11.535	103.597 ± 10.929
3620	2572.510 ± 408.611	2666.160 ± 418.392	33.433 ± 7.602	104.033 ± 11.621
3630	2617.230 ± 372.142	2693.440 ± 518.979	34.020 ± 11.867	104.823 ± 18.382
3640	2566.960 ± 355.526	2886.580 ± 614.855	31.857 ± 3.136	104.990 ± 13.717
3650	2707.520 ± 425.182	2801.380 ± 497.102	35.657 ± 9.054	107.850 ± 20.010
3660	2674.770 ± 438.375	2820.560 ± 474.737	36.620 ± 11.362	105.420 ± 11.082
3670	2575.380 ± 327.363	2938.210 ± 660.926	35.230 ± 9.493	107.210 ± 14.816
3680	2614.570 ± 391.527	2781.060 ± 404.686	35.843 ± 9.006	106.743 ± 16.982
3690	2653.700 ± 357.194	2726.840 ± 411.592	32.177 ± 4.145	110.857 ± 31.002
3700	2717.130 ± 358.279	2785.960 ± 397.008	35.620 ± 8.471	112.923 ± 32.623
3710	2846.480 ± 484.453	2883.550 ± 632.151	34.143 ± 6.918	106.490 ± 10.081
3720	2712.770 ± 369.945	2852.030 ± 467.397	37.973 ± 15.708	108.683 ± 20.524
3730	2851.600 ± 444.375	2906.700 ± 485.756	35.693 ± 9.863	110.603 ± 20.124
3740	2778.910 ± 394.709	2928.750 ± 476.242	35.740 ± 10.403	107.797 ± 14.174
3750	3139.430 ± 628.162	2990.210 ± 603.146	34.957 ± 8.123	108.257 ± 17.579
3760	3128.900 ± 704.789	3106.050 ± 719.378	37.220 ± 9.217	129.727 ± 72.916
3770	2775.080 ± 440.480	3003.870 ± 544.825	37.960 ± 15.191	108.710 ± 12.624
3780	2789.640 ± 441.690	3031.520 ± 543.545	34.793 ± 7.772	108.740 ± 19.812
3790	3083.400 ± 639.886	2871.870 ± 581.250	34.953 ± 8.657	106.000 ± 11.708
3800	2731.260 ± 321.035	2934.100 ± 478.257	32.987 ± 6.155	114.317 ± 38.497
3810	2770.700 ± 329.703	2985.280 ± 520.563	32.947 ± 5.671	110.403 ± 24.387
3820	2772.270 ± 358.294	3060.550 ± 438.052	34.227 ± 7.224	110.000 ± 16.229
3830	2928.700 ± 535.235	2988.800 ± 465.754	37.963 ± 10.885	112.610 ± 27.255
3840	3326.440 ± 471.583	3445.920 ± 682.915	36.987 ± 12.302	107.667 ± 10.800
3850	3242.020 ± 567.910	3015.530 ± 391.127	38.220 ± 28.937	113.060 ± 26.286
3860	2916.210 ± 435.566	3243.500 ± 605.047	37.757 ± 11.973	116.647 ± 28.464
3870	3393.870 ± 631.987	3179.510 ± 540.269	38.123 ± 11.886	110.770 ± 13.423
3880	3148.670 ± 526.227	3225.600 ± 582.852	38.650 ± 10.728	113.197 ± 13.524
3890	3015.520 ± 463.847	3312.800 ± 567.058	34.703 ± 5.828	109.257 ± 10.027
3900	3161.660 ± 467.660	3230.580 ± 580.346	33.843 ± 5.350	114.077 ± 20.449
3910	2989.320 ± 432.499	3503.800 ± 554.634	34.513 ± 7.692	114.820 ± 15.132
3920	3209.230 ± 682.705	3358.950 ± 557.937	38.767 ± 14.013	118.483 ± 19.499
3930	3242.770 ± 506.186	3140.010 ± 451.184	39.330 ± 11.677	125.020 ± 46.039
3940	2971.460 ± 332.920	3178.290 ± 387.724	34.480 ± 7.059	112.130 ± 16.172
3950	2977.800 ± 408.561	3313.880 ± 526.436	33.400 ± 5.543	112.093 ± 13.726
3960	2878.000 ± 256.203	3302.950 ± 531.928	37.870 ± 27.577	110.380 ± 25.743
3970	2929.060 ± 310.922	3382.970 ± 514.815	34.703 ± 7.698	115.027 ± 16.004
3980	3257.090 ± 570.863	3268.700 ± 456.716	36.330 ± 8.531	118.507 ± 21.861

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
3990	2966.680 $\pm$ 369.718	3477.610 $\pm$ 523.253	40.687 $\pm$ 12.878	116.040 $\pm$ 15.541
4000	3227.080 $\pm$ 437.329	3211.470 $\pm$ 457.035	33.120 $\pm$ 3.030	114.397 $\pm$ 24.366
4010	3099.160 $\pm$ 386.749	3220.470 $\pm$ 444.960	35.980 $\pm$ 7.661	113.107 $\pm$ 16.894
4020	2998.190 $\pm$ 306.326	3274.230 $\pm$ 444.340	34.740 $\pm$ 6.189	117.590 $\pm$ 16.837
4030	3562.600 $\pm$ 685.560	3604.250 $\pm$ 738.227	42.393 $\pm$ 14.966	142.377 $\pm$ 58.687
4040	3193.230 $\pm$ 584.236	3856.800 $\pm$ 450.600	39.513 $\pm$ 10.507	120.767 $\pm$ 18.508
4050	3747.550 $\pm$ 858.742	4026.170 $\pm$ 1294.040	36.160 $\pm$ 6.655	114.970 $\pm$ 14.409
4060	3983.080 $\pm$ 513.641	3087.240 $\pm$ 466.277	39.420 $\pm$ 10.793	122.000 $\pm$ 18.666
4070	3039.160 $\pm$ 525.609	3809.980 $\pm$ 336.945	40.230 $\pm$ 11.814	119.663 $\pm$ 15.287
4080	3436.990 $\pm$ 486.291	3378.130 $\pm$ 623.277	39.553 $\pm$ 14.111	122.707 $\pm$ 17.661
4090	3044.510 $\pm$ 283.167	3737.600 $\pm$ 447.897	33.283 $\pm$ 5.697	116.130 $\pm$ 14.233
4100	3622.470 $\pm$ 442.785	3365.580 $\pm$ 589.869	43.293 $\pm$ 19.091	129.053 $\pm$ 37.246
4110	3562.880 $\pm$ 613.023	3447.830 $\pm$ 527.529	50.820 $\pm$ 32.495	131.670 $\pm$ 33.834
4120	3462.320 $\pm$ 417.490	3279.240 $\pm$ 449.733	47.183 $\pm$ 29.012	134.097 $\pm$ 40.709
4130	3220.500 $\pm$ 319.526	3436.430 $\pm$ 527.224	34.477 $\pm$ 3.159	115.750 $\pm$ 15.723
4140	3519.810 $\pm$ 605.788	3746.420 $\pm$ 557.459	46.257 $\pm$ 36.185	130.937 $\pm$ 34.333
4150	3805.110 $\pm$ 588.272	3253.490 $\pm$ 519.480	44.380 $\pm$ 18.519	127.707 $\pm$ 27.955
4160	3731.440 $\pm$ 365.642	3084.680 $\pm$ 157.356	37.000 $\pm$ 7.650	114.790 $\pm$ 8.505
4170	3465.550 $\pm$ 556.674	3713.190 $\pm$ 525.300	43.997 $\pm$ 13.955	127.877 $\pm$ 24.958
4180	3598.830 $\pm$ 446.175	3385.330 $\pm$ 384.144	37.090 $\pm$ 6.940	116.453 $\pm$ 13.397
4190	3817.420 $\pm$ 450.071	3555.120 $\pm$ 503.770	40.013 $\pm$ 12.858	122.953 $\pm$ 17.513
4200	3698.060 $\pm$ 460.345	3820.130 $\pm$ 482.971	44.027 $\pm$ 20.899	123.117 $\pm$ 29.329
4210	3342.510 $\pm$ 317.106	3372.790 $\pm$ 325.460	34.953 $\pm$ 3.532	113.893 $\pm$ 6.105
4220	3640.250 $\pm$ 638.025	3542.400 $\pm$ 455.364	41.140 $\pm$ 10.480	122.173 $\pm$ 17.534
4230	3641.930 $\pm$ 505.072	3776.750 $\pm$ 440.930	40.730 $\pm$ 14.096	127.460 $\pm$ 36.809
4240	3595.430 $\pm$ 477.669	3622.270 $\pm$ 535.804	39.553 $\pm$ 18.314	117.100 $\pm$ 11.729
4250	3814.040 $\pm$ 1798.740	3965.890 $\pm$ 804.467	42.420 $\pm$ 15.310	125.150 $\pm$ 14.437
4260	4687.380 $\pm$ 1515.510	4641.410 $\pm$ 1540.810	56.697 $\pm$ 36.058	143.810 $\pm$ 43.315
4270	3754.860 $\pm$ 625.237	3865.300 $\pm$ 693.905	41.610 $\pm$ 14.160	128.470 $\pm$ 18.105
4280	3676.820 $\pm$ 450.399	3688.370 $\pm$ 458.502	40.973 $\pm$ 11.144	122.320 $\pm$ 18.454
4290	3674.730 $\pm$ 630.266	3728.970 $\pm$ 596.821	39.777 $\pm$ 9.616	123.400 $\pm$ 17.364
4300	3667.460 $\pm$ 617.521	3947.010 $\pm$ 688.833	44.643 $\pm$ 13.537	127.043 $\pm$ 23.231
4310	4033.350 $\pm$ 941.960	4015.870 $\pm$ 986.246	42.257 $\pm$ 11.072	129.607 $\pm$ 25.899
4320	3597.010 $\pm$ 619.866	3759.520 $\pm$ 727.083	43.753 $\pm$ 12.978	123.893 $\pm$ 16.171
4330	3696.750 $\pm$ 365.102	3684.870 $\pm$ 529.748	42.580 $\pm$ 14.840	129.363 $\pm$ 35.226
4340	4160.160 $\pm$ 1019.520	4015.770 $\pm$ 1064.090	44.660 $\pm$ 15.880	138.377 $\pm$ 49.751
4350	3866.100 $\pm$ 629.705	3694.430 $\pm$ 452.482	39.270 $\pm$ 7.459	125.303 $\pm$ 15.733
4360	3682.730 $\pm$ 421.069	3785.200 $\pm$ 394.967	41.227 $\pm$ 8.953	122.470 $\pm$ 11.191
4370	3694.680 $\pm$ 339.240	3802.760 $\pm$ 545.765	48.200 $\pm$ 29.780	130.540 $\pm$ 31.323
4380	3595.010 $\pm$ 314.143	3746.900 $\pm$ 260.666	45.980 $\pm$ 15.708	136.550 $\pm$ 36.889
4390	4004.740 $\pm$ 485.361	4115.660 $\pm$ 424.960	44.537 $\pm$ 13.121	135.373 $\pm$ 27.108
4400	4015.830 $\pm$ 452.892	4090.120 $\pm$ 518.909	47.293 $\pm$ 18.219	142.260 $\pm$ 51.196
4410	3929.680 $\pm$ 582.709	4246.450 $\pm$ 831.056	46.077 $\pm$ 16.920	132.493 $\pm$ 24.990
4420	3791.170 $\pm$ 486.868	3977.730 $\pm$ 375.673	39.057 $\pm$ 6.327	132.063 $\pm$ 32.224
4430	3924.780 $\pm$ 528.549	3973.340 $\pm$ 416.800	46.013 $\pm$ 21.336	131.570 $\pm$ 18.826
4440	4048.320 $\pm$ 930.274	4052.500 $\pm$ 630.440	46.297 $\pm$ 16.280	128.483 $\pm$ 15.377
4450	4181.100 $\pm$ 752.769	4094.450 $\pm$ 498.272	48.027 $\pm$ 21.693	133.243 $\pm$ 37.059
4460	4400.110 $\pm$ 701.074	4437.820 $\pm$ 633.022	47.823 $\pm$ 14.442	152.533 $\pm$ 77.626
4470	4041.810 $\pm$ 462.596	4163.530 $\pm$ 516.390	44.527 $\pm$ 10.008	136.143 $\pm$ 28.160
4480	4033.720 $\pm$ 349.860	4092.920 $\pm$ 367.624	43.137 $\pm$ 12.516	132.770 $\pm$ 15.675

Size (N)	First (μs)	Last (μs)	Middle (μs)	Random (μs)
4490	3986.880 ± 538.817	4112.740 ± 696.430	42.520 ± 9.311	130.177 ± 17.610
4500	4362.420 ± 740.249	4369.500 ± 586.133	46.937 ± 20.855	141.500 ± 49.267
4510	4174.830 ± 498.100	4346.770 ± 598.831	46.130 ± 20.393	140.403 ± 29.895
4520	3996.780 ± 380.963	4066.080 ± 428.536	44.030 ± 16.050	135.577 ± 32.123
4530	4150.070 ± 548.219	4078.630 ± 446.863	46.970 ± 27.780	139.220 ± 46.234
4540	4094.080 ± 521.180	4238.600 ± 462.269	43.097 ± 15.492	150.630 ± 64.677
4550	4344.790 ± 605.214	4444.940 ± 590.698	47.307 ± 17.549	147.073 ± 43.724
4560	4441.480 ± 634.577	4594.350 ± 666.766	47.750 ± 23.334	133.940 ± 20.585
4570	4473.260 ± 973.054	4322.140 ± 577.510	44.723 ± 9.476	132.633 ± 19.241
4580	4320.240 ± 490.388	4303.190 ± 511.403	53.760 ± 20.525	145.767 ± 37.431
4590	4119.660 ± 401.259	4277.160 ± 549.418	48.497 ± 23.394	136.037 ± 30.934
4600	4252.890 ± 588.942	4390.860 ± 611.818	45.863 ± 11.558	156.627 ± 86.666
4610	4371.360 ± 628.855	4386.720 ± 520.547	47.307 ± 11.894	135.513 ± 23.683
4620	4173.270 ± 419.256	4348.110 ± 466.018	47.067 ± 13.506	141.633 ± 33.735
4630	4135.230 ± 409.659	4344.990 ± 536.542	43.893 ± 10.222	132.500 ± 13.513
4640	4012.930 ± 321.612	4143.330 ± 370.259	42.557 ± 8.178	133.803 ± 21.789
4650	4280.510 ± 475.232	4484.340 ± 467.772	44.753 ± 11.153	136.130 ± 38.499
4660	4453.400 ± 532.138	4555.180 ± 579.705	48.663 ± 17.207	140.397 ± 20.220
4670	4636.240 ± 522.677	4586.070 ± 474.157	54.733 ± 36.029	150.150 ± 53.980
4680	4514.120 ± 540.590	4529.490 ± 526.671	52.070 ± 27.185	151.270 ± 39.905
4690	4268.930 ± 369.420	4454.690 ± 718.563	47.080 ± 19.007	144.360 ± 43.007
4700	5142.820 ± 940.844	5206.540 ± 1051.900	51.137 ± 12.603	144.927 ± 22.398
4710	4436.160 ± 929.168	4474.480 ± 518.355	42.517 ± 8.780	130.440 ± 11.585
4720	4231.040 ± 450.527	4318.570 ± 481.262	47.120 ± 16.564	138.040 ± 32.144
4730	4348.380 ± 675.532	4303.330 ± 387.287	46.590 ± 13.282	140.160 ± 25.108
4740	4407.910 ± 550.015	4693.510 ± 770.498	45.483 ± 10.855	134.840 ± 11.769
4750	4449.830 ± 503.037	4703.430 ± 594.414	48.350 ± 12.711	137.970 ± 20.468
4760	4916.610 ± 1284.050	5433.590 ± 1858.420	53.007 ± 25.744	164.510 ± 85.844
4770	4509.300 ± 501.084	4760.840 ± 737.453	53.743 ± 33.978	154.870 ± 75.803
4780	4428.150 ± 457.177	4658.070 ± 478.030	45.963 ± 10.139	138.523 ± 14.706
4790	4641.500 ± 769.504	4740.450 ± 565.466	47.900 ± 12.415	148.073 ± 45.419
4800	4482.370 ± 420.641	4652.470 ± 514.199	45.017 ± 9.604	143.903 ± 26.248
4810	4649.790 ± 618.054	4834.900 ± 754.034	48.480 ± 13.562	136.443 ± 16.486
4820	4554.770 ± 435.740	4738.940 ± 775.710	45.190 ± 9.539	135.310 ± 11.864
4830	4490.060 ± 378.425	4536.530 ± 399.515	44.420 ± 8.673	136.367 ± 10.768
4840	4762.340 ± 603.627	4875.160 ± 778.666	44.047 ± 9.418	142.773 ± 20.652
4850	4876.030 ± 458.574	4957.080 ± 563.678	48.927 ± 13.571	149.913 ± 23.076
4860	5105.860 ± 656.961	5256.910 ± 841.850	54.760 ± 26.576	157.717 ± 49.125
4870	4874.510 ± 1044.970	4764.320 ± 932.506	49.050 ± 16.328	146.560 ± 27.970
4880	4735.000 ± 848.301	4995.410 ± 651.624	48.910 ± 14.136	140.150 ± 19.148
4890	4916.870 ± 937.086	5725.000 ± 1938.450	52.830 ± 17.538	142.570 ± 22.924
4900	4718.850 ± 774.732	4632.960 ± 495.504	47.503 ± 10.238	140.040 ± 12.557
4910	4851.700 ± 664.144	4979.730 ± 499.535	51.420 ± 23.565	155.693 ± 54.651
4920	5368.850 ± 865.600	5557.230 ± 890.452	53.573 ± 14.384	159.157 ± 40.648
4930	5184.160 ± 717.349	5191.620 ± 632.355	56.010 ± 29.206	155.110 ± 28.618
4940	5050.350 ± 562.683	5013.340 ± 460.524	48.100 ± 11.175	146.267 ± 30.125
4950	5303.900 ± 1133.580	5785.220 ± 1712.120	57.070 ± 22.868	162.647 ± 33.867
4960	4978.310 ± 464.934	5054.180 ± 429.703	50.540 ± 12.945	160.500 ± 49.282
4970	5186.400 ± 694.623	5181.220 ± 804.594	52.620 ± 28.383	163.033 ± 62.105
4980	4728.360 ± 437.226	4890.970 ± 435.646	48.083 ± 10.335	143.480 ± 12.674

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
4990	5046.590 $\pm$ 505.688	4998.910 $\pm$ 438.909	45.877 $\pm$ 6.323	140.460 $\pm$ 11.284
5000	4802.530 $\pm$ 397.402	5150.450 $\pm$ 574.195	51.707 $\pm$ 40.510	143.730 $\pm$ 15.512

B.3 Pre-Sorted Descending Array Distribution (500 Data Points)

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
10	0.230 $\pm$ 0.084	0.227 $\pm$ 0.058	0.213 $\pm$ 0.078	0.410 $\pm$ 0.169
20	0.273 $\pm$ 0.105	0.257 $\pm$ 0.086	0.207 $\pm$ 0.083	0.630 $\pm$ 0.070
30	0.703 $\pm$ 0.201	0.760 $\pm$ 0.206	0.437 $\pm$ 0.122	1.063 $\pm$ 0.130
40	1.490 $\pm$ 4.275	0.713 $\pm$ 0.210	0.437 $\pm$ 0.145	1.123 $\pm$ 0.128
50	0.893 $\pm$ 0.098	0.950 $\pm$ 0.111	0.473 $\pm$ 0.114	1.420 $\pm$ 0.130
60	1.373 $\pm$ 0.069	1.477 $\pm$ 0.097	0.547 $\pm$ 0.114	1.567 $\pm$ 0.103
70	1.617 $\pm$ 0.166	1.643 $\pm$ 0.216	0.653 $\pm$ 0.114	1.900 $\pm$ 0.155
80	2.477 $\pm$ 0.844	2.540 $\pm$ 0.866	0.877 $\pm$ 0.265	2.340 $\pm$ 0.298
90	2.830 $\pm$ 0.989	2.677 $\pm$ 0.458	0.903 $\pm$ 0.241	3.100 $\pm$ 3.596
100	5.550 $\pm$ 0.449	12.720 $\pm$ 27.156	1.843 $\pm$ 0.415	4.660 $\pm$ 5.610
110	4.043 $\pm$ 1.119	9.167 $\pm$ 23.927	1.180 $\pm$ 0.343	3.213 $\pm$ 0.434
120	17.740 $\pm$ 51.230	4.383 $\pm$ 0.568	1.100 $\pm$ 0.136	3.930 $\pm$ 3.362
130	7.213 $\pm$ 12.780	5.140 $\pm$ 0.640	1.210 $\pm$ 0.121	3.433 $\pm$ 0.206
140	5.133 $\pm$ 0.281	5.263 $\pm$ 0.293	1.310 $\pm$ 0.142	3.760 $\pm$ 0.266
150	6.767 $\pm$ 3.512	6.827 $\pm$ 3.098	1.363 $\pm$ 0.171	3.897 $\pm$ 0.292
160	6.877 $\pm$ 0.848	7.397 $\pm$ 2.241	1.430 $\pm$ 0.124	4.180 $\pm$ 0.251
170	8.503 $\pm$ 3.815	7.997 $\pm$ 2.390	1.623 $\pm$ 0.322	4.387 $\pm$ 0.332
180	8.277 $\pm$ 2.048	11.823 $\pm$ 18.977	2.140 $\pm$ 2.659	4.713 $\pm$ 0.300
190	14.887 $\pm$ 11.426	25.883 $\pm$ 42.704	2.730 $\pm$ 1.623	9.193 $\pm$ 12.287
200	11.360 $\pm$ 3.063	11.740 $\pm$ 3.177	2.013 $\pm$ 0.551	6.360 $\pm$ 4.363
210	11.237 $\pm$ 3.730	10.753 $\pm$ 1.885	1.947 $\pm$ 0.281	5.587 $\pm$ 0.512
220	11.973 $\pm$ 1.645	12.733 $\pm$ 3.198	1.973 $\pm$ 0.114	5.710 $\pm$ 0.281
230	12.020 $\pm$ 0.815	12.963 $\pm$ 3.229	2.057 $\pm$ 0.155	5.910 $\pm$ 0.295
240	13.197 $\pm$ 2.425	14.657 $\pm$ 4.370	2.197 $\pm$ 0.380	6.250 $\pm$ 0.593
250	20.250 $\pm$ 9.506	25.383 $\pm$ 19.540	5.963 $\pm$ 14.888	8.987 $\pm$ 5.150
260	33.893 $\pm$ 48.727	25.413 $\pm$ 7.586	3.773 $\pm$ 0.964	10.040 $\pm$ 6.336
270	21.990 $\pm$ 7.991	23.020 $\pm$ 10.881	3.173 $\pm$ 1.214	8.823 $\pm$ 3.535
280	20.077 $\pm$ 5.444	19.880 $\pm$ 5.125	2.647 $\pm$ 0.508	7.517 $\pm$ 0.665
290	32.047 $\pm$ 25.386	31.767 $\pm$ 18.557	4.067 $\pm$ 1.933	9.900 $\pm$ 3.825
300	22.090 $\pm$ 5.186	23.853 $\pm$ 6.570	2.870 $\pm$ 0.485	8.953 $\pm$ 4.479
310	22.720 $\pm$ 6.512	22.770 $\pm$ 5.219	2.853 $\pm$ 0.369	8.093 $\pm$ 0.553
320	33.840 $\pm$ 15.224	33.393 $\pm$ 14.829	4.270 $\pm$ 1.586	11.377 $\pm$ 5.901
330	43.983 $\pm$ 3.752	44.613 $\pm$ 4.206	5.667 $\pm$ 3.430	10.940 $\pm$ 0.524
340	48.693 $\pm$ 17.811	47.623 $\pm$ 18.231	5.377 $\pm$ 1.806	12.773 $\pm$ 8.189
350	30.437 $\pm$ 9.167	29.067 $\pm$ 6.751	4.017 $\pm$ 2.845	9.927 $\pm$ 1.911
360	29.090 $\pm$ 5.553	28.563 $\pm$ 5.049	3.560 $\pm$ 0.733	9.783 $\pm$ 1.295
370	43.467 $\pm$ 26.460	41.930 $\pm$ 23.339	4.650 $\pm$ 2.366	11.507 $\pm$ 3.614
380	31.320 $\pm$ 5.975	31.950 $\pm$ 4.993	3.570 $\pm$ 0.156	10.267 $\pm$ 2.378
390	40.623 $\pm$ 18.576	42.077 $\pm$ 24.634	5.640 $\pm$ 4.032	13.717 $\pm$ 6.231
400	38.720 $\pm$ 9.710	44.947 $\pm$ 23.003	4.390 $\pm$ 2.083	12.467 $\pm$ 6.303
410	34.810 $\pm$ 4.163	36.560 $\pm$ 6.120	4.183 $\pm$ 2.633	10.663 $\pm$ 0.885
420	49.363 $\pm$ 19.814	51.070 $\pm$ 28.983	5.223 $\pm$ 2.811	12.750 $\pm$ 2.944
430	39.027 $\pm$ 6.015	40.767 $\pm$ 7.689	3.983 $\pm$ 0.158	11.140 $\pm$ 0.401
440	47.743 $\pm$ 17.401	51.360 $\pm$ 16.638	4.990 $\pm$ 2.930	13.113 $\pm$ 3.709
450	45.120 $\pm$ 7.963	45.383 $\pm$ 7.937	4.083 $\pm$ 0.151	11.580 $\pm$ 0.407
460	59.263 $\pm$ 22.789	59.110 $\pm$ 18.785	5.117 $\pm$ 1.623	13.637 $\pm$ 3.976
470	43.710 $\pm$ 4.078	44.910 $\pm$ 5.059	4.257 $\pm$ 0.189	12.550 $\pm$ 2.333
480	94.207 $\pm$ 23.956	89.343 $\pm$ 27.546	8.663 $\pm$ 2.570	18.057 $\pm$ 4.254

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
490	69.897 $\pm$ 38.959	63.363 $\pm$ 20.635	6.087 $\pm$ 4.619	13.837 $\pm$ 1.604
500	59.490 $\pm$ 16.492	66.657 $\pm$ 31.808	5.730 $\pm$ 3.082	15.203 $\pm$ 5.155
510	71.727 $\pm$ 21.089	62.140 $\pm$ 10.800	4.930 $\pm$ 0.439	13.987 $\pm$ 0.597
520	74.107 $\pm$ 31.919	69.103 $\pm$ 29.873	6.170 $\pm$ 3.381	15.307 $\pm$ 4.584
530	72.640 $\pm$ 44.443	65.657 $\pm$ 27.055	5.523 $\pm$ 1.694	15.307 $\pm$ 3.883
540	69.610 $\pm$ 16.111	67.010 $\pm$ 12.931	5.687 $\pm$ 1.596	15.700 $\pm$ 4.221
550	75.737 $\pm$ 23.931	68.060 $\pm$ 22.155	6.703 $\pm$ 7.387	15.597 $\pm$ 2.697
560	81.100 $\pm$ 25.337	80.263 $\pm$ 23.944	6.270 $\pm$ 2.839	16.040 $\pm$ 3.800
570	83.023 $\pm$ 43.006	76.493 $\pm$ 25.706	6.010 $\pm$ 1.945	15.557 $\pm$ 1.862
580	77.600 $\pm$ 16.934	83.750 $\pm$ 32.514	5.727 $\pm$ 1.099	15.707 $\pm$ 1.147
590	81.220 $\pm$ 13.460	79.863 $\pm$ 31.378	6.073 $\pm$ 1.416	16.137 $\pm$ 0.869
600	87.610 $\pm$ 22.447	89.870 $\pm$ 28.239	6.163 $\pm$ 1.484	16.677 $\pm$ 1.719
610	93.307 $\pm$ 40.626	84.653 $\pm$ 21.622	6.297 $\pm$ 1.906	17.467 $\pm$ 3.442
620	100.443 $\pm$ 38.792	97.210 $\pm$ 45.408	7.893 $\pm$ 6.135	19.420 $\pm$ 7.138
630	105.270 $\pm$ 53.570	97.700 $\pm$ 32.554	6.750 $\pm$ 1.925	18.280 $\pm$ 4.774
640	122.740 $\pm$ 53.537	104.040 $\pm$ 35.072	8.893 $\pm$ 9.095	20.210 $\pm$ 5.416
650	102.900 $\pm$ 35.590	100.393 $\pm$ 35.018	7.450 $\pm$ 3.059	19.613 $\pm$ 5.057
660	106.260 $\pm$ 38.492	107.587 $\pm$ 41.736	8.187 $\pm$ 6.178	19.907 $\pm$ 5.985
670	112.060 $\pm$ 27.970	112.033 $\pm$ 40.679	7.463 $\pm$ 3.179	19.407 $\pm$ 3.919
680	124.797 $\pm$ 53.695	118.777 $\pm$ 55.018	8.017 $\pm$ 3.616	21.383 $\pm$ 5.697
690	116.190 $\pm$ 35.794	115.587 $\pm$ 41.581	7.727 $\pm$ 2.641	20.977 $\pm$ 5.463
700	119.987 $\pm$ 61.440	121.423 $\pm$ 46.492	7.823 $\pm$ 2.294	21.493 $\pm$ 6.623
710	120.013 $\pm$ 42.064	118.330 $\pm$ 62.251	7.623 $\pm$ 1.974	19.657 $\pm$ 2.401
720	127.840 $\pm$ 46.276	115.737 $\pm$ 31.148	7.547 $\pm$ 1.586	23.017 $\pm$ 17.049
730	130.193 $\pm$ 56.256	128.783 $\pm$ 60.200	8.750 $\pm$ 4.161	21.563 $\pm$ 4.946
740	146.410 $\pm$ 81.427	137.107 $\pm$ 51.979	8.777 $\pm$ 3.419	22.613 $\pm$ 6.123
750	132.050 $\pm$ 35.517	139.243 $\pm$ 64.734	9.553 $\pm$ 8.938	22.210 $\pm$ 7.884
760	138.713 $\pm$ 36.701	146.413 $\pm$ 98.931	10.693 $\pm$ 8.816	22.703 $\pm$ 5.368
770	152.240 $\pm$ 57.779	138.763 $\pm$ 62.630	9.260 $\pm$ 3.549	22.843 $\pm$ 4.005
780	135.857 $\pm$ 28.585	139.833 $\pm$ 38.419	8.390 $\pm$ 2.215	22.950 $\pm$ 5.072
790	160.020 $\pm$ 66.145	158.800 $\pm$ 57.759	8.690 $\pm$ 1.959	22.710 $\pm$ 2.176
800	154.627 $\pm$ 47.859	150.113 $\pm$ 49.683	12.950 $\pm$ 18.315	23.850 $\pm$ 7.416
810	175.653 $\pm$ 61.224	169.880 $\pm$ 58.350	10.970 $\pm$ 5.503	25.810 $\pm$ 6.749
820	216.800 $\pm$ 111.145	203.593 $\pm$ 68.933	11.270 $\pm$ 3.047	25.663 $\pm$ 5.176
830	197.357 $\pm$ 109.170	182.783 $\pm$ 64.968	11.247 $\pm$ 5.260	27.480 $\pm$ 8.681
840	175.037 $\pm$ 49.577	172.287 $\pm$ 54.784	9.293 $\pm$ 2.320	23.817 $\pm$ 2.159
850	186.673 $\pm$ 74.184	174.447 $\pm$ 76.638	9.423 $\pm$ 2.780	24.280 $\pm$ 4.396
860	178.437 $\pm$ 44.495	174.320 $\pm$ 82.042	9.307 $\pm$ 2.194	24.547 $\pm$ 2.073
870	192.943 $\pm$ 56.760	178.663 $\pm$ 89.374	9.430 $\pm$ 2.732	24.823 $\pm$ 3.228
880	194.100 $\pm$ 65.507	178.980 $\pm$ 48.559	13.020 $\pm$ 16.068	26.183 $\pm$ 6.231
890	208.757 $\pm$ 88.824	203.843 $\pm$ 77.155	10.973 $\pm$ 3.450	26.420 $\pm$ 3.391
900	230.617 $\pm$ 72.856	229.033 $\pm$ 77.095	11.467 $\pm$ 3.468	26.780 $\pm$ 3.379
910	220.960 $\pm$ 94.470	208.500 $\pm$ 74.262	12.463 $\pm$ 13.016	26.343 $\pm$ 3.588
920	218.410 $\pm$ 67.355	204.217 $\pm$ 59.436	11.093 $\pm$ 3.192	27.793 $\pm$ 4.001
930	212.740 $\pm$ 62.283	210.260 $\pm$ 70.376	11.367 $\pm$ 3.421	27.720 $\pm$ 3.452
940	229.467 $\pm$ 75.279	231.880 $\pm$ 106.751	13.337 $\pm$ 7.628	30.137 $\pm$ 10.029
950	217.020 $\pm$ 65.324	245.437 $\pm$ 195.384	11.273 $\pm$ 3.454	29.743 $\pm$ 8.284
960	213.897 $\pm$ 60.656	216.430 $\pm$ 66.207	10.287 $\pm$ 2.289	28.953 $\pm$ 6.906
970	246.530 $\pm$ 113.138	251.470 $\pm$ 120.695	10.923 $\pm$ 3.072	29.227 $\pm$ 5.559
980	242.307 $\pm$ 116.344	218.133 $\pm$ 63.198	11.350 $\pm$ 3.123	28.350 $\pm$ 2.935

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
990	259.677 $\pm$ 87.178	258.213 $\pm$ 117.205	12.587 $\pm$ 4.499	31.593 $\pm$ 7.460
1000	274.673 $\pm$ 122.015	250.807 $\pm$ 82.421	11.970 $\pm$ 3.702	37.407 $\pm$ 23.685
1010	237.487 $\pm$ 77.763	238.163 $\pm$ 128.757	10.917 $\pm$ 2.128	29.690 $\pm$ 4.721
1020	244.867 $\pm$ 107.571	235.377 $\pm$ 62.032	11.867 $\pm$ 3.106	29.080 $\pm$ 3.426
1030	261.560 $\pm$ 80.749	260.517 $\pm$ 121.709	12.487 $\pm$ 4.866	29.930 $\pm$ 3.864
1040	251.683 $\pm$ 83.371	238.830 $\pm$ 63.745	11.557 $\pm$ 2.940	31.327 $\pm$ 5.946
1050	281.497 $\pm$ 116.174	271.987 $\pm$ 115.811	12.893 $\pm$ 5.674	37.557 $\pm$ 28.120
1060	259.323 $\pm$ 74.316	253.483 $\pm$ 63.399	13.163 $\pm$ 3.913	31.893 $\pm$ 5.423
1070	296.903 $\pm$ 92.136	289.763 $\pm$ 134.294	12.617 $\pm$ 3.197	37.430 $\pm$ 23.034
1080	285.443 $\pm$ 91.474	291.113 $\pm$ 106.926	12.037 $\pm$ 2.844	34.650 $\pm$ 13.491
1090	354.160 $\pm$ 134.252	304.947 $\pm$ 110.104	17.270 $\pm$ 16.997	38.363 $\pm$ 13.265
1100	277.947 $\pm$ 91.053	274.863 $\pm$ 78.463	12.013 $\pm$ 2.521	32.360 $\pm$ 7.799
1110	304.020 $\pm$ 105.830	266.960 $\pm$ 77.608	12.323 $\pm$ 2.390	31.553 $\pm$ 3.356
1120	278.863 $\pm$ 90.896	275.337 $\pm$ 66.695	14.597 $\pm$ 12.629	31.270 $\pm$ 2.267
1130	280.983 $\pm$ 64.199	286.390 $\pm$ 118.351	13.510 $\pm$ 5.207	32.123 $\pm$ 3.207
1140	326.180 $\pm$ 100.133	305.523 $\pm$ 114.415	14.217 $\pm$ 4.255	35.627 $\pm$ 7.077
1150	340.253 $\pm$ 107.929	334.020 $\pm$ 141.802	16.560 $\pm$ 8.584	34.633 $\pm$ 5.021
1160	302.827 $\pm$ 109.060	275.037 $\pm$ 68.739	12.513 $\pm$ 2.692	33.190 $\pm$ 5.170
1170	379.723 $\pm$ 121.809	382.277 $\pm$ 169.746	16.770 $\pm$ 6.658	37.940 $\pm$ 11.040
1180	380.250 $\pm$ 111.777	373.240 $\pm$ 129.505	15.913 $\pm$ 5.840	36.323 $\pm$ 5.439
1190	310.927 $\pm$ 70.665	300.643 $\pm$ 75.992	13.243 $\pm$ 2.851	35.553 $\pm$ 11.457
1200	331.960 $\pm$ 100.366	324.837 $\pm$ 131.080	15.940 $\pm$ 10.170	35.757 $\pm$ 5.974
1210	328.917 $\pm$ 120.606	319.113 $\pm$ 88.512	13.137 $\pm$ 2.953	35.163 $\pm$ 6.081
1220	343.073 $\pm$ 73.934	322.053 $\pm$ 59.993	13.683 $\pm$ 4.311	35.533 $\pm$ 6.663
1230	332.650 $\pm$ 73.825	329.553 $\pm$ 93.357	13.967 $\pm$ 3.519	35.957 $\pm$ 5.094
1240	359.533 $\pm$ 130.913	360.387 $\pm$ 147.158	15.177 $\pm$ 6.803	38.750 $\pm$ 8.498
1250	408.403 $\pm$ 133.985	403.347 $\pm$ 124.400	16.347 $\pm$ 5.052	40.347 $\pm$ 9.185
1260	351.270 $\pm$ 74.173	344.740 $\pm$ 74.941	15.143 $\pm$ 5.054	36.253 $\pm$ 4.621
1270	349.837 $\pm$ 69.576	337.830 $\pm$ 79.740	13.540 $\pm$ 2.900	34.827 $\pm$ 2.750
1280	368.847 $\pm$ 118.486	334.827 $\pm$ 75.666	13.960 $\pm$ 3.452	36.460 $\pm$ 7.407
1290	357.307 $\pm$ 92.064	364.067 $\pm$ 140.049	15.053 $\pm$ 7.296	37.333 $\pm$ 7.622
1300	386.037 $\pm$ 113.074	356.057 $\pm$ 85.458	13.773 $\pm$ 2.124	35.667 $\pm$ 2.338
1310	377.293 $\pm$ 103.927	357.210 $\pm$ 107.573	16.050 $\pm$ 6.544	37.797 $\pm$ 9.319
1320	379.050 $\pm$ 107.576	333.507 $\pm$ 53.896	14.133 $\pm$ 2.862	37.277 $\pm$ 4.207
1330	440.000 $\pm$ 161.224	378.460 $\pm$ 98.960	16.347 $\pm$ 5.804	39.613 $\pm$ 8.815
1340	373.890 $\pm$ 96.292	361.650 $\pm$ 83.306	14.297 $\pm$ 3.259	36.910 $\pm$ 4.589
1350	415.593 $\pm$ 98.379	448.257 $\pm$ 228.163	17.370 $\pm$ 7.639	41.647 $\pm$ 11.279
1360	396.340 $\pm$ 90.509	386.890 $\pm$ 101.552	14.750 $\pm$ 3.436	38.093 $\pm$ 5.439
1370	413.713 $\pm$ 121.549	365.937 $\pm$ 56.802	14.583 $\pm$ 2.650	37.517 $\pm$ 2.617
1380	403.103 $\pm$ 87.377	390.650 $\pm$ 79.852	15.620 $\pm$ 4.325	42.600 $\pm$ 13.338
1390	450.040 $\pm$ 131.341	418.987 $\pm$ 96.191	15.410 $\pm$ 3.788	41.203 $\pm$ 9.369
1400	456.817 $\pm$ 140.511	469.293 $\pm$ 193.461	18.327 $\pm$ 7.656	41.673 $\pm$ 6.161
1410	437.247 $\pm$ 107.944	402.777 $\pm$ 91.956	14.753 $\pm$ 2.653	38.460 $\pm$ 3.393
1420	426.747 $\pm$ 124.221	432.283 $\pm$ 144.974	22.133 $\pm$ 34.154	41.567 $\pm$ 14.978
1430	515.020 $\pm$ 221.191	489.373 $\pm$ 155.362	18.763 $\pm$ 5.742	43.277 $\pm$ 8.888
1440	590.107 $\pm$ 214.307	539.057 $\pm$ 205.693	19.997 $\pm$ 6.425	47.990 $\pm$ 12.303
1450	501.153 $\pm$ 154.914	480.897 $\pm$ 132.523	17.123 $\pm$ 4.339	42.963 $\pm$ 6.187
1460	484.630 $\pm$ 109.769	445.697 $\pm$ 106.573	16.073 $\pm$ 4.324	40.990 $\pm$ 3.571
1470	512.090 $\pm$ 187.569	453.657 $\pm$ 114.791	17.047 $\pm$ 4.138	46.563 $\pm$ 15.986
1480	497.727 $\pm$ 175.466	474.487 $\pm$ 140.212	20.317 $\pm$ 12.458	44.380 $\pm$ 9.304

Size (N)	First (μs)	Last (μs)	Middle (μs)	Random (μs)
1490	486.733 ± 126.003	476.463 ± 162.708	16.483 ± 3.327	43.797 ± 8.649
1500	532.643 ± 205.523	507.547 ± 203.631	17.050 ± 3.504	45.620 ± 11.723
1510	538.990 ± 178.732	534.450 ± 208.966	18.617 ± 7.033	50.637 ± 31.904
1520	541.707 ± 227.696	496.267 ± 148.562	16.820 ± 2.816	43.087 ± 3.840
1530	549.387 ± 209.777	487.480 ± 141.140	17.223 ± 3.938	44.933 ± 7.442
1540	520.270 ± 146.402	470.760 ± 80.112	16.957 ± 4.577	42.927 ± 4.897
1550	480.157 ± 93.126	487.743 ± 127.885	16.493 ± 3.432	44.877 ± 8.793
1560	513.003 ± 158.574	478.880 ± 89.930	18.510 ± 9.241	48.787 ± 19.187
1570	598.790 ± 224.092	512.687 ± 136.128	18.460 ± 5.485	47.117 ± 10.061
1580	556.687 ± 162.666	485.650 ± 106.459	16.723 ± 2.588	46.490 ± 8.860
1590	544.610 ± 104.355	550.810 ± 198.094	18.953 ± 6.541	45.763 ± 5.856
1600	574.787 ± 187.888	553.917 ± 172.083	19.877 ± 8.223	46.447 ± 6.158
1610	573.590 ± 154.185	536.610 ± 104.806	20.113 ± 10.164	48.403 ± 13.317
1620	521.020 ± 108.226	501.497 ± 72.132	16.380 ± 0.720	42.880 ± 1.231
1630	589.970 ± 135.856	535.883 ± 175.169	17.643 ± 3.569	46.720 ± 8.540
1640	600.483 ± 208.572	577.790 ± 214.593	20.683 ± 14.191	51.107 ± 14.354
1650	573.273 ± 128.646	591.250 ± 192.746	19.540 ± 6.511	53.317 ± 39.483
1660	659.030 ± 293.591	581.173 ± 172.692	18.340 ± 3.632	51.283 ± 11.274
1670	651.247 ± 188.003	615.223 ± 151.194	20.517 ± 6.678	49.400 ± 8.965
1680	612.180 ± 177.542	589.953 ± 229.592	20.467 ± 9.432	50.423 ± 12.419
1690	730.170 ± 283.720	710.900 ± 298.642	22.633 ± 9.109	59.557 ± 36.419
1700	1067.470 ± 560.194	917.237 ± 277.675	28.120 ± 6.452	61.727 ± 14.490
1710	701.630 ± 170.372	679.623 ± 228.088	22.353 ± 9.827	56.163 ± 21.802
1720	745.577 ± 278.672	712.283 ± 234.456	23.093 ± 7.480	59.217 ± 21.864
1730	669.920 ± 163.410	612.713 ± 219.611	20.780 ± 6.704	54.910 ± 17.277
1740	629.257 ± 179.104	627.260 ± 197.389	21.550 ± 9.555	54.033 ± 16.633
1750	671.397 ± 195.425	668.450 ± 243.623	22.737 ± 16.343	51.357 ± 13.771
1760	741.550 ± 234.859	744.563 ± 252.808	23.220 ± 9.426	56.240 ± 14.574
1770	665.203 ± 164.973	701.047 ± 265.803	21.530 ± 6.250	51.593 ± 6.901
1780	704.217 ± 187.555	694.940 ± 228.315	22.873 ± 8.539	53.023 ± 7.501
1790	757.107 ± 277.369	705.760 ± 189.754	24.657 ± 12.706	55.710 ± 17.060
1800	781.470 ± 291.319	710.687 ± 215.320	25.977 ± 21.700	55.190 ± 13.552
1810	731.480 ± 175.249	700.547 ± 212.964	21.163 ± 7.221	54.067 ± 10.459
1820	767.503 ± 267.704	699.230 ± 235.424	22.040 ± 6.444	55.517 ± 13.464
1830	765.690 ± 236.989	791.860 ± 258.269	25.120 ± 13.006	58.100 ± 13.196
1840	778.683 ± 247.063	718.210 ± 190.106	22.227 ± 5.426	55.343 ± 13.665
1850	738.680 ± 184.362	703.740 ± 194.799	23.660 ± 12.094	55.093 ± 14.975
1860	722.980 ± 182.656	750.587 ± 256.947	21.947 ± 5.985	56.713 ± 15.413
1870	821.240 ± 273.234	726.667 ± 179.315	21.837 ± 7.591	59.103 ± 35.128
1880	792.180 ± 271.167	709.387 ± 146.943	22.130 ± 10.479	51.357 ± 2.443
1890	767.437 ± 218.272	739.093 ± 168.288	23.883 ± 10.114	59.200 ± 16.973
1900	789.047 ± 257.761	746.383 ± 238.052	24.480 ± 10.953	55.957 ± 14.861
1910	779.227 ± 153.925	784.757 ± 220.592	23.440 ± 9.153	57.813 ± 13.876
1920	849.527 ± 293.961	813.757 ± 194.581	24.843 ± 12.375	59.027 ± 14.852
1930	836.430 ± 235.914	748.307 ± 126.086	24.653 ± 17.430	59.227 ± 15.794
1940	851.967 ± 196.727	791.097 ± 198.128	23.533 ± 9.794	59.960 ± 18.294
1950	918.667 ± 351.606	881.347 ± 318.830	23.597 ± 5.723	61.840 ± 12.078
1960	801.087 ± 256.779	761.807 ± 222.679	21.310 ± 4.809	56.157 ± 8.028
1970	814.543 ± 193.350	835.853 ± 203.874	23.177 ± 8.975	60.200 ± 14.707
1980	792.000 ± 161.842	765.553 ± 192.898	21.153 ± 4.411	55.897 ± 4.837

Size (N)	First (μs)	Last (μs)	Middle (μs)	Random (μs)
1990	795.567 ± 190.343	748.190 ± 94.530	21.797 ± 6.194	59.140 ± 17.583
2000	805.717 ± 137.601	813.080 ± 145.481	23.713 ± 9.996	61.140 ± 16.999
2010	798.920 ± 160.911	810.523 ± 148.328	23.647 ± 6.817	58.753 ± 7.281
2020	860.593 ± 205.715	800.447 ± 161.557	21.767 ± 3.493	58.540 ± 9.729
2030	958.893 ± 265.801	868.337 ± 261.264	25.050 ± 9.366	74.667 ± 74.421
2040	845.490 ± 178.210	852.100 ± 180.763	24.430 ± 6.786	59.617 ± 10.305
2050	909.893 ± 233.404	839.397 ± 154.503	22.620 ± 3.484	60.283 ± 11.908
2060	861.830 ± 158.743	828.370 ± 162.594	22.223 ± 4.483	56.943 ± 4.420
2070	907.403 ± 176.963	853.723 ± 195.594	22.587 ± 4.247	59.367 ± 8.919
2080	960.150 ± 228.598	875.207 ± 187.152	24.857 ± 7.591	61.387 ± 11.450
2090	917.173 ± 218.515	843.017 ± 130.758	23.770 ± 5.411	62.273 ± 17.156
2100	951.990 ± 233.288	928.013 ± 241.465	26.410 ± 9.986	63.660 ± 16.441
2110	964.757 ± 187.800	921.497 ± 271.559	23.897 ± 4.377	61.987 ± 7.682
2120	957.633 ± 241.479	887.397 ± 206.874	24.290 ± 5.376	62.780 ± 12.982
2130	937.140 ± 156.783	897.117 ± 154.877	23.003 ± 4.754	61.127 ± 8.105
2140	943.807 ± 214.669	856.877 ± 92.114	22.327 ± 2.846	58.443 ± 4.085
2150	941.293 ± 210.404	859.940 ± 106.397	23.530 ± 4.276	62.157 ± 7.538
2160	1025.800 ± 246.201	874.823 ± 128.985	26.567 ± 10.321	67.563 ± 16.994
2170	971.197 ± 177.218	980.277 ± 266.499	25.160 ± 7.666	62.303 ± 9.047
2180	1013.170 ± 231.875	989.617 ± 287.219	26.280 ± 10.574	65.657 ± 15.439
2190	1057.760 ± 264.063	971.693 ± 198.237	26.680 ± 8.308	66.013 ± 11.190
2200	991.037 ± 192.711	949.433 ± 218.743	25.733 ± 6.742	65.803 ± 25.952
2210	1097.330 ± 315.212	1065.600 ± 328.178	23.383 ± 0.996	61.707 ± 5.113
2220	1074.390 ± 259.878	1004.140 ± 228.576	31.823 ± 22.166	70.473 ± 21.506
2230	1041.400 ± 252.965	963.613 ± 159.893	25.513 ± 6.330	77.080 ± 64.054
2240	1067.300 ± 279.737	1008.260 ± 182.292	28.123 ± 13.271	66.327 ± 14.678
2250	1078.810 ± 283.195	970.810 ± 159.965	26.157 ± 14.020	72.477 ± 42.327
2260	1026.580 ± 161.270	1018.360 ± 228.340	26.030 ± 6.431	73.800 ± 45.570
2270	1125.990 ± 301.454	1085.130 ± 324.035	27.580 ± 10.588	68.287 ± 16.502
2280	1143.700 ± 277.233	1215.600 ± 344.924	35.880 ± 21.598	78.560 ± 29.735
2290	1202.390 ± 367.143	1051.530 ± 236.922	25.910 ± 5.002	66.247 ± 9.083
2300	1224.120 ± 392.564	1040.680 ± 184.717	29.793 ± 20.951	70.140 ± 17.994
2310	1191.210 ± 313.091	1079.990 ± 215.877	32.787 ± 14.830	69.553 ± 12.283
2320	1265.040 ± 421.130	1267.890 ± 495.331	28.557 ± 8.968	67.523 ± 8.727
2330	1191.680 ± 306.390	1151.250 ± 274.781	28.697 ± 9.388	75.080 ± 44.771
2340	1144.350 ± 301.085	1165.530 ± 291.883	27.983 ± 9.352	69.340 ± 13.689
2350	1265.610 ± 462.550	1220.240 ± 384.461	29.637 ± 10.829	71.853 ± 14.837
2360	1384.570 ± 460.525	1346.350 ± 433.435	34.447 ± 13.377	74.510 ± 11.116
2370	1181.260 ± 274.048	1346.870 ± 463.588	32.143 ± 12.891	76.663 ± 27.221
2380	1269.110 ± 358.145	1221.620 ± 318.891	32.930 ± 15.143	73.450 ± 16.151
2390	1252.620 ± 340.124	1211.530 ± 237.406	31.520 ± 11.979	74.950 ± 19.032
2400	1205.210 ± 270.183	1173.380 ± 285.102	28.823 ± 7.842	80.563 ± 49.557
2410	1225.790 ± 262.150	1080.580 ± 149.903	27.907 ± 5.930	68.463 ± 7.078
2420	1154.720 ± 172.268	1121.360 ± 197.466	27.540 ± 7.543	67.440 ± 6.470
2430	1347.130 ± 338.518	1135.190 ± 171.150	26.170 ± 4.040	70.580 ± 19.500
2440	1148.780 ± 173.965	1163.740 ± 208.698	28.640 ± 7.666	71.027 ± 8.899
2450	1290.360 ± 322.895	1284.560 ± 414.190	36.593 ± 26.853	74.067 ± 13.482
2460	1430.980 ± 506.858	1249.110 ± 330.319	28.067 ± 6.592	70.080 ± 8.574
2470	1274.940 ± 322.897	1205.660 ± 218.771	28.153 ± 5.822	70.607 ± 7.694
2480	1264.560 ± 304.995	1257.160 ± 292.154	27.593 ± 5.273	69.553 ± 6.430

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
2490	1252.570 $\pm$ 308.043	1258.560 $\pm$ 246.264	29.443 $\pm$ 8.719	74.820 $\pm$ 16.743
2500	1211.160 $\pm$ 172.258	1175.680 $\pm$ 181.021	27.560 $\pm$ 5.476	71.843 $\pm$ 8.397
2510	1256.400 $\pm$ 207.510	1175.810 $\pm$ 156.261	28.013 $\pm$ 6.231	71.637 $\pm$ 8.282
2520	1204.300 $\pm$ 173.163	1173.530 $\pm$ 182.926	25.950 $\pm$ 1.371	69.903 $\pm$ 6.371
2530	1362.350 $\pm$ 337.179	1290.560 $\pm$ 320.072	28.947 $\pm$ 6.414	71.913 $\pm$ 8.656
2540	1356.860 $\pm$ 320.280	1251.400 $\pm$ 192.726	30.830 $\pm$ 9.706	75.373 $\pm$ 13.066
2550	1279.360 $\pm$ 213.158	1296.560 $\pm$ 233.170	29.273 $\pm$ 6.690	74.690 $\pm$ 13.504
2560	1330.130 $\pm$ 209.595	1255.000 $\pm$ 228.477	31.903 $\pm$ 12.825	77.647 $\pm$ 20.728
2570	1318.850 $\pm$ 242.763	1228.620 $\pm$ 156.395	30.407 $\pm$ 9.210	75.780 $\pm$ 8.763
2580	1386.030 $\pm$ 262.969	1272.480 $\pm$ 246.714	28.683 $\pm$ 5.658	73.197 $\pm$ 8.941
2590	1315.600 $\pm$ 158.437	1325.820 $\pm$ 231.485	29.220 $\pm$ 5.366	76.300 $\pm$ 9.595
2600	1401.820 $\pm$ 262.922	1366.960 $\pm$ 277.991	34.943 $\pm$ 15.873	87.233 $\pm$ 34.895
2610	1369.920 $\pm$ 279.813	1401.200 $\pm$ 358.574	33.673 $\pm$ 11.618	81.320 $\pm$ 20.318
2620	1620.040 $\pm$ 542.580	1510.190 $\pm$ 457.283	33.643 $\pm$ 9.497	82.467 $\pm$ 17.195
2630	1444.610 $\pm$ 363.655	1340.320 $\pm$ 225.164	29.010 $\pm$ 4.522	75.217 $\pm$ 13.498
2640	1447.860 $\pm$ 326.780	1408.040 $\pm$ 252.862	34.947 $\pm$ 11.426	78.633 $\pm$ 12.539
2650	1715.070 $\pm$ 556.107	1758.270 $\pm$ 575.709	36.470 $\pm$ 14.241	86.567 $\pm$ 21.258
2660	1488.710 $\pm$ 415.029	1455.870 $\pm$ 344.347	34.993 $\pm$ 15.704	78.880 $\pm$ 10.856
2670	1369.420 $\pm$ 168.629	1364.350 $\pm$ 239.467	30.663 $\pm$ 6.634	76.083 $\pm$ 6.512
2680	1417.340 $\pm$ 194.030	1378.770 $\pm$ 190.758	30.043 $\pm$ 6.203	78.730 $\pm$ 13.102
2690	1472.640 $\pm$ 302.838	1475.880 $\pm$ 298.562	31.903 $\pm$ 7.594	79.763 $\pm$ 14.300
2700	1500.240 $\pm$ 302.509	1504.310 $\pm$ 453.708	30.687 $\pm$ 5.268	78.103 $\pm$ 7.091
2710	1508.760 $\pm$ 228.149	1501.120 $\pm$ 320.012	39.463 $\pm$ 38.450	81.837 $\pm$ 13.627
2720	1645.110 $\pm$ 452.635	1500.690 $\pm$ 309.000	38.583 $\pm$ 18.960	87.040 $\pm$ 27.223
2730	1559.890 $\pm$ 394.369	1660.570 $\pm$ 435.693	36.670 $\pm$ 14.925	82.837 $\pm$ 11.762
2740	1530.540 $\pm$ 322.302	1467.040 $\pm$ 243.852	30.413 $\pm$ 6.130	77.073 $\pm$ 7.104
2750	1648.340 $\pm$ 389.546	1466.880 $\pm$ 223.006	30.467 $\pm$ 4.978	77.713 $\pm$ 6.708
2760	1587.240 $\pm$ 266.966	1532.320 $\pm$ 303.176	39.360 $\pm$ 36.435	86.593 $\pm$ 29.835
2770	1659.630 $\pm$ 330.527	1587.760 $\pm$ 370.321	38.530 $\pm$ 21.034	89.477 $\pm$ 32.141
2780	1620.580 $\pm$ 327.384	1694.340 $\pm$ 476.660	33.623 $\pm$ 7.329	80.857 $\pm$ 8.071
2790	2134.780 $\pm$ 1355.290	1907.880 $\pm$ 743.432	50.657 $\pm$ 58.735	106.683 $\pm$ 78.868
2800	1539.970 $\pm$ 213.349	1526.800 $\pm$ 262.157	34.617 $\pm$ 10.069	83.270 $\pm$ 12.707
2810	1628.070 $\pm$ 337.499	1520.290 $\pm$ 287.434	32.543 $\pm$ 7.838	80.843 $\pm$ 10.047
2820	1546.160 $\pm$ 200.851	1592.240 $\pm$ 288.847	31.483 $\pm$ 6.006	80.857 $\pm$ 10.057
2830	1604.050 $\pm$ 244.059	1560.360 $\pm$ 236.231	35.847 $\pm$ 14.700	89.557 $\pm$ 33.021
2840	1589.150 $\pm$ 309.920	1556.090 $\pm$ 264.048	31.830 $\pm$ 5.741	82.217 $\pm$ 12.906
2850	1638.270 $\pm$ 309.342	1646.660 $\pm$ 384.179	30.827 $\pm$ 4.508	80.970 $\pm$ 20.304
2860	1839.200 $\pm$ 416.290	1573.480 $\pm$ 300.780	31.673 $\pm$ 3.235	85.137 $\pm$ 26.445
2870	1553.530 $\pm$ 171.157	1834.400 $\pm$ 385.633	36.767 $\pm$ 14.524	90.463 $\pm$ 24.522
2880	1756.560 $\pm$ 405.075	1799.260 $\pm$ 399.556	34.097 $\pm$ 6.982	93.913 $\pm$ 26.579
2890	1943.640 $\pm$ 453.614	1852.140 $\pm$ 476.865	38.110 $\pm$ 10.148	94.250 $\pm$ 26.292
2900	1807.450 $\pm$ 306.483	1869.100 $\pm$ 414.667	48.100 $\pm$ 34.341	88.713 $\pm$ 12.873
2910	1915.710 $\pm$ 459.926	1814.180 $\pm$ 342.157	38.243 $\pm$ 12.864	92.467 $\pm$ 23.497
2920	1906.640 $\pm$ 429.473	1899.290 $\pm$ 493.112	40.577 $\pm$ 31.482	99.223 $\pm$ 55.266
2930	1872.100 $\pm$ 370.280	1828.750 $\pm$ 372.404	36.313 $\pm$ 14.779	89.397 $\pm$ 20.677
2940	1833.160 $\pm$ 343.926	1755.520 $\pm$ 303.084	39.223 $\pm$ 11.673	86.493 $\pm$ 10.649
2950	1796.470 $\pm$ 270.581	1804.490 $\pm$ 370.567	39.997 $\pm$ 16.995	92.730 $\pm$ 25.246
2960	1803.980 $\pm$ 392.529	1774.980 $\pm$ 249.046	37.810 $\pm$ 18.837	88.830 $\pm$ 16.723
2970	1760.170 $\pm$ 291.784	1668.490 $\pm$ 223.183	35.957 $\pm$ 11.830	86.367 $\pm$ 9.333
2980	1739.300 $\pm$ 312.489	1782.670 $\pm$ 324.802	36.983 $\pm$ 8.911	88.873 $\pm$ 17.710

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
2990	1900.270 $\pm$ 464.689	1875.010 $\pm$ 401.499	38.433 $\pm$ 12.444	99.033 $\pm$ 45.756
3000	1809.040 $\pm$ 351.828	1799.000 $\pm$ 343.440	40.470 $\pm$ 37.132	90.657 $\pm$ 24.695
3010	2112.940 $\pm$ 447.761	2139.740 $\pm$ 588.507	57.260 $\pm$ 97.226	98.050 $\pm$ 19.659
3020	2002.910 $\pm$ 526.016	2053.840 $\pm$ 458.053	40.977 $\pm$ 12.382	96.807 $\pm$ 21.166
3030	2193.150 $\pm$ 639.703	2207.890 $\pm$ 638.664	41.123 $\pm$ 13.648	93.813 $\pm$ 16.865
3040	2077.260 $\pm$ 473.588	1932.400 $\pm$ 504.364	45.030 $\pm$ 29.928	91.937 $\pm$ 16.416
3050	1874.400 $\pm$ 342.828	1768.810 $\pm$ 237.976	35.913 $\pm$ 8.379	91.813 $\pm$ 15.896
3060	2060.650 $\pm$ 489.665	2029.420 $\pm$ 560.650	36.050 $\pm$ 7.657	94.340 $\pm$ 28.550
3070	2088.800 $\pm$ 565.459	2104.450 $\pm$ 520.436	35.477 $\pm$ 5.819	91.880 $\pm$ 13.102
3080	2088.940 $\pm$ 433.775	2032.150 $\pm$ 520.226	37.663 $\pm$ 9.677	98.500 $\pm$ 42.672
3090	1970.050 $\pm$ 474.687	2002.390 $\pm$ 387.537	36.707 $\pm$ 11.192	92.430 $\pm$ 26.170
3100	2110.210 $\pm$ 486.898	2117.590 $\pm$ 531.102	37.707 $\pm$ 8.484	90.587 $\pm$ 8.719
3110	2208.460 $\pm$ 541.509	2071.980 $\pm$ 394.786	36.740 $\pm$ 7.249	96.467 $\pm$ 27.571
3120	2073.730 $\pm$ 381.190	2140.730 $\pm$ 479.304	37.847 $\pm$ 10.445	93.740 $\pm$ 11.880
3130	2014.460 $\pm$ 300.922	2027.550 $\pm$ 407.183	43.783 $\pm$ 32.010	97.707 $\pm$ 25.443
3140	2179.040 $\pm$ 575.999	2203.120 $\pm$ 681.739	44.243 $\pm$ 19.507	101.500 $\pm$ 37.970
3150	2103.380 $\pm$ 430.623	2081.130 $\pm$ 465.444	41.130 $\pm$ 17.394	98.690 $\pm$ 20.868
3160	2108.010 $\pm$ 411.143	1975.450 $\pm$ 386.834	40.593 $\pm$ 18.820	93.433 $\pm$ 20.057
3170	2094.730 $\pm$ 387.967	2050.590 $\pm$ 428.855	40.037 $\pm$ 9.107	97.220 $\pm$ 23.198
3180	2152.860 $\pm$ 471.173	2075.000 $\pm$ 394.660	42.200 $\pm$ 14.840	97.590 $\pm$ 16.349
3190	2110.390 $\pm$ 428.244	2158.640 $\pm$ 517.582	40.247 $\pm$ 16.572	94.323 $\pm$ 11.829
3200	2086.440 $\pm$ 410.091	2019.530 $\pm$ 291.216	36.690 $\pm$ 5.794	93.600 $\pm$ 8.626
3210	2108.880 $\pm$ 412.486	2120.930 $\pm$ 367.669	40.143 $\pm$ 16.467	100.707 $\pm$ 22.629
3220	2152.260 $\pm$ 334.633	2139.190 $\pm$ 405.655	41.090 $\pm$ 11.697	95.967 $\pm$ 15.940
3230	2162.840 $\pm$ 511.378	2183.610 $\pm$ 488.078	36.670 $\pm$ 5.200	97.120 $\pm$ 24.783
3240	2301.930 $\pm$ 479.041	2146.640 $\pm$ 471.557	39.957 $\pm$ 9.319	94.827 $\pm$ 10.396
3250	2114.030 $\pm$ 337.888	2111.600 $\pm$ 303.815	46.587 $\pm$ 45.646	103.633 $\pm$ 39.727
3260	2205.580 $\pm$ 438.049	2200.620 $\pm$ 484.876	39.137 $\pm$ 10.050	97.230 $\pm$ 14.893
3270	2132.500 $\pm$ 362.578	2063.470 $\pm$ 308.880	38.957 $\pm$ 12.334	94.400 $\pm$ 16.563
3280	2120.470 $\pm$ 283.673	2015.000 $\pm$ 246.618	38.640 $\pm$ 9.904	92.583 $\pm$ 8.449
3290	2143.190 $\pm$ 320.898	2072.540 $\pm$ 287.685	36.467 $\pm$ 6.983	97.047 $\pm$ 14.643
3300	2227.640 $\pm$ 426.474	2144.840 $\pm$ 373.199	36.823 $\pm$ 5.955	94.683 $\pm$ 8.195
3310	2189.420 $\pm$ 288.945	2108.120 $\pm$ 288.813	50.757 $\pm$ 60.957	99.100 $\pm$ 28.340
3320	2282.650 $\pm$ 360.018	2242.280 $\pm$ 394.858	39.387 $\pm$ 8.655	99.977 $\pm$ 16.402
3330	2393.750 $\pm$ 520.316	2284.700 $\pm$ 504.642	41.143 $\pm$ 11.633	97.833 $\pm$ 10.430
3340	2312.000 $\pm$ 362.046	2178.010 $\pm$ 261.213	40.343 $\pm$ 15.781	103.160 $\pm$ 18.674
3350	2254.320 $\pm$ 365.049	2282.140 $\pm$ 544.454	38.700 $\pm$ 9.911	99.690 $\pm$ 16.660
3360	2261.400 $\pm$ 431.991	2248.690 $\pm$ 268.855	38.507 $\pm$ 8.934	99.000 $\pm$ 13.111
3370	2202.800 $\pm$ 238.759	2240.310 $\pm$ 334.408	44.423 $\pm$ 22.196	100.310 $\pm$ 11.408
3380	2317.690 $\pm$ 374.548	2321.420 $\pm$ 438.212	40.197 $\pm$ 8.992	97.087 $\pm$ 9.528
3390	2316.530 $\pm$ 344.931	2300.150 $\pm$ 351.883	40.113 $\pm$ 9.060	98.363 $\pm$ 13.976
3400	2454.100 $\pm$ 442.159	2378.220 $\pm$ 524.008	40.250 $\pm$ 10.240	105.020 $\pm$ 17.121
3410	2405.150 $\pm$ 501.335	2492.660 $\pm$ 546.869	39.267 $\pm$ 8.187	101.187 $\pm$ 17.165
3420	2304.490 $\pm$ 317.242	2281.990 $\pm$ 313.260	41.303 $\pm$ 12.676	107.270 $\pm$ 26.227
3430	2340.630 $\pm$ 478.426	2264.760 $\pm$ 265.007	39.803 $\pm$ 9.598	101.787 $\pm$ 13.612
3440	2297.600 $\pm$ 302.381	2245.740 $\pm$ 318.796	42.787 $\pm$ 13.138	99.367 $\pm$ 12.199
3450	2310.760 $\pm$ 305.205	2228.220 $\pm$ 235.527	39.173 $\pm$ 8.578	98.963 $\pm$ 10.608
3460	2492.620 $\pm$ 566.520	2514.870 $\pm$ 511.689	41.013 $\pm$ 10.393	101.880 $\pm$ 14.673
3470	2380.140 $\pm$ 479.916	2281.100 $\pm$ 363.415	41.290 $\pm$ 10.760	102.557 $\pm$ 18.970
3480	2345.680 $\pm$ 319.620	2254.530 $\pm$ 229.170	42.093 $\pm$ 21.647	108.143 $\pm$ 42.559

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
3490	2578.190 $\pm$ 497.463	2350.720 $\pm$ 308.923	41.257 $\pm$ 8.930	99.723 $\pm$ 8.585
3500	2397.010 $\pm$ 339.887	2408.160 $\pm$ 354.485	40.700 $\pm$ 8.174	99.023 $\pm$ 8.618
3510	2472.740 $\pm$ 455.254	2388.120 $\pm$ 303.000	41.420 $\pm$ 9.687	100.663 $\pm$ 11.657
3520	2414.850 $\pm$ 305.537	2445.260 $\pm$ 295.314	46.647 $\pm$ 18.762	101.787 $\pm$ 12.937
3530	2450.590 $\pm$ 280.113	2399.680 $\pm$ 310.494	40.060 $\pm$ 7.937	101.380 $\pm$ 10.981
3540	2447.460 $\pm$ 288.329	2473.480 $\pm$ 375.658	41.147 $\pm$ 10.452	106.030 $\pm$ 33.054
3550	2492.180 $\pm$ 351.288	2460.940 $\pm$ 332.448	40.400 $\pm$ 6.941	104.387 $\pm$ 12.222
3560	2429.770 $\pm$ 262.172	2457.600 $\pm$ 302.458	42.873 $\pm$ 13.605	108.530 $\pm$ 26.097
3570	2542.460 $\pm$ 414.582	2458.840 $\pm$ 332.067	39.307 $\pm$ 5.949	102.013 $\pm$ 10.207
3580	2574.500 $\pm$ 501.061	2543.710 $\pm$ 487.468	43.143 $\pm$ 9.144	106.550 $\pm$ 19.996
3590	2462.970 $\pm$ 296.686	2530.240 $\pm$ 346.025	39.447 $\pm$ 5.403	100.907 $\pm$ 9.276
3600	2569.570 $\pm$ 368.129	2682.250 $\pm$ 411.574	45.653 $\pm$ 21.196	109.180 $\pm$ 24.711
3610	2764.750 $\pm$ 488.713	2694.500 $\pm$ 417.092	43.517 $\pm$ 9.553	107.433 $\pm$ 17.389
3620	2662.040 $\pm$ 439.409	2656.160 $\pm$ 405.506	40.250 $\pm$ 3.453	109.963 $\pm$ 25.689
3630	2962.030 $\pm$ 682.959	2661.720 $\pm$ 461.131	44.323 $\pm$ 10.124	113.220 $\pm$ 31.266
3640	2736.770 $\pm$ 547.536	2627.690 $\pm$ 373.830	46.340 $\pm$ 28.163	114.213 $\pm$ 39.955
3650	2729.430 $\pm$ 452.396	2648.700 $\pm$ 399.575	46.927 $\pm$ 30.298	110.383 $\pm$ 24.791
3660	2864.070 $\pm$ 513.299	2714.180 $\pm$ 435.122	49.993 $\pm$ 33.435	108.427 $\pm$ 17.972
3670	2843.470 $\pm$ 489.497	2905.520 $\pm$ 562.474	41.570 $\pm$ 6.205	106.760 $\pm$ 14.944
3680	2889.940 $\pm$ 491.563	2871.060 $\pm$ 473.471	47.533 $\pm$ 18.976	113.310 $\pm$ 22.026
3690	2960.020 $\pm$ 484.819	2844.590 $\pm$ 505.276	45.047 $\pm$ 10.328	123.150 $\pm$ 46.438
3700	2755.100 $\pm$ 462.915	2881.290 $\pm$ 555.788	42.873 $\pm$ 9.083	113.243 $\pm$ 23.561
3710	2810.850 $\pm$ 393.435	2738.130 $\pm$ 364.492	47.670 $\pm$ 20.912	110.063 $\pm$ 16.337
3720	2783.400 $\pm$ 395.700	2761.700 $\pm$ 333.510	60.740 $\pm$ 53.572	124.057 $\pm$ 55.124
3730	2950.560 $\pm$ 501.431	2760.810 $\pm$ 622.521	43.863 $\pm$ 12.037	109.613 $\pm$ 10.965
3740	2831.170 $\pm$ 388.731	2767.060 $\pm$ 438.306	49.747 $\pm$ 30.953	118.503 $\pm$ 54.883
3750	3209.150 $\pm$ 689.082	3004.080 $\pm$ 547.686	46.920 $\pm$ 15.866	116.390 $\pm$ 31.205
3760	3001.130 $\pm$ 545.595	2881.320 $\pm$ 541.230	45.843 $\pm$ 10.685	112.813 $\pm$ 17.850
3770	2701.770 $\pm$ 223.013	2745.490 $\pm$ 356.493	42.910 $\pm$ 9.485	108.143 $\pm$ 12.405
3780	2699.330 $\pm$ 308.430	2725.250 $\pm$ 318.281	41.740 $\pm$ 5.062	108.480 $\pm$ 14.773
3790	2849.460 $\pm$ 402.842	2709.440 $\pm$ 301.902	44.957 $\pm$ 10.101	110.603 $\pm$ 10.605
3800	2900.300 $\pm$ 419.896	2728.430 $\pm$ 300.123	43.703 $\pm$ 7.852	111.547 $\pm$ 20.504
3810	3068.440 $\pm$ 558.930	3001.770 $\pm$ 588.975	47.227 $\pm$ 17.025	115.580 $\pm$ 25.328
3820	2882.890 $\pm$ 468.357	2858.500 $\pm$ 348.205	41.253 $\pm$ 5.453	110.677 $\pm$ 15.783
3830	2959.940 $\pm$ 392.373	2894.820 $\pm$ 421.072	47.850 $\pm$ 16.079	123.360 $\pm$ 49.707
3840	2929.990 $\pm$ 412.576	2854.170 $\pm$ 320.228	42.017 $\pm$ 5.579	110.027 $\pm$ 11.636
3850	2944.470 $\pm$ 389.684	2930.920 $\pm$ 364.270	43.787 $\pm$ 9.921	111.567 $\pm$ 11.841
3860	2958.860 $\pm$ 333.603	2911.550 $\pm$ 297.773	48.543 $\pm$ 18.077	117.430 $\pm$ 18.113
3870	2951.260 $\pm$ 361.922	2926.680 $\pm$ 476.724	47.353 $\pm$ 16.069	115.357 $\pm$ 17.365
3880	2922.980 $\pm$ 321.449	3051.230 $\pm$ 467.407	45.753 $\pm$ 13.119	113.717 $\pm$ 13.014
3890	3002.470 $\pm$ 322.281	2893.320 $\pm$ 297.319	48.467 $\pm$ 15.683	125.167 $\pm$ 46.395
3900	3066.780 $\pm$ 359.734	3063.190 $\pm$ 658.007	47.547 $\pm$ 12.611	113.553 $\pm$ 12.406
3910	3001.380 $\pm$ 377.389	2992.550 $\pm$ 435.293	42.173 $\pm$ 3.775	113.737 $\pm$ 14.920
3920	3307.270 $\pm$ 526.943	3034.430 $\pm$ 399.197	42.890 $\pm$ 4.748	112.840 $\pm$ 7.904
3930	3012.430 $\pm$ 357.529	3389.180 $\pm$ 516.463	55.470 $\pm$ 24.217	126.690 $\pm$ 31.814
3940	3244.390 $\pm$ 537.856	3071.130 $\pm$ 466.400	43.330 $\pm$ 4.003	113.610 $\pm$ 12.855
3950	3243.010 $\pm$ 508.692	3199.610 $\pm$ 467.443	43.743 $\pm$ 6.303	119.440 $\pm$ 30.311
3960	3216.880 $\pm$ 484.881	3338.840 $\pm$ 561.398	43.453 $\pm$ 4.245	114.873 $\pm$ 9.037
3970	3266.450 $\pm$ 484.061	3452.680 $\pm$ 599.828	49.457 $\pm$ 9.699	114.820 $\pm$ 9.893
3980	2967.040 $\pm$ 257.634	3767.120 $\pm$ 322.167	51.897 $\pm$ 21.089	122.500 $\pm$ 17.820

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
3990	3255.230 $\pm$ 564.460	3480.590 $\pm$ 434.379	50.273 $\pm$ 19.242	124.943 $\pm$ 42.504
4000	3334.560 $\pm$ 542.722	3195.160 $\pm$ 371.870	52.790 $\pm$ 21.953	133.080 $\pm$ 38.013
4010	3176.350 $\pm$ 440.980	3248.640 $\pm$ 418.414	52.343 $\pm$ 23.631	122.737 $\pm$ 31.965
4020	3418.830 $\pm$ 506.090	3092.250 $\pm$ 437.566	47.117 $\pm$ 13.255	118.190 $\pm$ 21.607
4030	3967.330 $\pm$ 1023.790	3529.800 $\pm$ 795.472	53.117 $\pm$ 12.563	131.533 $\pm$ 37.872
4040	3244.680 $\pm$ 717.245	3472.910 $\pm$ 385.759	50.867 $\pm$ 18.775	129.473 $\pm$ 28.031
4050	3171.150 $\pm$ 355.112	3036.570 $\pm$ 247.036	43.057 $\pm$ 2.692	116.863 $\pm$ 14.503
4060	3091.400 $\pm$ 367.755	3124.510 $\pm$ 285.481	46.073 $\pm$ 7.743	121.243 $\pm$ 34.855
4070	3080.700 $\pm$ 272.088	3225.780 $\pm$ 373.132	43.277 $\pm$ 1.227	116.193 $\pm$ 9.349
4080	3740.550 $\pm$ 910.118	3825.510 $\pm$ 915.122	60.213 $\pm$ 37.568	160.503 $\pm$ 124.888
4090	4369.850 $\pm$ 1492.760	4383.570 $\pm$ 2328.860	68.370 $\pm$ 42.789	149.437 $\pm$ 70.397
4100	3457.140 $\pm$ 426.021	3061.840 $\pm$ 211.794	45.673 $\pm$ 7.937	115.153 $\pm$ 8.821
4110	3536.100 $\pm$ 511.174	3372.630 $\pm$ 461.309	47.380 $\pm$ 7.407	124.567 $\pm$ 46.110
4120	3649.570 $\pm$ 417.282	3242.240 $\pm$ 421.546	45.337 $\pm$ 6.377	123.900 $\pm$ 27.436
4130	4006.880 $\pm$ 617.641	3876.290 $\pm$ 820.690	56.490 $\pm$ 17.963	141.443 $\pm$ 47.546
4140	3309.500 $\pm$ 472.344	3162.850 $\pm$ 272.911	49.967 $\pm$ 12.532	125.080 $\pm$ 21.053
4150	3206.870 $\pm$ 264.346	3423.430 $\pm$ 360.353	51.810 $\pm$ 17.518	123.650 $\pm$ 22.820
4160	3592.950 $\pm$ 458.716	3522.790 $\pm$ 495.536	48.013 $\pm$ 7.685	118.097 $\pm$ 7.887
4170	3848.840 $\pm$ 1228.660	4034.800 $\pm$ 955.336	54.623 $\pm$ 16.625	130.263 $\pm$ 22.549
4180	3276.570 $\pm$ 306.682	3257.770 $\pm$ 294.561	50.333 $\pm$ 13.109	118.503 $\pm$ 9.170
4190	3373.730 $\pm$ 358.736	3292.500 $\pm$ 317.103	46.837 $\pm$ 9.061	119.137 $\pm$ 23.735
4200	3282.700 $\pm$ 474.807	3532.190 $\pm$ 330.072	45.013 $\pm$ 2.530	118.930 $\pm$ 6.625
4210	3846.850 $\pm$ 549.689	3811.730 $\pm$ 491.089	55.067 $\pm$ 17.165	135.237 $\pm$ 30.995
4220	4133.700 $\pm$ 622.953	4104.620 $\pm$ 879.883	67.363 $\pm$ 50.619	140.707 $\pm$ 39.875
4230	3868.450 $\pm$ 538.744	3773.780 $\pm$ 536.737	59.957 $\pm$ 22.784	136.473 $\pm$ 39.005
4240	3522.930 $\pm$ 378.240	3870.010 $\pm$ 601.517	48.537 $\pm$ 8.864	125.120 $\pm$ 14.357
4250	3550.020 $\pm$ 472.985	3620.180 $\pm$ 393.349	47.277 $\pm$ 7.203	123.390 $\pm$ 11.417
4260	3589.480 $\pm$ 435.713	3530.080 $\pm$ 402.699	60.703 $\pm$ 26.534	135.250 $\pm$ 33.859
4270	3725.910 $\pm$ 541.233	3654.390 $\pm$ 643.523	50.760 $\pm$ 12.010	127.873 $\pm$ 21.144
4280	4073.300 $\pm$ 782.008	3823.070 $\pm$ 564.709	49.983 $\pm$ 9.562	131.583 $\pm$ 17.852
4290	3858.110 $\pm$ 816.442	3836.210 $\pm$ 647.693	57.797 $\pm$ 17.937	132.857 $\pm$ 20.989
4300	3589.400 $\pm$ 351.166	3536.160 $\pm$ 358.424	51.427 $\pm$ 13.784	122.667 $\pm$ 14.301
4310	3536.630 $\pm$ 403.712	3663.400 $\pm$ 359.095	53.733 $\pm$ 17.947	129.027 $\pm$ 17.901
4320	3749.830 $\pm$ 404.859	3877.590 $\pm$ 421.840	52.213 $\pm$ 11.833	134.203 $\pm$ 44.532
4330	3901.620 $\pm$ 437.128	3897.070 $\pm$ 422.154	55.757 $\pm$ 14.648	133.640 $\pm$ 23.710
4340	4077.530 $\pm$ 592.558	3979.820 $\pm$ 629.637	63.643 $\pm$ 30.149	136.883 $\pm$ 35.469
4350	3980.890 $\pm$ 592.037	4036.780 $\pm$ 437.009	54.810 $\pm$ 13.779	141.127 $\pm$ 61.588
4360	4446.210 $\pm$ 712.936	4186.480 $\pm$ 680.994	55.590 $\pm$ 19.287	137.253 $\pm$ 35.999
4370	4445.410 $\pm$ 1024.970	4286.130 $\pm$ 785.047	61.213 $\pm$ 20.371	138.487 $\pm$ 29.206
4380	3897.740 $\pm$ 475.941	3703.250 $\pm$ 394.512	53.553 $\pm$ 11.486	127.483 $\pm$ 12.061
4390	3817.040 $\pm$ 447.959	3731.860 $\pm$ 383.113	53.260 $\pm$ 21.320	133.397 $\pm$ 30.262
4400	3782.460 $\pm$ 471.840	3839.490 $\pm$ 400.247	56.107 $\pm$ 24.461	138.377 $\pm$ 41.628
4410	3858.990 $\pm$ 372.970	3689.810 $\pm$ 390.358	56.340 $\pm$ 22.425	134.533 $\pm$ 29.144
4420	3835.390 $\pm$ 343.496	3819.100 $\pm$ 326.368	55.317 $\pm$ 18.167	138.847 $\pm$ 35.696
4430	4003.500 $\pm$ 478.416	4109.480 $\pm$ 666.568	57.370 $\pm$ 24.843	133.927 $\pm$ 25.113
4440	4288.750 $\pm$ 708.434	4275.600 $\pm$ 652.948	59.120 $\pm$ 20.341	144.823 $\pm$ 42.179
4450	4249.040 $\pm$ 467.659	4224.220 $\pm$ 595.972	53.240 $\pm$ 11.345	139.443 $\pm$ 40.589
4460	4066.450 $\pm$ 575.387	4069.170 $\pm$ 448.928	64.713 $\pm$ 33.549	150.433 $\pm$ 54.141
4470	4134.440 $\pm$ 662.539	3984.310 $\pm$ 405.805	53.300 $\pm$ 13.798	129.283 $\pm$ 14.870
4480	4114.050 $\pm$ 457.823	4023.710 $\pm$ 442.528	54.737 $\pm$ 11.830	129.453 $\pm$ 12.310

Size (N)	First (μs)	Last (μs)	Middle (μs)	Random (μs)
4490	4300.280 ± 788.525	4525.120 ± 941.083	55.740 ± 16.824	134.820 ± 17.371
4500	4467.790 ± 745.998	4173.990 ± 742.291	59.133 ± 19.406	145.200 ± 33.041
4510	4017.110 ± 531.271	3898.370 ± 374.562	51.147 ± 8.765	135.087 ± 29.470
4520	4166.860 ± 708.640	4036.780 ± 611.665	56.260 ± 16.747	130.377 ± 10.912
4530	4339.580 ± 763.346	4310.390 ± 566.094	57.690 ± 15.989	144.977 ± 68.772
4540	4371.380 ± 527.537	4412.430 ± 583.144	61.517 ± 21.085	148.617 ± 49.927
4550	4458.960 ± 558.030	4494.060 ± 646.832	54.977 ± 14.845	134.520 ± 17.590
4560	4584.660 ± 826.102	4399.850 ± 710.781	56.623 ± 13.624	143.170 ± 27.820
4570	4602.330 ± 804.878	4699.130 ± 760.663	63.690 ± 38.678	148.400 ± 47.722
4580	4392.350 ± 514.756	4380.970 ± 537.079	64.327 ± 27.209	144.860 ± 43.324
4590	4115.770 ± 381.381	4166.850 ± 386.667	51.833 ± 10.033	136.750 ± 28.949
4600	4445.150 ± 877.029	4452.060 ± 844.226	56.583 ± 16.209	141.897 ± 26.300
4610	4321.130 ± 534.745	4338.010 ± 490.449	57.230 ± 19.993	136.530 ± 13.709
4620	4659.260 ± 1082.380	4499.400 ± 873.257	63.753 ± 27.411	144.357 ± 34.700
4630	4426.850 ± 490.094	4434.340 ± 534.565	54.327 ± 10.241	133.950 ± 9.899
4640	4518.920 ± 515.941	4392.960 ± 575.227	52.180 ± 9.669	144.833 ± 36.095
4650	4499.450 ± 570.251	4534.140 ± 452.768	61.617 ± 30.569	141.023 ± 27.637
4660	4460.340 ± 601.328	4440.110 ± 793.677	58.080 ± 15.710	138.500 ± 17.631
4670	4326.680 ± 630.796	4323.970 ± 578.641	54.237 ± 10.365	138.120 ± 22.042
4680	4275.050 ± 503.313	4286.590 ± 424.895	59.287 ± 14.430	138.967 ± 17.200
4690	4697.910 ± 827.485	4584.190 ± 763.800	64.213 ± 27.045	140.240 ± 18.939
4700	4366.720 ± 589.553	4330.810 ± 417.600	53.220 ± 9.278	135.697 ± 13.197
4710	4551.040 ± 550.516	4382.320 ± 472.758	57.867 ± 13.052	152.630 ± 55.799
4720	4497.660 ± 851.969	4215.520 ± 478.634	57.060 ± 18.381	146.887 ± 38.435
4730	4319.550 ± 436.799	4353.260 ± 406.099	54.217 ± 12.903	139.690 ± 18.224
4740	4959.420 ± 1676.900	4762.020 ± 1034.970	57.850 ± 13.453	143.670 ± 25.148
4750	4654.440 ± 547.355	4943.900 ± 1376.030	66.637 ± 26.084	155.683 ± 49.166
4760	4403.330 ± 376.993	4359.120 ± 373.355	53.583 ± 7.950	135.460 ± 11.083
4770	4582.780 ± 414.927	4709.810 ± 570.832	60.647 ± 26.662	147.817 ± 25.641
4780	4694.480 ± 551.792	4725.220 ± 540.777	66.540 ± 28.098	150.007 ± 28.256
4790	4562.880 ± 534.227	4615.380 ± 476.213	65.100 ± 43.781	139.810 ± 14.479
4800	4861.210 ± 516.053	4735.980 ± 551.732	60.073 ± 18.443	153.370 ± 31.672
4810	4601.500 ± 414.764	4583.180 ± 542.240	62.117 ± 27.376	145.790 ± 37.059
4820	4723.590 ± 496.831	4683.360 ± 517.275	58.990 ± 20.998	150.377 ± 48.128
4830	4870.930 ± 978.933	4828.550 ± 637.455	58.863 ± 23.597	145.193 ± 23.766
4840	4697.200 ± 436.258	4603.810 ± 424.938	56.607 ± 9.947	141.623 ± 36.815
4850	4412.760 ± 322.875	4637.010 ± 360.339	64.183 ± 29.188	145.720 ± 17.587
4860	4566.540 ± 530.476	4501.020 ± 428.237	57.653 ± 12.762	139.687 ± 14.246
4870	4702.860 ± 445.692	4691.690 ± 477.894	60.403 ± 19.137	143.847 ± 24.029
4880	5240.290 ± 744.001	5159.600 ± 719.602	65.447 ± 48.194	153.847 ± 55.161
4890	4966.610 ± 610.585	5051.290 ± 484.545	54.610 ± 6.648	143.490 ± 13.055
4900	4845.800 ± 470.814	4728.090 ± 475.645	57.210 ± 13.784	145.083 ± 18.113
4910	5121.750 ± 784.710	5453.310 ± 1059.670	71.213 ± 30.420	156.583 ± 30.167
4920	4863.430 ± 489.561	4976.380 ± 421.143	65.497 ± 23.860	156.447 ± 40.521
4930	5284.030 ± 642.041	5312.510 ± 509.353	72.243 ± 29.396	172.283 ± 53.890
4940	5164.870 ± 813.835	5081.240 ± 529.378	58.577 ± 12.203	144.897 ± 15.676
4950	4994.750 ± 589.593	5132.220 ± 898.728	59.473 ± 13.985	148.660 ± 20.669
4960	6486.030 ± 1566.750	6298.600 ± 1277.870	72.110 ± 23.824	157.070 ± 21.024
4970	6664.270 ± 1439.030	7057.530 ± 1255.100	84.327 ± 29.347	174.537 ± 37.771
4980	6435.360 ± 1374.500	6780.680 ± 1111.520	74.860 ± 29.541	172.133 ± 33.430

Size (N)	First (μ s)	Last (μ s)	Middle (μ s)	Random (μ s)
4990	6508.900 $\pm$ 867.419	6412.880 $\pm$ 948.084	73.560 $\pm$ 19.262	184.590 $\pm$ 52.846
5000	10575.000 $\pm$ 2249.230	10972.500 $\pm$ 2222.310	134.820 $\pm$ 64.123	277.293 $\pm$ 99.333